

Thin Sets Are Not Equally Thin: Minimax Learning of Submanifold Integrals*

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Abstract

Many economic parameters are identified by “thin sets” (submanifolds with Lebesgue measure zero) and hence difficult to recover from data in an ambient space. This paper provides a unified theory for estimation and inference of such “thin-set” identified functionals. We show that thin sets are *not* equally thin: their intrinsic dimensionality m matters in a precise manner. For a nonparametric regression h_0 with Hölder smoothness s and d -dimensional covariates in the ambient space, we show that $n^{-\frac{s}{2s+d-m}}$ is the minimax optimal rate of estimating linear and nonlinear (e.g., quadratic, upper contour) integrals of h_0 on an m -dimensional submanifold ($0 \leq m < d$), which is the fastest possible attainable rate among all estimators. The minimax lower bound rate result is generalized to estimating submanifold integrals when h_0 is a nonparametric density and a nonparametric instrumental variable function. The asymptotic normality of t statistics is established via sieve Riesz representation, and the corresponding inference is computed using Sobol points.

Keywords: submanifold, contour set, level set, Hausdorff measure, differential geometry, minimax rate, sieve Riesz representer, asymptotic normality.

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1 Introduction

Many parameters of interest in economics are identified by information contained in lower-dimensional *thin sets*, i.e., subsets of the covariate space that have Lebesgue measure zero yet may still carry economic meanings. [Khan and Tamer \(2010\)](#) coined the term *thin-set identification* to describe settings in which identifying information is concentrated on such measure-zero sets, and showed that the resulting parameters are *irregular* in the sense that they cannot be estimated at the parametric $n^{-1/2}$ -rate, where n denotes the sample size.

In this paper, we provide a more nuanced and, in some sense, refined view about *thin sets*. While all thin sets possess Lebesgue measure zero in their ambient space, we show that they may differ substantively in terms of their intrinsic dimensions and geometric structures. These refined differences lead to quantitatively different convergence rate of estimation and different forms of first-order expansions for inference on aggregate parameters (integrals) defined on such thin sets.

Specifically, this paper considers semiparametric estimation and inference for general integral functionals on submanifolds of the following form:

$$\Gamma(h_0) := \int_{\mathcal{M}} \phi(h_0(x), x) w(x) d\mathcal{H}^m(x), \quad (1)$$

where $\phi : \mathbb{R} \times \mathcal{X} \mapsto \mathbb{R}$ is a known transformation of an unknown function h_0 that can be estimated nonparametrically from data, $w : \mathcal{X} \mapsto \mathbb{R}$ is a known (weight) function, and $\mathcal{X}$ is a known closed convex subset (with positive Lebesgue measure) in $\mathbb{R}^d$. Here $\mathcal{H}^m$ denotes the m -dimensional Hausdorff measure in $\mathbb{R}^d$, which coincides with the Lebesgue measure in $\mathbb{R}^m$ (and hence has Lebesgue measure zero in $\mathbb{R}^d$ whenever $m < d$).¹ In this paper,

$$\mathcal{M} := \{x \in \mathcal{X} : g(x) = \mathbf{0}\} \quad (2)$$

denotes an m -dimensional ($0 \leq m < d$) submanifold, where $g : \mathcal{X} \mapsto \mathbb{R}^{d-m}$ may be known, or unknown but can be estimated parametrically, semiparametrically or nonparametrically from data. By definition, the m -dimensional submanifold $\mathcal{M}$ has positive m -dimensional Hausdorff volume but zero d -dimensional Lebesgue volume, and hence is a thin set in the covariate ambient space $\mathcal{X} \subset \mathbb{R}^d$.

Submanifold integrals emerge naturally in many economic settings where we take

¹See Section 2 for a precise definition of $\mathcal{H}^m$. For $m < d$, the m -dimensional Hausdorff measure $\mathcal{H}^m$ can be thought as a generalization of a variety of “uniform” measures for lower-dimensional subsets in $\mathbb{R}^d$ such as point, line, area, surface, and volume measures.

derivatives of a standard Lebesgue integral (dx in $\mathbb{R}^d$), such as an expectation, with respect to a changing domain of integration:

$$\left. \frac{d}{dt} \int_{\Omega_t} \omega_t(x) dx \right|_{t=0} = \left. \frac{d}{dt} \int_{\mathbb{R}^d} \mathbb{1}\{x \in \Omega_t\} \omega_t(x) dx \right|_{t=0}. \quad (3)$$

Mathematically, the derivative of an integral with respect to its domain Ω_t translates into an integral over the *boundary* $\partial\Omega_t$ of its domain, and the boundary is often a lower-dimensional submanifold of the original domain. As we show in Section 2, many parameters of interest are identified by solutions to first-order condition for optimization over subpopulation, which often take the form (3).

Differentiation of an integral with respect to its domain also shows up in asymptotic analysis. For example, consider estimation of the following integral functional on upper contour set of the unknown h_0

$$V(h_0) := \int \mathbb{1}\{h_0(x) \geq 0\} w(x) dx = \int_{\{x \in \mathcal{X} : h_0(x) \geq 0\}} w(x) dx, \quad (4)$$

where $w(x)$ is the marginal density of x with support $\mathcal{X} \subset \mathbb{R}^d$. When $h_0(x) = \text{CATE}(x) := \mathbb{E}[Y_i(1) - Y_i(0) | X_i = x]$ is the conditional average treatment effect (CATE) for type $x \in \mathcal{X}$, we call $V(h_0)$ a value functional, which is a policy parameter of interest in many applications. If h_0 is nonparametrically estimated by $\hat{h}$, the impact of the estimation error on the plug-in estimation of $V(h_0)$ can be analyzed using the pathwise derivative of $V(h) - V(h_0)$ with respect to h_0 in the direction $[h - h_0]$, which becomes a submanifold integral of the form:

$$DV(h_0)[h - h_0] := \left. \frac{dV(h_0 + t[h - h_0])}{dt} \right|_{t=0} = \int_{\mathcal{M}_0} \frac{h(x) - h_0(x)}{\|\nabla_x h_0(x)\|} w(x) d\mathcal{H}^{d-1}(x),$$

where the level set² $\mathcal{M}_0 := \{x \in \mathcal{X} : h_0(x) = 0\}$ is a $m = (d - 1)$ -dimensional submanifold in $\mathbb{R}^d$.

In this paper, we assume that a data set $\mathcal{D}_n := \{(Y_i, X_i)\}_{i=1}^n$ is a random sample of size n drawn from an unknown probability distribution of (Y, X) , in which the unknown density of X is bounded away from zero and infinity on its support $\mathcal{X}$ in $\mathbb{R}^d$. We also assume that the unknown function $h_0 : \mathcal{X} \mapsto \mathbb{R}$ and/or the unknown submanifold mapping $g : \mathcal{X} \mapsto \mathbb{R}^{d-m}$ can be identified and estimated from the data $\mathcal{D}_n$ in the ambient space.³ Our parameters of interest are the m -dimensional submanifold

²In this paper we use $\mathcal{M}_0$ to highlight the level set when it is an unknown function h_0 .

³Our framework is different from the literature that assumes the data is sampled directly from lower dimensional submanifolds.

integrals $\Gamma(h_0)$ and the upper contour integrals $V(h_0)$. In Section 2 we show that many economic functionals of interest can be represented as integrals of the forms $\Gamma(h_0)$ and $V(h_0)$.

We first establish the minimax lower bound rates of estimation for the linear integral functional on a submanifold:

$$L(h_0) := \int_{\mathcal{M}} h_0(x) w(x) d\mathcal{H}^m(x), \quad (5)$$

a core special case of $\Gamma(h_0)$ in (1) with $\phi = h_0$ and $\mathcal{M}$ a known m -dimensional submanifold ($m < d$). For the sake of concreteness we assume that the unknown function $h_0 : \mathcal{X} \mapsto \mathbb{R}$ belongs to a Hölder class of finite smoothness $s > 0$. When h_0 is respectively a nonparametric regression $\mathbb{E}[Y|X]$, a nonparametric density of X and a nonparametric instrumental variables (NPIV) regression $\mathbb{E}[Y - h_0(X)|Z] = 0$, we establish the minimax lower bound rates for estimating $L(h_0)$, which are the *fastest* possible convergence rates among all estimators for $L(h_0)$. These lower bound rates are all *slower* than the parametric convergence rate of $n^{-1/2}$ whenever $m < d$, confirming that $L(h_0)$ is an *irregular* functional whenever $m < d$. Specifically, we show that $r_n^* := n^{-\frac{s}{2s+d-m}}$ is the minimax lower bound rate for estimating $L(h_0)$ when h_0 is a nonparametric regression and a density. Interestingly, this rate coincides with the famous lower bound rate of Stone (1980) for the pointwise estimation of a nonparametric regression with $(d-m)$ -dimensional covariates. In Subsection 3.2 we also establish that $r_{NPIV,n}$ is the minimax lower bound rate for estimating $L(h_0)$ when h_0 satisfies a NPIV restriction but does not need to be point-identified. Importantly, the rate $r_{NPIV,n}$ for $L(h_0)$ coincides with the previously established minimax lower bound rate for estimating a $(d-m)$ -dimensional point-identified NPIV function (Chen and Reiss (2011)).

Under some mild regularity conditions, we show that the minimax lower bound rates for $L(h_0)$ are also the minimax lower bounds rates for nonlinear integrals $\Gamma(h_0)$ with possibly *unknown* submanifolds, upper contour integrals $V(h_0)$, and surface integrals of the form

$$S(h_0) := \int_{\{x \in \mathcal{X}: h_0(x)=c\}} w(x) d\mathcal{H}^{d-1}(x). \quad (6)$$

For example, when h_0 is a nonparametric regression or a density, the rate $r_n^* = n^{-\frac{s}{2s+d-m}}$ is the minimax lower bound rate for estimating nonlinear integral $\Gamma(h_0)$ on m -dimensional possibly unknown submanifolds. In particular, when $m = d - 1$

the minimax rate r_n^* becomes $n^{-\frac{s}{2s+1}}$, which is the *fastest* possible convergence rate for estimating $V(h_0)$, $S(h_0)$ and $\Gamma(h_0)$ on $m = (d - 1)$ -dimensional $\mathcal{M}$ among all estimators. This result recovers the minimax lower bound of Horowitz (1993) for smooth maximum score estimation that corresponds to an $m = (d - 1)$ -dimensional parametric submanifold $\{x \in \mathcal{X} : x'\beta_0 = 0\}$.⁴

We then show that the above lower bound rates are attainable, and hence minimax-optimal, by presenting sieve-based estimators for $\theta_0 = L(h_0)$, $\Gamma(h_0)$ and $V(h_0)$ for the case when h_0 is a nonparametric regression with Hölder smoothness s and d -dimensional covariates. For the linear integral $L(h_0)$, we consider the plug-in sieve estimator. For the nonlinear integrals $\Gamma(h_0)$ and $V(h_0)$, we consider plug-in, split-sample and leave-one-out sieve estimators. We provide low level sufficient conditions under which they achieve the optimal rate $r_n^* = n^{-\frac{s}{2s+d-m}}$, with the smoothness requirement for the split-sample and leave-one-out debiased estimators weaker than that for the plug-in estimators for $\Gamma(h_0)$ and $V(h_0)$.

Given the irregularity of submanifold integral functionals, they do not admit well-defined Riesz representers; however, the *sieve* Riesz representers remain well-defined and computable in closed form. Following Chen, Liao and Sun (2014); Chen and Liao (2014) and Chen and Pouzo (2015), we construct valid confidence intervals via sieve student- t statistics. By exploiting the submanifold structure, we characterize the growth rate of the sieve Riesz representer norm and obtain tighter control of nonlinear remainders. For the upper contour integral $V(h_0)$, its pathwise derivatives are computed using the calculus of moving submanifolds.

Monte Carlo simulations confirm our theoretical results: the sieve estimators produce reasonably small RMSEs that shrink with sample sizes, and the realized confidence interval coverage is close to the nominal 95% level. The submanifold integrals in the simulations are numerically computed using Sobol quasi-random sequences (Sobol, 1967) for its better numerical performance than uniform random sampling.

In a companion paper Chen, Chen and Gao (2025a), we apply the theory developed here to inference on value functionals of the CATE under first-best nonparametric treatment assignment. Using the Job Training Partnership Act (JTPA) data set, we compute confidence intervals for the nonparametric first-best welfare and the treatment share for the JTPA job training program, two parameters estimated in

⁴Our minimax lower bound proof follows the nonparametric literature such as Tsybakov (2009), which differs from the proof of Horowitz (1993) for his smooth maximum score model.

Kitagawa and Tetenov (2018) without reported confidence intervals.

Related Literature

Our paper contributes to the literature on semiparametric estimation and inference on irregular integral functionals. To our best knowledge, our paper is the first to provide a unified theory on general submanifold integral functionals of an unknown function h_0 with Hölder smoothness $s > 0$. We establish the minimax-optimal estimation rate for linear and nonlinear submanifold integral functionals of h_0 and for integrals of upper contour set on h_0 , in which h_0 could be a regression, a density, and a NPIV function. In addition, we provide simple asymptotic normality based confidence intervals for these irregular integral functionals using sieve Riesz representation.

Our minimax lower bound estimation rate results can be viewed as quantitative refinements of the famous singular semiparametric information bound results of Chamberlain (1986) and Khan and Tamer (2010) on “thin-set” identified parameters. In Section 2 we show many semiparametric “thin-set” identified parameters can be viewed as integral functionals on m -dimensional submanifolds (for $m < d$). Previously, Kim and Pollard (1990) shows that the first-order condition for the maximum score criterion of Manski (1975) is a submanifold integral with a $m = (d - 1)$ -dimensional hyperplane, and derives the famous $n^{-1/3}$ convergence rate using empirical process theory.⁵ Sasaki (2015) also notes that differentiation with respect to the domain of integration produces a $m = (d - 1)$ -dimensional submanifold integrals in his identification paper on quantile functions in nonseparable structural models.

We view our minimax lower bound results for the large class of submanifold functionals $L(h_0)$, $\Gamma(h_0)$, $V(h_0)$ and $S(h_0)$ *important*, as they provide a minimax informational criterion to compare many different machine learning estimators. Our paper presents various sieve estimators for $L(h_0)$, $\Gamma(h_0)$ and $V(h_0)$ and establishes that they can attain the minimax lower bound rate of $r_n^* = n^{-\frac{s}{2s+d-m}}$ when h_0 is a nonparametric regression. Other estimators for these functionals can also be presented and be verified if they are minimax rate optimal. For example, Qiao (2021) proposes a kernel plug-in density estimator of the surface integral $S(h_0)$ of the form (6) and establishes a convergence rate $n^{-s/(2s+1)}$ when $s \geq d+1$. Qiao (2021) concludes his paper by stating that “Another open problem is the minimax rates of estimating the surface integrals

⁵Horowitz (1992, 1993) obtains the upper and lower bound rates of $n^{-s/(2s+1)}$ for the smoothed maximum score estimator without mentioning $(d - 1)$ -dimensional submanifold.

on level sets". Notice that the surface integral $S(h_0)$ is a $m = (d-1)$ -dimensional non-linear submanifold functional, his kernel estimator matches our minimax lower bound rate r_n^* provided $s \geq d+1$. In works that are concurrent to ours, Cattaneo, Titiunik and Yu (2025a,b) consider estimation of linear integrals over $m = 1$ -dimensional known submanifolds that arise in the boundary discontinuity designs using local polynomial regressions and obtain a minimax optimal convergence rate of $n^{-\frac{s}{2s+d-1}}$ in their contexts (they mostly consider $d = 2, m = 1$). Due to the lack of space, we leave it to future work to compare finite sample performance of different minimax rate-optimal estimators for these submanifold integrals.

Technically, Qiao (2021), Cattaneo, Titiunik and Yu (2025a,b) and our paper all utilize some mathematical tools in differential geometry and geometric measure theory to establish the asymptotic properties of different estimators of different *irregular* submanifold integrals. Differential geometry tools have also been used in asymptotic analysis of *regular* functionals (i.e., the ones that can be estimated at the parametric convergence rate of $n^{-1/2}$). Chernozhukov, Fernández-Val and Luo (2018) uses integrals on level sets to study sorted partial effects in heterogeneous coefficient models and establishes the convergence rate of $n^{-1/2}$ for their regular functionals. Feng, Hong and Nekipelov (2025) shows how Hausdorff integrals can be used in the analysis of regular integral functionals.

Organization of the Paper Section 2 provides motivating examples for submanifold integrals in econometrics. Section 3 establishes minimax lower bounds rates for estimating linear and nonlinear submanifold integrals $L(h_0)$, $\Gamma(h_0)$ when h_0 is a nonparametric regression, a nonparametric density and a NPIV function respectively. Section 4 shows that the minimax lower bound rate is attainable by sieve estimators for estimating $L(h_0)$, $\Gamma(h_0)$ and $V(h_0)$ when h_0 is a nonparametric regression. Section 5 provides the asymptotic normality of the sieve estimators proposed in Section 4, along with consistency sieve variance estimators, for inference on both linear and nonlinear submanifold integrals. Section 6 presents Monte Carlo simulation results. The Appendix contains sections on mathematical tools in differential geometry used in this paper, technical lemmas, as well as all the proofs.

2 Setup and Examples

2.1 Setup

Throughout this paper, we let $\{Y_i, X_i\}_{i=1}^n$ be a random sample of size n drawn from a unknown joint probability distribution $P_{(Y,X)}$, where Y_i is a scalar-valued outcome variable, and X_i is a vector of observed covariates with a convex and compact support $\mathcal{X} \subseteq \mathbb{R}^d$. Let $h_0 : \mathcal{X} \rightarrow \mathbb{R}$ be a nonparametric function⁶ that is directly identified from the data and can be estimated using standard nonparametric estimation methods. A leading example of h_0 is the conditional expectation (nonparametric regression) function $h_0(x) = \mathbb{E}[Y_i | X_i = x]$, which will be our main focus in the paper. That said, h_0 may also take the form of density functions, conditional quantiles and structural regression functions in NPIV models.

We consider submanifolds $\mathcal{M} := \{x \in \mathcal{X} : g(x) = \mathbf{0}\}$ that take the form of level sets of functions g . Depending on the problem setup, g may be known or unknown, parametric or nonparametric, and it may be taken to be different from or the same as h_0 , which will be illustrated in the examples below and treated in subsequent sections. We maintain the following standard regularity condition in this paper:

Assumption 1 (Regular Level Set). *(i) $g : \mathcal{X} \rightarrow \mathbb{R}^{d-m}$ is a continuously differentiable function with $\mathbf{0} \in \text{int}(g(\mathcal{X})) \subseteq \mathbb{R}^{d-m}$; (ii) $\nabla_x g(x)$ has full rank $d - m$ for every $x \in \mathcal{M} := \{x \in \mathcal{X} : g(x) = \mathbf{0}\}$.*⁷

Let $\mathcal{J}_g(x)$ denote the Jacobian of $g : \mathbb{R}^d \rightarrow \mathbb{R}^{d-m}$ defined by

$$\mathcal{J}_g(x) := \sqrt{\sum_{B(x)} \det(B(x))^2} = \sqrt{\det(\nabla_x g(x) \nabla_x g(x)')},$$

where B indexes all $(d - m) \times (d - m)$ minors of $\nabla_x g(x)$. Under Assumption 1 and compactness of $\mathcal{X}$, we have $\mathcal{J}_g(x) \geq \text{Const.} > 0$ for all $x \in \mathcal{M}$.

Under Assumption 1, the set $\mathcal{M}$ defined in (2) is an m -dimensional submanifold of $\mathbb{R}^d$ (e.g. by Theorem 12.1 of Loomis and Sternberg, 2014). We note that $\mathcal{M}$ has zero Lebesgue measure on $\mathbb{R}^d$ for all $0 \leq m < d$, although it has positive m -dimensional Hausdorff measure. In this paper we use $\mathcal{H}^m(x)$ to denote the m -dimensional Hausdorff measure on $\mathbb{R}^d$, which is defined as follows: For a set $A \subseteq \mathbb{R}^d$,

⁶More generally, h_0 may be a vector of nonparametric functions.

⁷Note that $\mathbf{0}$ may be replaced with any other constant vector $\mathbf{c} \in \text{int}(g(\mathcal{X})) \subseteq \mathbb{R}^{d-m}$ without affecting the results in this paper.

define $\mathcal{H}^m(A) := \lim_{\delta \rightarrow 0} \mathcal{H}_\delta^m$, where, for any $\delta \in (0, \infty)$,

$$\mathcal{H}_\delta^m(A) := \inf \left\{ \sum_{j=1}^{\infty} \alpha(m) \left(\frac{\text{diam}(C_j)}{2} \right)^m : A \subseteq \cup_{j=1}^{\infty} C_j, \text{diam}(C_j) \leq \delta \right\},$$

with $\alpha(m) = \frac{\pi^{m/2}}{\Gamma((m/2)+1)} = \frac{\pi^{m/2}}{\int_0^\infty e^{-x} x^{m/2} dx}$. The m -dimensional Hausdorff measure becomes the standard m -dimensional Lebesgue measure in the lower dimensional $\mathbb{R}^m$; see, for example, [Evans and Gariepy \(2015\)](#) for more details on the Hausdorff measure.

2.2 Motivating Examples

In this subsection, we provide some econometric examples for submanifold integrals, which roughly belong to two categories.

Category 1: Examples in which researchers are interested in some aggregate parameters of **estimated or optimized subpopulations**. Then, either the first-order expansion in asymptotic analysis (of estimators), or the first-order condition for optimality, often takes the form of the time derivative of an integral with a changing region of integration Ω_t , which produces a submanifold integral term by the generalized Leibniz rule:⁸ under regularity conditions, (3) can be expressed as

$$\left. \frac{d}{dt} \int_{\Omega_t} \omega_t(x) dx \right|_{t=0} = \underbrace{\left. \int_{\Omega_t} \frac{\partial}{\partial t} \omega_t(x) dx \right|_{t=0}}_{\text{(I)}} + \underbrace{\left. \int_{\partial\Omega_t} \langle \mathbf{n}_t(x), \mathbf{v}_t(x) \rangle \omega_t(x) d\mathcal{H}^{d-1}(x) \right|_{t=0}}_{\text{(II)}} \quad (7)$$

where term (I) captures the effect of the change in the integrand $w_t(x)$ with the region of integration Ω_t held fixed, while term (II) captures the effect of the change in the region of integration Ω_t with the integrand $w_t(x)$ held fixed. Importantly, $\partial\Omega_t$, the boundary of Ω_t , is often a submanifold of dimension $d-1$, and thus term (II) takes the form of an integral over the submanifold $\partial\Omega_t$ with respect to the $(d-1)$ -dimensional Hausdorff measure, which can also be viewed as the surface measure on the boundary submanifold $\partial\Omega_t$. For the integrand terms, $\mathbf{n}_t(x)$ is the outward-pointing unit normal vector, $\mathbf{v}_t(x)$ is the velocity vector associated with the time movement in the $\partial\Omega_t$. For example, consider a simple domain $\Omega_t = [a_t, b_t] \subset \mathbb{R}$, its boundary $\partial\Omega_t$ is a set of two points $\{a_t, b_t\}$, which is the 0-dimensional submanifold, and the 0-dimensional

⁸See, for example, [Theorem 4.2 in Chapter 9 of Delfour and Zolésio \(2011\)](#).

Hausdorff measure is simply the point counting measure. When x is one-dimensional, (7) becomes the standard Leibniz rule for univariate calculus:

$$\left. \frac{d}{dt} \int_{a_t}^{b_t} \omega_t(x) dx \right|_{t=0} = \left. \int_{a_t}^{b_t} \frac{\partial}{\partial t} \omega_t(x) dx \right|_{t=0} + \left(\omega_t(b_t) \frac{d}{dt} b_t - \omega_t(a_t) \frac{d}{dt} a_t \right) \Big|_{t=0}.$$

Throughout this paper we focus on situations where term (II) in (7) is not vanishing.

Category 2: Researchers are interested (for some other reasons than above) in some aggregate parameters of certain **boundary or marginal subpopulations** with the boundary or margin characterized by a lower-dimensional submanifold.

Roughly speaking, Examples 1-5 are of Category 1, Examples 6-8 are of Category 2, while Examples 9-10 can be of both categories. We emphasize that we do not intend the categorization above to be exact nor exhaustive, but more to provide a high-level summary of the origins of submanifold integrals.

Example 1 (Maximum Score Estimation of Binary Choice Models). Consider any model that satisfies the following sign alignment restriction

$$h_0(x) \geq 0 \Leftrightarrow x' \beta_0 \geq 0 \quad (8)$$

where h_0 is a nonparametrically identified and estimable function of x and β_0 is a d -dimensional parameter normalized to lie on the unit sphere, i.e., $\beta_0 \in \mathbb{S}^{d-1} := \{\beta \in \mathbb{R}^d : \|\beta\| = 1\}$. For example, in the following binary choice model with a conditional median independence restriction as in Manski (1975),

$$y_i = \mathbb{1} \{X_i' \beta_0 + \epsilon_i \geq 0\}, \quad \text{med}(\epsilon_i | X_i) = 0,$$

the sign alignment restriction (8) is satisfied with $h_0(x) = \mathbb{E}[Y_i | X_i = x] - \frac{1}{2}$. The population criterion function for maximum score estimator can then be written as

$$W(\beta) := \int h_0(x) \mathbb{1} \{x' \beta \geq 0\} p(x) dx, \quad (9)$$

where $p(\cdot)$ denotes the density of X_i . Under appropriate conditions, β_0 can be point identified under a scale normalization, $\beta_0 = \arg \max_{\beta \in \mathbb{S}^{d-1}} W(\beta)$. By the generalized Leibniz rule, the first order condition (FOC) for the optimality of β_0 is given by

$$\mathbf{0} = \nabla_{\beta} W(\beta_0) := \int_{\mathcal{M}_0} h_0(x) x p(x) d\mathcal{H}^{d-1}(x), \quad (10)$$

where $\mathcal{M} = \{x \in \mathcal{X} : x' \beta_0 = 0\}$ is a $m = (d-1)$ -dimensional hyperplane (see, e.g.,

Kim and Pollard (1990)). We note that the hyperplane $\mathcal{M}_0$ has Lebesgue measure 0 in $\mathbb{R}^d$, which is why the identification of β_0 is referred to as a type of “thin-set identification” in Khan and Tamer (2010). Consequently, the right-hand side of (10) cannot be represented by a regular Lebesgue integral, but instead by a Hausdorff integral over a $m = (d - 1)$ -dimensional manifold (hyperplane) in $\mathbb{R}^d$.

Example 2 (Optimal Linear Treatment Assignment). It has been recognized, say in Kitagawa and Tetenov (2018), that optimal linear treatment assignment problem shares some similarity with maximum score estimation. Specifically, consider the problem of optimizing over a parametric family of treatment assignment rules that assigns the treatment status 0/1 according to $\mathbb{1}\{x'\beta \geq 0\}$ for a given observed type x , where β is a d -dimensional choice parameter. Then the welfare function of the assignment rule parameter β is given by

$$W(\beta) := \int \mathbb{1}\{x'\beta \geq 0\} h_0(x) p(x) dx, \quad (11)$$

where $h_0(x) = \mathbb{E}[Y_i(1) - Y_i(0) | X_i = x]$ is the conditional average treatment effect (CATE) for type x . Hence (11) is of exactly the same form as the maximum score criterion function (9). As a result, the FOC of (11) for the optimal β_0 (under scale normalization) is again given by the submanifold integral (10). Often times, researchers conduct (costly) experiments on and estimate CATE from a sample of moderate sample size, but the target population on which the treatment in question might be implemented can be of a much larger scale. In such settings $p(\cdot)$ may be known or estimable using a much larger sample size than that used to estimate h_0 , and thus we may focus on the estimation error for h_0 in the optimization of $W(\beta)$.

Example 3 (Aggregate Parameter over Estimated Subpopulation). Consider the estimation of the following general welfare or value functional parameter

$$W(h_0, f_0) := \int \mathbb{1}\{h_0(x) \geq 0\} f_0(x) p(x) dx,$$

where h_0 and f_0 may be both unknown but nonparametrically estimable. For example, if we set $h_0(x) = \text{CATE}(x)$ and $f_0(x) = 1$, then $W(h_0, f_0)$ becomes the value functional $V(h_0)$ defined in (4). If we set $h_0(x) = \text{CATE}(x)$ and $f_0(x) = \lambda'x$ for some fixed $\lambda \neq \mathbf{0}$, then $W(h_0, f_0)$ becomes the average characteristics of the subpopulation with nonnegative CATE. If we set $h_0(x) = f_0(x) = \text{CATE}(x)$, then $W(h_0, f_0)$ becomes $W(\text{CATE}, \text{CATE})$ —the welfare under the “first-best” treatment assignment. Alternatively, we may take $f_0(x)$ to be any other value/cost function associated with

type x (see (Chen, Chen and Gao, 2025a)). When h_0 and f_0 are real-valued functions and are nonparametrically estimated by $\hat{h}$ and $\hat{f}$, the pathwise derivative of $W(h_0, f_0)$ with respect to (h_0, f_0) is a key object in the characterization of the asymptotic behaviors of the plug-in estimator $W(\hat{h}, \hat{f})$. Generally, the pathwise derivative of W in the direction of $[h - h_0, f - f_0]$ is given by

$$\begin{aligned} & DW(h_0, f_0)[h - h_0, f - f_0] \\ &:= \lim_{t \searrow 0} \frac{W(h_0 + t(h - h_0), f_0 + t(f - f_0)) - W(h_0, f_0)}{t} \\ &= \int_{\{h_0(x) \geq 0\}} (f(x) - f_0(x)) p(x) dx + \int_{\{h_0(x) = 0\}} (h(x) - h_0(x)) \frac{f_0(x) p(x)}{\|\nabla_x h_0(x)\|} d\mathcal{H}^{d-1}(x) \end{aligned} \quad (12)$$

which consists of two terms: the first is the perturbation of the integrand f_0 in the direction of $[f - f_0]$ over the true region of integral $\{h_0(x) \geq 0\}$, while the second is the perturbation of (the boundary) region of integration induced by the perturbation of h_0 in the direction of $[h - h_0]$. While the first term is standard in the semiparametric estimation literature, the second term takes the nonstandard form of a Hausdorff integral over the $m = (d - 1)$ -dimensional submanifold $\mathcal{M}_0 := \{x \in \mathcal{X} : h_0(x) = 0\}$.⁹ Hence, as long as $f_0(x)p(x)$ does not vanish on the submanifold $\mathcal{M}_0 := \{x \in \mathcal{X} : h_0(x) = 0\}$, the second term in (12) will remain as the leading term in the asymptotic behavior of $W(\hat{h}, \hat{f}) - W(h_0, f_0)$. See our companion paper (Chen, Chen and Gao, 2025a) for a detailed analysis of this example.

Example 4 (Average Treatment Effects under Propensity Score or Density Trimming).¹⁰ Crump, Hotz, Imbens and Mitnik (2009, CHIM thereafter) proposes as a systematic approach to deal with limited overlap problems in the estimation of ATEs, and shows that the optimal subpopulation that minimizes the asymptotic variance of ATE under homoskedastic errors takes the form of propensity score trimming:

$$\text{ATE}_{p\text{-trimmed}} := \mathbb{E}[\text{CATE}(X_i) | \alpha \leq p_0(X_i) \leq 1 - \alpha].$$

In practice, the propensity score $p_0(x)$ might require nonparametric estimation. CHIM did not provide theoretical results on the asymptotic distribution of their pro-

⁹In particular, $\|\nabla_x h_0(x)\|$, which measures the “thinness” of the level set, enters into the derivative formula explicitly here. In the previous examples with hyperplane boundaries, we took $g(x) = x' \beta$ with $\beta \in \mathbb{S}^{d-1}$, and thus $\|\nabla_x g(x)\| = \|\beta\| = 1$, which is why this term becomes implicit in formula (10).

¹⁰We thank Tim Armstrong for suggesting this example and for sending us his note on this.

posed estimators with the first-stage nonparametric estimation error of $p_0(x)$ taken into account. Clearly $\text{ATE}_{p\text{-trimmed}}$ is a two-sided version of Example 3 with the region of integration defined by the two-sided inequality $\alpha \leq p_0(X_i) \leq 1 - \alpha$ on the propensity score function, and the directional derivative of $\text{ATE}_{p\text{-trimmed}}$ with respect to p_0 will again features submanifold integrals on the level sets of $\{x : p_0(x) = \alpha\}$ and $\{x : p_0(x) = 1 - \alpha\}$.

Example 5 (ReLU-Based Generalized Maximum Score Estimator under Multi-Index Single-Crossing Conditions). In a follow-up paper (Chen, Gao and Wen, 2025c), we investigate a ReLU-based generalization of the maximum score estimator for multi-index single-crossing condition (MISC) models (Gao and Li, 2024). Formally, consider a random sample $(Y_i, X_i)_{i=1}^n$ where Y_i is an outcome with support $\mathcal{Y} \subseteq \mathbb{R}^{d_y}$ and $X_i := (X_{i1}, \dots, X_{iJ}) \in \mathbb{R}^{d \times J}$ is a $d \times J$ random matrix with support $\mathcal{X} \subset \mathbb{R}^{d \times J}$. Let $h_0 : \mathcal{X} \rightarrow \mathbb{R}$ be a real-valued functional of the conditional distribution of Y_i given X_i that is directly identified and nonparametrically estimable from the data, and $\beta_0 \in \mathbb{S}^{d-1}$ be a finite-dimensional parameter. We say that (h_0, β_0) satisfies the (*multi-index single-crossing*) MISC condition if, for all $x = (x_1, \dots, x_J) \in \mathcal{X}$,

$$\begin{aligned} x'_j \beta_0 &> 0, \forall j = 1, \dots, J &\Rightarrow h_0(x) &\geq 0, \\ x'_j \beta_0 &< 0, \forall j = 1, \dots, J &\Rightarrow h_0(x) &\leq 0. \end{aligned} \quad (13)$$

The condition is said to be *strict* if the inequalities on the right-hand side of (13) are strict, i.e., $h_0(x) > 0$ whenever $x'_j \beta_0 > 0$ for all j , and $h_0(x) < 0$ whenever $x'_j \beta_0 < 0$ for all j . Note that, when $J = 1$, (13) reduces exactly to the sign-alignment restriction in the original maximum score formulation (Manski, 1975, 1985). However, the maximum score criterion function in (Manski, 1975, 1985) does not generalize to multi-index settings with $J \geq 2$.

Chen, Gao and Wen (2025c) proposes a novel criterion function the encodes the sign restrictions in the MISC condition framework using ReLU functions (instead of indicator functions as Manski's maximum score criterion function). Specifically, given $h : \mathcal{X} \rightarrow \mathbb{R}$ and $\beta \in \mathbb{S}^{d-1}$, define

$$\begin{aligned} g_{+, \beta, h}(x) &:= \left[h(x) - \min_{1 \leq j \leq J} (-x'_j \beta) \right]_+, \\ g_{-, \beta, h}(x) &:= \left[-h(x) - \min_{1 \leq j \leq J} (x'_j \beta) \right]_+. \end{aligned} \quad (14)$$

Chen, Gao and Wen (2025c) then constructs the following criterion function

$$W(\beta) := W_+(\beta) + W_-(\beta), \quad W_{\pm}(\beta) := \mathbb{E}[g_{\pm, \beta, h_0}(X_i)].$$

which is shown to ensure $\beta_0 \in \arg \max_{\beta \in \mathbb{S}^{d-1}} W(\beta)$ under the MISC condition. Chen, Gao and Wen (2025c) then considers two estimation procedures: a two-step procedure in which a first-stage nonparametric estimator of h_0 is plugged into (14) as well as a joint machine learning procedure by formulating the criterion W as a special layer in a deep neural network (DNN) training problem. On a theoretical level, Chen, Gao and Wen (2025c) shows that the MISC criterion $W(\beta)$ also features “thin-set” identification with information concentrated on a $(d-1)$ -dimensional submanifold that piecewisely takes the form of a hyperplane defined by $\{x'_j \beta_0 = 0\}$ across $j = 1, \dots, J$. Then the submanifold integral results in this paper are utilized to establish the convergence rate and asymptotic normality of the ReLU-based generalized maximum score estimator in Chen, Gao and Wen (2025c).

Example 6 (Generalized Partial Means and Nonparametric Regression with Generated Covariates). Suppose that we are interested in the conditional mean of $\psi(Y_i, X_i)$ given $g(X_i) = c \in \mathbb{R}^{d-m}$, where ψ is some known transformation of Y_i and X_i :

$$\mathbb{E}[\psi(Y_i, X_i) | g(X_i) = c] = \int_{\{g(x)=c\}} h_0(x) w_0(x) d\mathcal{H}^m(x)$$

with $h_0(x) := \mathbb{E}[\psi(Y_i, X_i) | X_i = x]$, $w_0(x) := p_0(x | g(x) = c) = \frac{p_0(x)}{\mathcal{J}_g(x) p_{g(X)}(c)}$, $p_0(x)$ being the density of X_i , and $p_{g(X)}$ is the density of $g(X_i)$. In particular, when g extracts a subvector of $x \in \mathbb{R}^d$, say, $g(x) = (x_1, \dots, x_{d-m})$ for some $1 \leq m < d$, the above specializes to the partial mean functional of the form $\mathbb{E}[Y_i | X_{i,1} = c_1, \dots, X_{i,d-m} = c_{d-m}]$ as studied in Newey (1994). The weighted boundary average treatment effect considered in Cattaneo, Titiunik and Yu (2025a) is another example with the level set function g given by a known distance function. When g is nonparametrically estimated, the above can be also viewed as the nonparametric regression function with nonparametrically generated covariates as considered in Mammen, Rothe and Schienle (2012).¹¹

Example 7 (Marginal Treatment Effects and Policy Relevant Treatment Effects).

¹¹Mammen, Rothe and Schienle (2012) develops the asymptotic theory for generated covariates using kernel first-stage estimators, and establishes that, under appropriate conditions, the estimation errors from the first-stage are “smoothed out” partially, which can be viewed as a special case of the rate acceleration result developed in our paper.

Heckman and Vytlacil (2005, 2007) proposes the marginal treatment effect (MTE) as a unifying concept that underlies a wide variety of treatment effects studied in causal inference and program evaluations. It is well-known that MTE can be written as a derivative with respect to the propensity score:

$$\begin{aligned} \text{MTE}(x, p) &:= \mathbb{E}[Y_i(1) - Y_i(0) | X_i = x, p_0(Z_i) = p], \\ &= \frac{\partial}{\partial p} \mathbb{E}[Y_i | X_i = x, p_0(Z_i) = p] \end{aligned} \quad (15)$$

where $p_0(z) := \mathbb{E}[D_i = 1 | Z_i = z]$ is the propensity score (for the binary treatment $D \in \{0, 1\}$) given the instruments Z_i , $f_0(z|x)$ is the conditional density of Z_i given the observed covariates $X_i = x$, and Y_i denotes the observed outcome. It is clear from (15) that MTE takes the form of the derivative of a submanifold integral, where the submanifold is given by the propensity score level set $p_0(z) = p$. Here, both the CATE function and the propensity score function can be nonparametrically estimated.

As argued in Carneiro, Heckman and Vytlacil (2010), in many scenarios it is useful to consider incremental policy reforms and focus on the analysis of marginal policy changes, whose effects are usually concentrated on individuals “at the margin”. The average marginal treatment effect (AMTE) summarizes the mean benefit of the treatment for the subpopulation who is indifferent between participation in the treatment and nonparticipation. Formally, this “at-the-margin” population $\{(z, u) \in \mathbb{R}^{d_z+1} : p_0(z) - u = 0\}$ is a submanifold defined by the level set of the propensity score function p , and the average over this subpopulation can again be represented by a submanifold integral in the form of

$$\text{AMTE}(x) := \int_{\{p_0(z)=u\}} \text{MTE}(x, u) f_0(z|x) d\mathcal{H}_{(z,u)}^{d_z}.$$

In addition, MTE is also used to define a wide range of causal parameters, notably the policy-relevant treatment effect (PRTE) proposed in Heckman and Vytlacil (2001). Let F_P denote the CDF of $P_i := p_0(Z_i)$ and suppose that a proposed policy changes this CDF from F_P to $\tilde{F}$. Then the PRTE is defined by

$$\text{PRTE}(x, \tilde{F}) := \int_0^1 \text{MTE}(x, u) \omega_{\text{PRTE}}(u, \tilde{F}) du, \quad \omega_{\text{PRTE}}(u, \tilde{F}) := \frac{F_P(u) - \tilde{F}(u)}{\mathbb{E}_{\tilde{F}}(P_i) - \mathbb{E}_{F_P}(P_i)}.$$

Since MTE is defined as a derivative, PRTE takes the form of an average derivative parameter, for which a parametric estimation rate of $n^{-1/2}$ is possible under a certain “vanishing-on-boundary” condition (see Example 8 for a related discussion).

Carneiro, Heckman and Vytlacil (2010) pointed out that this type of “vanishing-on-boundary” condition may be violated in plausible economic policy scenarios in the analysis of PRTE, which makes it not estimable at a parametric rate $n^{-1/2}$ in general. Our theoretical results can be useful in scenarios where “vanishing-on-boundary” conditions are not imposed in estimation and testing PRTE in various applications.

Example 8 (Weighted Average Derivatives). Related to Example 7 above, suppose that we are interested in the following weighted average derivative of h_0 :

$$\text{WAD}(h_0) := \int \nabla_x h_0(x) w(x) dx.$$

For example, Powell, Stock and Stoker (1989, PSS thereafter) considers a density-weighted average derivative, which is a special case of the above with $w(x) := p^2(x)$, with $p(x)$ denoting the marginal density of x on its support $\mathcal{X}$. The standard analysis of $\text{WAD}(\hat{h})$ exploits the following integration-by-parts formula (a special case of the Divergence Theorem):

$$\int_{\mathcal{X}} \nabla_x h_0(x) w(x) dx = \int_{\partial\mathcal{X}} \vec{\mathbf{n}}(x) h_0(x) w(x) d\mathcal{H}^{d-1}(x) - \int_{\mathcal{X}} h_0(x) \nabla_x w(x) dx, \quad (16)$$

where the first term is a Hausdorff integral over the submanifold defined by the support boundary $\partial\mathcal{X}$. The standard analysis of $\text{WAD}(h_0)$, such as in PSS and Newey and Stoker (1993), exploits a “vanishing-on-boundary” assumption, say, $h_0(x)w(x) = 0$ for all $x \in \partial\mathcal{X}$, so that the first term in (16) degenerates to 0. For example, PSS assumes that $p(x) = 0$ on $\partial\mathcal{X}$ (Assumption 2 in their paper), and hence $\text{WAD}(h_0) = -2 \int h_0(x) \nabla_x p(x) p(x) dx$. PSS then proceeds to establish that $\sqrt{n}(\text{WAD}(\hat{h}) - \text{WAD}(h_0)) = O_p(1)$ and is asymptotic normal. As clear from (16), the parametric convergence rate of $n^{-1/2}$ relies crucially on the “vanishing on boundary” assumption, without which the leading asymptotic term in $\text{WAD}(\hat{h}) - \text{WAD}(h_0)$ would be the first term (the submanifold integral) that converges at a slower-than- $n^{-1/2}$ rate, as established in our paper.

Example 9 (Structural Functions in NPIV Regression). The previous examples focus on *exogenous* nonparametric regression functions. Here we present a canonical example of a nonparametric function with endogeneity: the structural functions in the nonparametric instrumental variables (NPIV) model, i.e., the h_0 function below:

$$Y_i = h_0(X_i) + \epsilon_i, \quad \mathbb{E}[\epsilon_i | Z_i] = 0.$$

Previous work by, for example, Chen and Pouzo (2015), Chen and Christensen (2018),

and [Chen, Christensen and Kankanala \(2025b\)](#), provides theoretical results on the estimation and inference on h_0 , as well as various (families of) linear and nonlinear functionals of h_0 , including point evaluation, average derivatives/elasticities, and consumer surplus/welfare functionals. In principle, the h_0 function in all the submanifold functionals defined in Examples 1-6 above could be replaced by the structural function in the NPIV model here, and the theory about submanifold integrals developed in the current paper would still be relevant.

Example 10 (Nonparametric Quantile). Similarly, the nonparametric function h_0 need not be restricted as a conditional expectation function; it can be broadly defined as any estimable nonparametric function, for example, nonparametric quantile and density functions. These general nonparametric estimation problems (without endogeneity) have been studied widely in statistics, econometrics, and beyond in various settings. In the presence of endogeneity issues, [Horowitz and Lee \(2007\)](#), [Chen and Pouzo \(2015\)](#), and [Chen, Pouzo and Powell \(2019\)](#) have also provided theoretical results for the nonparametric quantile IV regression. The established theoretical results from these previous studies can be combined with the theory of submanifold integrals developed here for the study of new functional parameters of interest.

In summary, we hope that the above examples illustrate the need for estimation and inference on general submanifold integrals of unknown functions, which has not been systematically studied in the existing literature. In the rest of the paper, we shall focus on the case where h_0 is a nonparametric regression function, and provide a general analysis of the estimation and inference for linear submanifold integrals (5), nonlinear submanifold integrals (1) whose first-order linear approximation takes the form of (5), as well as integrals on upper contour set of h_0 (4). We present additional results when h_0 is a nonparametric density function and a nonparametric instrumental variables regression in the Appendix.

3 Minimax Rate of Estimation: Lower Bound

We first establish minimax lower bound rates for estimation of linear ($L(h_0)$) and nonlinear ($\Gamma(h_0)$) submanifold integrals when h_0 is a nonparametric regression $\mathbb{E}[Y|X]$ or a nonparametric density of X in Section 3.1, and establish similar lower bound results when h_0 is an NPIV structural function in Subsection 3.2.

Throughout this paper we assume that h_0 belongs to Hölder class of functions. We now recall the definition of Hölder class of functions. Let $\mathcal{X} = \mathcal{X}_1 \times \dots \times \mathcal{X}_d$ be the Cartesian product of compact intervals $\mathcal{X}_1, \dots, \mathcal{X}_d$, say, $\mathcal{X} = [0, 1]^d$ for simplicity. A real-valued function h on $\mathcal{X}$ is said to satisfy a Hölder condition with exponent $\gamma \in (0, 1]$ if there is a positive number c such that $|h(x) - h(y)| \leq c \|x - y\|^\gamma$ for all $x, y \in \mathcal{X}$; here $\|x - y\| = \left(\sum_{l=1}^d x_l^2\right)^{1/2}$ is the Euclidean norm of $x = (x_1, \dots, x_d) \in \mathcal{X}$. Given a d -tuple $\alpha = (\alpha_1, \dots, \alpha_d)$ of nonnegative integers, set $[\alpha] = \alpha_1 + \dots + \alpha_d$ and let ∇^α denote the differential operator defined by

$$\nabla^\alpha := \nabla_x^\alpha = \frac{\partial^{[\alpha]}}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}}.$$

Let $\lfloor s \rfloor$ be a nonnegative integer that is smaller than s , and set $s = \lfloor s \rfloor + \gamma$ for some $\gamma \in (0, 1]$. A real-valued function h on $\mathcal{X}$ is said to be s -smooth if it is $\lfloor s \rfloor$ times continuously differentiable on $\mathcal{X}$ and $\nabla^\alpha h$ satisfies a Hölder condition with exponent γ for all α with $[\alpha] = \lfloor s \rfloor$. Denote the class of all s -smooth real-valued functions on $\mathcal{X}$ by $\Lambda^s(\mathcal{X})$ (called a Hölder class), and the space of all $\lfloor s \rfloor$ -times continuously differentiable real-valued functions on $\mathcal{X}$ by $C^{\lfloor s \rfloor}(\mathcal{X})$. Define a Hölder ball with smoothness $s = \lfloor s \rfloor + \gamma$ as

$$\Lambda_c^s(\mathcal{X}) = \left\{ h \in C^{\lfloor s \rfloor}(\mathcal{X}) : \sup_{[\alpha] \leq \lfloor s \rfloor} \sup_{x \in \mathcal{X}} |\nabla^\alpha h(x)| \leq c, \sup_{[\alpha] = \lfloor s \rfloor} \sup_{x, y \in \mathcal{X}, x \neq y} \frac{|\nabla^\alpha h(x) - \nabla^\alpha h(y)|}{\|x - y\|^\gamma} \leq c \right\}.$$

Remark 1. If $h_0 \in \Lambda^s$ (for Hölder smooth $s > 0$), then $h_0|_{\mathcal{M}}$ is pointwise well-defined and hence the submanifold integrals $L(h_0)$ and $\Gamma(h_0)$ are well-defined. If h_0 is only assumed to belong to a Sobolev/Besov space of smoothness $s > 0$ but not sup-norm bounded, then one needs to impose a Sobolev/Besov trace condition $s > (d - m)/2$ to ensure that $L(h_0)$ and $\Gamma(h_0)$ are well defined. This is why we assume that $h_0 \in \Lambda^s$ for $s > 0$ instead of a Sobolev/Besov space of smoothness $s > (d - m)/2$ in this paper.

In the proofs of the minimax lower bound rate results, especially in the proofs of lower bounds for nonlinear submanifold integrals with possibly unknown submanifolds that could depend on h_0 , we assume that the submanifold function g is Hölder smooth.

Assumption 2 (Smoothness of Submanifolds). $g \in \Lambda^{s_1}(\mathcal{X})$ for some $s_1 \geq \max\{1, s\}$.

We wish to stress that the minimax lower bound rates established in our paper are still valid lower bound rates without imposing the extra smoothness Assumption 2. Nevertheless, Assumption 2 is satisfied in typical economics applications with unknown submanifolds.

3.1 Case 1: h_0 is a Regression or a Density

In Subsection 3.1.1, we first establish a minimax rate lower bound for estimating *linear* integrals on submanifolds. In Subsection 3.1.2 we then present a general rate lower bound for estimating general *nonlinear* integrals over submanifolds whose first derivatives take the form of the linear integrals on submanifolds.

3.1.1 For Linear Functional $\theta_0 = L(h_0)$

In this subsection we present the minimax lower bound rates for estimation of $\theta_0 = L(h_0)$ when $h_0(\cdot)$ is the unknown regression function $\mathbb{E}[Y_i | X_i = \cdot]$ (a nonparametric regression) or the unknown density of X .

Recall that $L : \Lambda^s \mapsto \mathbb{R}$ is the linear submanifold integral functional

$$L(h) := \int_{\mathcal{M}} h(x) w(x) d\mathcal{H}^m(x), \quad (17)$$

where both the level set function g (consequently the manifold $\mathcal{M}$) and the weight function w are assumed to be known. We first establish lower bounds for the minimax convergence rates for estimating $\theta_0 = L(h_0)$ when h_0 is assumed to belong to the Hölder class of functions with smoothness $s > 0$. The established lower bound rates hold for any possible consistent estimators of $L(h_0)$.

We impose the following standard assumptions on the density of the covariates, the known weight function and the regression error term.

Assumption 3 (Density on $\mathcal{X}$). $\mathcal{X} = [0, 1]^d$, and the density p_0 of X_i on $\mathcal{X}$ is uniformly bounded away from zero and infinity.

Assumption 4 (Weight Regularity). The weight function $w : \mathcal{X} \mapsto \mathbb{R}$ is uniformly bounded on $\mathcal{M}$ and not identically zero on $\mathcal{M}$.

Assumption 5 (Nondegeneracy). $\inf_{x \in \mathcal{X}} \mathbb{E}[\epsilon_i^2 | X_i = x] > 0$ with $\epsilon_i := Y_i - h_0(X_i)$.

Theorem 1 (Lower Bound Rate for $\theta_0 = L(h_0)$ When h_0 is a Regression). Under Assumptions 1, 2, 3, 4 and 5, the rate $r_n^* = n^{-\frac{s}{2s+d-m}}$ is the minimax rate lower bound for the estimation of $\theta_0(P, w) := \int_{\mathcal{M}} h_0(x) w(x) d\mathcal{H}^m(x)$, i.e.

$$\liminf_{n \rightarrow \infty} \inf_{\tilde{\theta}} \sup_{P, w} \mathbb{E}_P \left[n^{\frac{2s}{2s+d-m}} \left(\tilde{\theta} - \theta_0(P, w) \right)^2 \right] \geq c, \quad \text{for some constant } c > 0.$$

where P is any joint probability distribution of (X_i, Y_i) that satisfies $h_0(\cdot) := \mathbb{E}_P[Y_i | X_i = \cdot] \in \Lambda_c^s(\mathcal{X})$ along with Assumptions 3 and 5, w is any weight function satisfying Assumption 4, and $\tilde{\theta}$ is any estimator of $\theta_0 := \theta_0(P, w)$.

Note that the lower bound rate $r_n^* = n^{-\frac{s}{2s+d-m}}$ reproduces several well-known results in the literature as special cases: When $m = d$, the lower bound rate becomes $r_n^* = n^{-\frac{1}{2}}$, reproducing the standard parametric convergence rate for regular (full-dimensional) integral functionals.¹² When $m = 0$, the lower bound becomes $r_n^* = n^{-\frac{s}{2s+d}}$, reproducing the well-known (Stone, 1980) minimax optimal rate for point evaluation functionals of the d -dimensional nonparametric regression.¹³

Theorem 1 can be easily adapted to establish a minimax lower bound for $L(h_0)$ for other nonparametric object h_0 such as nonparametric density function of X or nonparametric quantile regression of Y on X . The following result shows that r_n^* is also a minimax lower bound rate when h_0 is a nonparametric density function of X .

Corollary 1 (Lower Bound Rate for $\theta_0 = L(h_0)$ When h_0 is a Density). *Under Assumptions 1, 3 and 4, the rate $r_n^* = n^{-\frac{s}{2s+d-m}}$ is the minimax rate lower bound for the estimation of $\theta_0(P, w) := \int_{\mathcal{M}} h_0(x) w(x) d\mathcal{H}^m(x)$, i.e.*

$$\liminf_{n \rightarrow \infty} \inf_{\tilde{\theta}_n} \sup_{P, w} \mathbb{E}_P \left[n^{\frac{2s}{2s+d-m}} \left(\tilde{\theta} - \theta_0(P, w) \right)^2 \right] \geq c, \quad \text{for some constant } c > 0.$$

where P is any probability distribution of X_i with its density satisfying $h_0 \in \Lambda_c^s(\mathcal{X})$ and Assumption 3, w is any uniformly bounded weight function, and $\tilde{\theta}_n$ is any estimator of $\theta_0 := \theta_0(P, w)$.

3.1.2 For Nonlinear Functional $\theta_0 = \Gamma(h_0)$

In the previous subsection we focus on the linear integral functionals over a known submanifold, but the key minimax rate $r_n^* = n^{-\frac{s}{2s+d-m}}$ continues to be relevant for a general class of nonlinear submanifold integral functionals.

Specifically, we consider the class of nonlinear integral functionals $\Gamma(h_0)$ whose pathwise derivative takes the form of a linear submanifold integral of form (17).

¹²When $m = d$, the Hausdorff measure $\mathcal{H}^d$ coincides with Lebesgue measure in $\mathbb{R}^d$.

¹³When $m = 0$, the Hausdorff measure $\mathcal{H}^0$ becomes a point counting measure.

Assumption 6 (Linearization). (i) $\Gamma : \mathbb{H} \rightarrow \mathbb{R}$ is pathwise differentiable with

$$D\Gamma(h_0)[v] := \left. \frac{\partial}{\partial t} \Gamma(h_0 + tv) \right|_{t=0} = \int_{\{x:g(x)=\mathbf{0}\}} v(x) \bar{w}(x) \mathcal{H}^m(x), \quad \forall v \in \mathcal{V} := \mathbb{H} - \{h_0\}, \quad (18)$$

for some uniformly bounded function $\bar{w}$ with $\int_{\{x:g(x)=\mathbf{0}\}} \bar{w}^2(x) d\mathcal{H}^m(x) \in (0, \infty)$, some level set function $g : \mathcal{X} \rightarrow \mathbb{R}_{d-m}$ satisfying Assumption 1 uniformly in $h_0 \in \Lambda_c^s$, where both $\bar{w}$ and g may depend on the true (and unknown) h_0 ;

(ii) There is a constant $0 < C_r < 1$ such that for any $u \in \Lambda_c^s$ with small $\|u\|_\infty$,

$$|\Gamma(h_0 + u) - \Gamma(h_0) - D\Gamma(h_0)[u]| \leq C_r |D\Gamma(h_0)[u]|.$$

Assumption 6(ii) is used to prevent the linearization remainder from *canceling* the first-order separation in the lower bound proof. It is easily satisfied by a Taylor series expansion to second order. It is also satisfied if Γ is Fréchet differentiable at h_0 with respect to the sup-norm, i.e., there exists a function $\omega : (0, \infty) \rightarrow (0, \infty)$ with $\omega(t) \rightarrow 0$ as $t \downarrow 0$ such that for all $u \in \Lambda_c^s$ with $\|u\|_\infty \leq t$,

$$|\Gamma(h_0 + u) - \Gamma(h_0) - D\Gamma(h_0)[u]| \leq \omega(t) \|u\|_\infty.$$

Assumption 6 for nonlinear functional $\Gamma(h_0)$ replaces the role of Assumption 4 for the linear functional $L(h_0)$. We emphasize that the level set function g (and thus the submanifold) is *not* necessarily known in Assumption 6. For instance, Examples 3-6 in Section 2.2 feature unknown and estimated submanifolds. Also, the upper contour integral $V(h_0)$ of the form (4) and the surface integral $S(h_0)$ of the form (6) are examples in which $g = h_0$ (nonparametrically defined).

Under Assumption 6, the minimax rate lower bound established in Theorem 1 for linear submanifold integrals continues to be valid for nonlinear submanifold functionals Γ as well. This is because the pathwise derivative $D\Gamma(h_0)[\cdot]$ is a linear submanifold integral with minimax convergence rate given by r_n^* in Theorem 1, and the linearization remainder can only (potentially) slow down the rate. We summarize this observation in the following Corollary.

Corollary 2 (Lower Bound Rate for $\theta_0 = \Gamma(h_0)$ When h_0 is a Regression or a Density). *Let Γ be a potentially nonlinear functional satisfying Assumption 6, and h_0 be a nonparametric regression or a density of X . Let Assumptions 2, 3, and 5 (when h_0 is a regression) hold. Then, a minimax rate lower bound for estimating $\theta_0 := \Gamma(h_0)$*

is given by $r_n^* = n^{-s/(2s+d-m)}$, i.e.,

$$\liminf_{n \rightarrow \infty} \inf_{\tilde{\theta}} \sup_P \mathbb{E}_P \left[n^{\frac{2s}{2s+d-m}} \left(\tilde{\theta} - \theta_0(P) \right)^2 \right] \geq c, \quad \text{for some constant } c > 0.$$

When $m = d-1$, the lower bound in Corollary 2 becomes $r_n^* = n^{-\frac{s}{2s+1}}$ for nonlinear submanifold integrals. This could be viewed as an extension of Horowitz (1993)'s result on the minimax lower bound rate for smoothed maximum score estimation.

3.2 Case 2: When h_0 is an NPIV Structural Function

3.2.1 For Linear Functional $\theta_0 = L(h_0)$

We now consider the rate lower bound for estimating the linear $\theta_0 = L(h_0)$ when h_0 is the structural function in the nonparametric instrumental variable (NPIV) model:

$$Y_i = h_0(X_i) + \epsilon_i, \quad \mathbb{E}[\epsilon_i | Z_i] = 0, \quad (19)$$

where $X_i \in \mathcal{X} \subset \mathbb{R}^d$ are endogenous regressors and $Z_i \in \mathcal{Z} \subset \mathbb{R}^{d_z}$ are instruments.

Let $T : L^2(\mathcal{X}) \rightarrow L^2(\mathcal{Z})$ be the conditional expectation operator $(Th)(z) = \mathbb{E}[h(X)|Z = z]$. We now introduce some basic conditions on the NPIV model (19).

Assumption 7. (i) X_i has compact rectangular support $\mathcal{X} \subset \mathbb{R}^d$ with nonempty interior and the density of X_i is uniformly bounded away from 0 and ∞ on $\mathcal{X}$; (ii) Z_i has compact rectangular support $\mathcal{Z} \subset \mathbb{R}^{d_z}$ and the density of Z_i is uniformly bounded away from 0 and ∞ on $\mathcal{Z}$; (iii) $\inf_{z \in \mathcal{Z}} \mathbb{E}[\epsilon_i^2 | Z_i = z] \geq \underline{\sigma}^2 > 0$ with $\epsilon_i := Y_i - h_0(X_i)$ satisfying NPIV model (19); (iv) $h_0 \in \Lambda_c^s$.

Assumption 8. For any $h_1, h_2 \in \Lambda_c^s$, $Th_1 = Th_2$ in $L^2(\mathcal{Z})$ implies $L(h_1) = L(h_2)$.

To establish a lower bound, we require a *link condition* that relates smoothness of T to the parameter space for $h_0 \in \Lambda_c^s$. For simplicity we use the condition similar to that used in Chen and Reiss (2011) and Chen and Christensen (2018). Let $\{\psi_{\tilde{j}, k, \tilde{G}}\}$ denote a tensor-product CDV wavelet basis for $L^2(\mathcal{X}) \cap L^\infty(\mathcal{X})$ of regularity strictly larger than s . See, e.g., Appendix A of Chen and Christensen (2018) for details on the construction and properties of this basis.

Assumption 9 (NPIV Link Condition). *There is a positive decreasing function $\nu : (0, \infty) \rightarrow (0, \infty)$ such that*

$$\|Th\|_{L^2(\mathcal{Z})}^2 \lesssim \sum_{\tilde{j}, \tilde{G}, k} \nu(2^{\tilde{j}})^2 \left\langle h, \psi_{\tilde{j}, k, \tilde{G}} \right\rangle_{L^2(\mathcal{X})}^2, \quad \forall h \in \Lambda_c^s(\mathcal{X}). \quad (20)$$

In the NPIV literature (see [Hall and Horowitz \(2005\)](#) and [Chen and Reiss \(2011\)](#)), the mildly ill-posed case corresponds to $\nu(t) = t^{-\varsigma}$, roughly saying that the operator T converts s -smooth functions of X into $(\varsigma + s)$ -smooth functions of Z . The severely ill-posed case, which corresponds to choosing $\nu(t) = \exp(-\frac{1}{2}t^\varsigma)$ and roughly says that T maps smooth functions of X into “supersmooth” functions of Z .

Theorem 2 (Lower Bound Rate for $\theta_0 = L(h_0)$ when h_0 is a NPIV). *Let Assumptions 1, 7, 8 and 9 hold for the NPIV model. We have*

$$\liminf_{n \rightarrow \infty} \inf_{\tilde{\theta}_n} \sup_{h \in \Lambda_c^s(\mathcal{X})} \mathbb{E}_{P_h} \left[r_{NPIV,n}^{-2} (\tilde{\theta}_n - L(h))^2 \right] \gtrsim C > 0$$

where

$$r_{NPIV,n} := \begin{cases} n^{-\frac{s}{2(s+\varsigma)+d-m}} & \text{if mildly ill-posed,} \\ (\log n)^{-\frac{s}{\varsigma}} & \text{if severely ill-posed,} \end{cases} \quad (21)$$

and $\inf_{\tilde{\theta}_n}$ denotes the infimum over all estimators of $L(h)$ based on the sample of size n , $\sup_{h \in \Lambda_c^s(\mathcal{X})} \mathbb{E}_{P_h}$ denotes the sup over $h \in \Lambda_c^s(\mathcal{X})$ and probability distributions of (X_i, Z_i, u_i) that satisfy Assumptions 7 and 9 with fixed ν , and the finite constant $C > 0$ does not depend on n .

According to Theorem 2, for severely ill-posed NPIV problems, the minimax lower bound for estimating linear submanifold integral $L(h_0)$ coincides with those in $L^2(X)$ ([Chen and Reiss \(2011\)](#)) and in sup-norm ([Chen and Christensen \(2018\)](#)) estimation of the NPIV function h_0 itself, which is also the same as estimating a NPIV function in $L^2(X)$ estimation of a NPIV function with $(d-m)$ dimensional endogenous regressors. For mildly ill-posed NPIV problems, the minimax lower bound for estimating $L(h_0)$ coincides with those in $L^2(X)$ estimation of a NPIV function with $(d-m)$ dimensional endogenous regressors.

Remark 2 (Bounded completeness is not needed for the minimax lower bound for estimating $L(h_0)$). *Note that Assumption 8 is equivalent to*

$$L(h) = 0 \text{ for all } h \in \ker(T) \cap \Lambda_c^s(\mathcal{X}), \text{ with } \ker(T) := \{h \in L^2(\mathcal{X}) : Th = 0 \text{ in } L^2(\mathcal{Z})\},$$

which is strictly weaker than bounded completeness of T . Indeed, bounded completeness implies $\ker(T) = \{0\}$ and hence implies Assumption 8, but the proof of Theorem 2 does not use injectivity of T : it only constructs two alternatives h_0, h_1 such that (a) the induced distributions are close in Kullback–Leibler distance (through small $\|T(h_1 - h_0)\|_{L^2(\mathcal{Z})}$), while (b) $|L(h_1) - L(h_0)|$ is separated at the stated rate.

Some identification is nevertheless needed to make estimation of $L(h_0)$ nontrivial. If Assumption 8 fails, then there exists $h^\dagger \in \ker(T) \cap \Lambda_c^s(\mathcal{X})$ with $L(h^\dagger) \neq 0$. For any $h_0 \in \Lambda_c^s(\mathcal{X})$ and sufficiently small $t \neq 0$ such that $h_1 := h_0 + th^\dagger \in \Lambda_c^s(\mathcal{X})$, we have $Th_1 = Th_0$ and hence the models indexed by h_0 and h_1 are observationally indistinguishable, but $L(h_1) - L(h_0) = tL(h^\dagger) \neq 0$. Hence, for any estimator $\tilde{\theta}_n$,

$$\sup_{h \in \{h_0, h_1\}} \mathbb{E}_{P_h} [(\tilde{\theta}_n - L(h))^2] \geq \frac{(L(h_1) - L(h_0))^2}{4} = \frac{t^2 L(h^\dagger)^2}{4},$$

so no consistent estimation of $L(h_0)$ is possible. Thus, Assumption 8 is a natural minimal identification condition for $L(h_0)$, whereas bounded completeness of T is unnecessary for establishing the lower bound rate in Theorem 2.

3.2.2 For Nonlinear Functional $\theta_0 = \Gamma(h_0)$

Corollary 3 (Lower Bound Rate for $\theta_0 = \Gamma(h_0)$ when h_0 is a NPIV). *Let Γ be a potentially nonlinear functional satisfying Assumption 6. Let Assumptions 7, 8 (for $D\Gamma(h_0)[\cdot]$) and 9 hold for the NPIV model. We have*

$$\liminf_{n \rightarrow \infty} \inf_{\tilde{\theta}_n} \sup_{h \in \Lambda_c^s(\mathcal{X})} \mathbb{E}_{P_h} [r_{NPIV,n}^{-2} (\tilde{\theta}_n - \Gamma(h))^2] \gtrsim C > 0$$

where $r_{NPIV,n}$ is the same minimax lower bound rate given in Theorem 2.

We note that Assumption 6(ii) and Assumption 8 (for $D\Gamma(h_0)[\cdot]$) together implies local identification of $\Gamma(h_0)$ only. We could impose a global identification assumption of $\Gamma(h_0)$ as follows:

Assumption 10 (Nonlinear analogue of Assumption 8: global reduced-form identification). *For any $h_1, h_2 \in \Lambda_c^s$, if $Th_1 = Th_2$ in $L^2(P_Z)$ then $\Gamma(h_1) = \Gamma(h_2)$.*

Assumption 10 states that $\Gamma : \Lambda_c^s \mapsto \mathbb{R}$ is constant on each observational equivalence class $[h] = h + \ker(T) := \{h' \in L^2(\mathcal{X}) \cap \Lambda_c^s : Th' = Th \text{ in } L^2(\mathcal{Z})\}$, so $\Gamma(h)$ is point-identified from the reduced-form regression model. This Assumption 10 is automatically satisfied if the conditional expectation operator T is bounded complete. It is interesting to note that, for the minimax lower bound rate calculation, it suffices to have the local identification of $\Gamma(h_0)$ in a sup-norm neighborhood of $h_0 \in \Lambda_c^s$.

4 Rate-Optimal Estimation

In this section we present simple sieve estimators that achieve the minimax lower bound rate of $r_n^* = n^{-\frac{s}{2s+d-m}}$ for $\theta_0 = L(h_0), \Gamma(h_0), V(h_0)$ when $h_0 \in \Lambda^s$ is a nonparametric regression function.¹⁴ We impose the following regularity condition on a nonparametric regression model in this and the next sections.

Assumption 11 (Bounded Variance of Errors). $\sup_{x \in \mathcal{X}} \mathbb{E}[\epsilon_i^2 | X_i = x] < \infty$.

For any $h \in \Lambda^s$ we let $\|h\|_\infty := \sup_{x \in \mathcal{X}} |h(x)|$ denote the sup-norm and $\|h\|_2 := \sqrt{\mathbb{E}[h(X)^2]}$ denote the $L_2(X)$ -norm. Here $L_2(X) := L_2(P_X)$ denote the space of square integrable functions, which is a Hilbert space under the $L_2(X)$ -norm with inner product $\langle h_1, h_2 \rangle_2 \equiv \mathbb{E}[h_1(X_i) h_2(X_i)]$.

4.1 Minimax Rate-Optimal Estimation of $L(h_0)$

We show that the lower bound $r_n^* = n^{-\frac{s}{2s+d-m}}$ established in Theorem 1 for $\theta_0 = L(h_0)$ can be attained by the following plug-in sieve estimator:

$$\hat{\theta} := L(\hat{h}) = \int_{\mathcal{M}} \hat{h}(x) w(x) d\mathcal{H}^m(x). \quad (22)$$

Hence we conclude that the rate r_n^* is minimax rate optimal, and plug-in estimator with sieve first stage attains this optimal rate.¹⁵

For the sake of concreteness, let $\mathbb{H}_K$ denote the closed linear span (clsp) in $L_2(X)$ -norm of a sieve basis functions $b^K(x) := (b_1^K(x), \dots, b_K^K(x))'$. By default we use a tensor product basis (see, e.g., Chen (2007)): $b^K(x) = \text{vec}(\bigotimes_{\ell=1}^d (b_1^K(x_\ell), \dots, b_{J_n}^K(x_\ell)))$ so $K = J^d$. The sieve dimension is chosen to increase with n , with $J \equiv J_n \nearrow \infty$ and thus $K \equiv K_n \nearrow \infty$ as $n \rightarrow \infty$, though we will suppress the subscript n in J and K for notational simplicity. Let $G := \mathbb{E}[b^K(X_i) b^K(X_i)']$ denote the (population) Gram matrix with its minimum eigenvalue $\lambda_{\min}(G) > 0$. Let $\bar{b}^K(x) := G^{-1/2} b^K(x)$ denote the orthonormalized vector of basis functions of b^K , then $\mathbb{H}_K = \text{clsp}(b^K) = \text{clsp}(\bar{b}^K)$. The sieve LS estimator $\hat{h}$ for $h_0 = \mathbb{E}[Y|X]$ is given by

$$\hat{h}(x) := b^{K_n}(x)' [\hat{G}]^{-1} \frac{1}{n} \sum_{i=1}^n b^{K_n}(X_i) Y_i = \bar{b}^{K_n}(x)' [\bar{G}_n]^{-1} \frac{1}{n} \sum_{i=1}^n \bar{b}^{K_n}(X_i) Y_i, \quad (23)$$

¹⁴We could also present additional estimators to achieve the minimax lower bound rates for θ_0 when h_0 is a nonparametric density of X or a nonparametric instrumental variables regression. We skip those due to the lack of space.

¹⁵In Appendix G, we show that this optimal rate is also attained by a plug-in kernel estimator.

where $\hat{G} := \frac{1}{n} \sum_{i=1}^n b^{K_n}(X_i) b^{K_n}(X_i)'$ and $\bar{G}_n := \frac{1}{n} \sum_{i=1}^n \bar{b}^{K_n}(X_i) \bar{b}^{K_n}(X_i)'$ denote the empirical Gram matrices. Let

$$h_{0,n} := \arg \min_{h \in \mathbb{H}_{K_n}} \|h - h_0\|_2 = \bar{b}^K(x)' \mathbb{E}[\bar{b}^K(X_i) h_0(X_i)] = \bar{b}^K(x)' \mathbb{E}[\bar{b}^K(X_i) Y_i]$$

be the population LS projection of h_0 onto the sieve space $\mathbb{H}_K = \text{clsp}(\bar{b}^K)$. Following [Chen, Liao and Sun \(2014\)](#), we define a *sieve Riesz representer* $v_{K_n}^* \in \mathbb{H}_K$ as follows:¹⁶

$$L[\nu] := \left. \frac{dL(h_0 + t\nu)}{dt} \right|_{t=0} = \langle v_{K_n}^*, \nu \rangle_2 \equiv \mathbb{E}[v_{K_n}^*(X_i) \nu(X_i)], \quad \text{for all } \nu \in \mathbb{H}_{K_n}. \quad (24)$$

Moreover, it can be solved in closed form as:

$$v_{K_n}^*(\cdot) = b^{K_n}(\cdot)' G^{-1} L[b^{K_n}] = \bar{b}^{K_n}(\cdot)' L[\bar{b}^{K_n}], \quad (25)$$

where $L[\bar{b}^{K_n}] := (L[\bar{b}_1^{K_n}], \dots, L[\bar{b}_{K_n}^{K_n}])'$. Define the sieve variance as

$$\|v_{K_n}^*\|_{sd}^2 := \mathbb{E} \left[\left[v_{K_n}^*(X_i) (Y_i - h_0(X_i)) \right]^2 \right] \equiv \mathbb{E} \left[\left[v_{K_n}^*(X_i) \epsilon_i \right]^2 \right]. \quad (26)$$

The following lemma establishes the rates at which $\|v_{K_n}^*\|_2$ and $\|v_{K_n}^*\|_{sd}$ grow to infinity. Its proof utilizes the decomposition (41) in Appendix A, which converts the m -dimensional Hausdorff integral to a finite sum of Lebesgue integrals on $\mathbb{R}^m$.

Lemma 1 (Growth Rate of Sieve Riesz Representer for $L(h_0)$). *Under Assumptions 1, 3, 4 and $\lambda_{\min}(G) > 0$ for each K , we have:*

$$\|v_{K_n}^*\|_2^2 = \|L[\bar{b}^{K_n}]\|^2 \equiv \sum_{k=1}^{K_n} (L[\bar{b}_k^{K_n}])^2 \asymp K_n^{\frac{d-m}{d}} = J_n^{d-m}. \quad (27)$$

Further, under Assumption 5 and 11, we have:

$$\|v_{K_n}^*\|_{sd}^2 \asymp \|v_{K_n}^*\|_2^2 \asymp K_n^{\frac{d-m}{d}}.$$

Lemma 1 is the core result that demonstrates the rate acceleration provided by the submanifold integral. It shows that the growth rates of the norm of the sieve Riesz representer and of the sieve variance are asymptotically proportional to $J^{(d-m)}$, where

¹⁶Since the minimax lower bound rate r_n^* for estimating $\theta_0 = L(h_0)$ is slower than $n^{-1/2}$, there does not exist a Riesz representer for $\theta_0 = L(h_0)$ in $L^2(X)$. Following [Chen, Liao and Sun \(2014\)](#), a sieve Riesz representer for a linear functional is always well-defined and has a closed-form expression in any finite dimensional linear sieve space $\mathcal{V}_{K_n}$. The sieve Riesz representer is not unique as it depends on the choice of the linear sieve space, and is used for the variance characterization for $L(\hat{h})$ as well as the sieve influence function. Since we are presenting a linear sieve plug-in estimator $L(\hat{h})$ for $\theta_0 = L(h_0)$ in this Section, we simply use the same linear sieve space $\mathcal{V}_{K_n} = \mathbb{H}_{K_n}$ for our sieve variance characterization.

the exponent $(d - m)$ is the codimension of the m -dimensional manifold, but, importantly, not d , the dimension of the ambient space $\mathbb{R}^d$ in which the manifold is embedded. In other words, even though the dimensionality of the first-stage estimation of h_0 is d , it is reduced by the integration over the m -dimensional submanifold.

For any $h \in \Lambda^s(\mathcal{X})$, we let $P_{K,n}h$ be the empirical LS projection of h onto the sieve space $\mathbb{H}_K$, which is given by $P_{K,n}h := b^K(x)' \hat{G}^{-1} \frac{1}{n} \sum_{i=1}^n b^K(X_i) h(X_i)$ with the sup operator norm defined by $\|P_{K,n}\|_\infty := \sup_{h: 0 < \|h\|_\infty < \infty} \frac{\|P_{K,n}h\|_\infty}{\|h\|_\infty}$. Let $\zeta_K := \sup_{x \in \mathcal{X}} \|b^K(x)\|$. We next impose some mild conditions on the linear sieve b^K that are satisfied by some commonly used sieve bases such as B-splines and wavelets; see, e.g., [Chen and Christensen \(2015\)](#).

Assumption 12 (Conditions on Linear Sieves). *The linear sieve b^K satisfies (i) $\lambda_{\min}(G) \geq \text{const.} > 0$ uniformly in K , (ii) $\zeta_K = O(\sqrt{K})$, (iii) $K \log K/n = o(1)$, (iv) $\|P_{K,n}\|_\infty = O_p(1)$, and (v) $\inf_{h \in \mathbb{H}_K} \|h - h_0\|_\infty \leq K^{-s/d}$ for $h_0 \in \Lambda^s(\mathcal{X})$.*

Theorem 3 (Convergence Rate under Sieve First Stage). *Let Assumptions [1](#), [3](#), [4](#), [5](#), [11](#) and [12](#)(i)(ii)(iii) hold. Then: (1)*

$$\frac{\sqrt{n}L(\hat{h} - P_{K,n}h_0)}{\|v_{K_n}^*\|_{sd}} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{v_{K_n}^*(X_i)}{\|v_{K_n}^*\|_{sd}} \epsilon_i + o_p(1) = O_p(1), \quad \text{with} \quad \|v_{K_n}^*\|_{sd} \asymp \sqrt{K_n^{\frac{d-m}{d}}}.$$

(2) *In addition, if Assumption [12](#)(iv)(v) holds, then:*

$$\hat{\theta} - \theta_0 \equiv L(\hat{h} - h_0) = O_p\left(\sqrt{K_n^{\frac{d-m}{d}}/n} + K_n^{-s/d}\right).$$

Further, if $s > m/2$, then by setting $K_n^* \asymp n^{\frac{d}{2s+d-m}}$, we obtain

$$\hat{\theta} - \theta_0 = O_p\left(n^{-\frac{s}{2s+d-m}}\right) = O_p(r_n^*),$$

which attains the lower bound rate r_n^* in [Theorem 1](#) and is thus minimax rate-optimal.

Theorems [1](#) and [3](#) together demonstrate that the problem of estimating linear integrals over an m -dimensional submanifold is akin to the $(d - m)$ -dimensional nonparametric regression problem in terms of convergence rates. Intuitively, the integration over the m -dimensional submanifold effectively “aggregates out” those m dimensions, leaving only $d - m$ effective dimensions in the nonparametric estimation problem.

In many economic applications as discussed in [Section 2](#), the level set function g is scalar-valued. Correspondingly the level set submanifold is of dimension $m = d - 1$

and co-dimension $d - m = 1$. Here all but one dimensions are “aggregated out” and the minimax optimal convergence rate becomes $L(h_0)$ is $n^{-s/(2s+1)}$, the same rate as a 1-dimensional nonparametric regression problem. This illustrates the (potentially) significant “dimension reduction” achieved through integration.

Remark 3. *In the proof of Theorem 1 on the lower bound r_n^* for $L(h_0)$, it suffices to assume that the density p_0 of $X \in \mathcal{X} \subset \mathbb{R}^d$ is known, and also that level set function is smooth $g \in \Lambda^{s_1}(\mathcal{X})$ for some $s_1 \geq \max\{1, s\}$ so that Assumption 1 is satisfied. Theorem 3 provides a feasible sieve estimator for $L(h_0)$ that attains the minimax lower bound rate of r_n^* when p_0 is unknown and without imposing the extra smoothness condition $g \in \Lambda^{s_1}(\mathcal{X})$. In Appendix Subsection C.1.1 we present an oracle estimator that uses the extra information of a known density p_0 and the extra smooth submanifold condition $g \in \Lambda^{s_1}(\mathcal{X})$. It is interesting that the oracle estimator attains the same lower bound rate of r_n^* as our feasible sieve estimator presented here. This means that in terms of minimax optimal rate of estimating $L(h_0)$, the extra information of knowing/using the marginal density of X as well as the smoothness of submanifold function g does not improve the speed of convergence. This is consistent with the fact that the optimal rate $r_n^* = n^{-s/(2s+d-m)}$ only depends on the smoothness of h_0 and the codimension $c := d - m$ of the submanifold $\mathcal{M}$.*

4.2 Minimax Rate-Optimal Estimation of $\Gamma(h_0)$

In general, whether the lower bound rate $r_n^* = n^{-s/(2s+d-m)}$ can be attained by any consistent estimator for the nonlinear integral functional $\Gamma(h_0)$ depends on the specific functional form of Γ . Here we consider a specific but still quite general class of submanifold integrals of nonlinear transformations of h_0 . We verify Assumption 6 in this context, and provide lower-level conditions for the attainability of the lower bound rate r_n^* , which then implies minimax optimality of the rate r_n^* for the estimation of $\Gamma(h_0)$. Specifically, consider Γ defined by

$$\Gamma(h) := \int_{\mathcal{M}} \phi(h(x), x) w(x) d\mathcal{H}^m(x), \quad (28)$$

where $\phi(t, x)$ is a known nonlinear transformation that is L -Lipschitz in t (so that its partial derivative $\phi_1(t, x) := \frac{\partial}{\partial t} \phi(t, x)$ is well-defined almost everywhere and uniformly bounded by the Lipschitz constant L whenever well-defined). We assume that $\phi_{11}(t, x) := \frac{\partial^2}{\partial t^2} \phi(t, x)$ is well-defined.

We now provide an umbrella theorem for $\Gamma(h)$, which relates it to the linear submanifold integral we studied earlier, and establishes the attainability of the linear minimax optimal rate r_n^* under appropriate smoothness requirement.

Assumption 13 (Regularity Conditions for Submanifold Integral of Nonlinear Transformations). (a) $\phi(\cdot, x)$ is Lipschitz uniformly in x , (b) $\phi_1(t, x)$ is Lipschitz, and (c) Assumption 4 holds.

Lemma 2 (Pathwise Derivative of Γ). Let $\theta_0 = \Gamma(h_0)$ be defined in (28) with a known m -dimensional submanifold $\mathcal{M}$ satisfying Assumption 1. If Assumption 13(a) holds, then Assumption 6 holds with

$$D\Gamma(h_0)[v] = \int_{\mathcal{M}} v(x) \phi_1(h_0(x), x) w(x) d\mathcal{H}^m(x). \quad (29)$$

Theorem 4 (Minimax Rate Optimal Estimation of $\Gamma(h_0)$). Let $\theta_0 = \Gamma(h_0)$ be defined in (28) with a known m -dimensional submanifold $\mathcal{M}$. Let Assumptions 1, 3, 5, 11, 12 and 13 hold.

(1) If $s > m/2$, then $\hat{\theta}_{SS} - \theta_0 = O_p(r_n^*)$, where $\hat{\theta}_{SS}$ is a Split-Sample estimator:

$$\hat{\theta}_{SS} := \Gamma(\bar{h}) - \frac{1}{8} \int_{\mathcal{M}} \phi_{11}(\bar{h}(x), x) (\hat{h}_1(x) - \hat{h}_2(x))^2 w(x) d\mathcal{H}^m(x), \quad (30)$$

where $\hat{h}_1$ and $\hat{h}_2$ are split-sample sieve estimators with sieve dimension $K_n^* \asymp n^{\frac{d}{2s+d-m}}$, and $\bar{h} := (\hat{h}_1 + \hat{h}_2)/2$.

(2) If $s > m/2$, then $\hat{\theta}_{LOO} - \theta_0 = O_p(r_n^*)$, where $\hat{\theta}_{LOO}$ is a Leave-One-Out estimator:

$$\hat{\theta}_{LOO} := \Gamma(\hat{h}) - \frac{1}{2n^2} \sum_{i=1}^n \int_{\mathcal{M}} \phi_{11}(\hat{h}(x), x) \hat{s}_i(x)^2 (\hat{\epsilon}_i^{(-i)})^2 w(x) d\mathcal{H}^m(x), \quad (31)$$

where $\hat{h}$ is the sieve LS estimator (23) with sieve dimension $K_n^* \asymp n^{\frac{d}{2s+d-m}}$; for i -th observation, $\hat{s}_i(x) := b^{K_n}(x)' \hat{G}^{-1} b^{K_n}(X_i)$, and $\hat{\epsilon}_i^{(-i)} := Y_i - \hat{h}^{(-i)}(X_i)$ is the LOO residual with the LOO sieve LS estimator $\hat{h}^{(-i)}$ computed using the sample $\{(Y_j, X_j) : j \neq i\}_{j=1}^n$.

(3) If $s \geq m$, then $\hat{\theta} - \theta_0 = O_p(r_n^*)$, where $\hat{\theta} = \Gamma(\hat{h})$ is a plug-in sieve LS estimator with sieve dimension $K_n^* \asymp n^{\frac{d}{2s+d-m}}$.

In either (1) or (2) or (3), the rate $r_n^* = n^{-s/(2s+d-m)}$ is minimax-optimal.

Both the SS and the LOO sieve debiased estimators are constructed by subtracting the diagonal parts of the quadratic remainder in the Taylor expansion of $\Gamma(\hat{h})$ at h_0 . Both estimators are shown to attain the minimax lower bound rate r_n^* . We note that the split-sample debiased sieve estimator $\hat{\theta}_{SS}$ for $\Gamma(h_0)$ is used to attain an upper bound rate that matches the lower bound rate r_n^* under the mild smoothness condition $s > m/2$. The plug-in sieve estimator $\hat{\theta} = \Gamma(\hat{h})$ is extremely easy to compute, but attains the minimax optimal rate only under stronger smoothness condition $s \geq m$.

Theorem 4 applies to quadratic submanifold functional Q as a special case, where

$$\Gamma(h) = Q(h) := \int_{\mathcal{M}} h^2(x) w(x) d\mathcal{H}^m(x), \quad (32)$$

and the split-sample debiased estimator for $Q(h_0)$ becomes $\hat{\theta}_{SS}^Q = \int_{\mathcal{M}} \hat{h}_1(x) \hat{h}_2(x) w(x) d\mathcal{H}^m(x)$. The study of $Q(h_0)$ is of conceptual importance given that it represents the simplest type of nonlinear integral functionals: when the standard Taylor expansion of a nonlinear function applies, the quadratic term in the expansion often constitutes the first-order approximation of the form of nonlinearity. For this reason, (full-dimensional) quadratic functionals have long been studied in statistics and econometrics as a leading example of nonlinear integral functionals: for a few examples out of many, see [Bickel and Rosenblatt \(1973\)](#), [Bickel and Ritov \(1988\)](#), [Fan \(1991\)](#), [Efromovich and Low \(1996\)](#), [Cai and Low \(2005\)](#) for earlier work in statistics on quadratic functionals of nonparametric density and regression functions, as well as [Chen and Christensen \(2018\)](#) and [Breunig and Chen \(2019\)](#) for more recent work on quadratic functionals of NPIV structural functions in econometrics. It has also been noted that the estimation of a quadratic functional is tightly related to hypothesis testing; see, for example, [Cai and Low \(2005, 2006\)](#) in the statistics literature, and [Breunig and Chen \(2024\)](#) in the adaptive minimax testing for NPIV models.

4.3 Minimax Rate-Optimal Estimation of $V(h_0)$

We now turn to another specific type of nonlinear integral functional, defined as an integral on the upper contour set of h_0 , i.e.

$$V(h_0) := \int_{\{x \in \mathcal{X}: h_0(x) \geq 0\}} w(x) dx, \quad (33)$$

where $h_0 : \mathbb{R}^d \rightarrow \mathbb{R}$ is an unknown (scalar-valued) function and $w(x)$ is known.

Even though here V is a full-dimensional (Lebesgue) integral per se, its pathwise derivative w.r.t. h becomes a submanifold integral over the level set $\{x \in \mathcal{X} : h_0(x) = 0\}$ as we show below. Hence, in this case the “level set function” $g = h_0$ is taken to be unknown and needs to be nonparametrically estimated. Notice that such “upper contour integrals” show up in several examples in Section 2 that feature subpopulations defined through inequalities, such as the welfare/value under optimal treatment assignment. We provide a corresponding theorem for upper contour integrals below.

Assumption 14 (Regularity Conditions for $V(h_0)$). *(a) h_0 is continuously differentiable and $\|\nabla_x h_0(x)\| \geq \underline{\epsilon} > 0$ on the level set $\mathcal{M}_0 := \{x \in \mathcal{X} : h_0(x) = 0\}$, (b) w , $\nabla_x w$, and the second-order (partial) derivatives of h_0 are all uniformly bounded.*

Lemma 3 (Pathwise Derivative of V). *Let $\theta_0 = V(h_0)$ be defined in (33). If Assumption 14(a) holds, then $\mathcal{M}_0$ is an $m = (d - 1)$ dimensional submanifold satisfying Assumption 1, and Assumption 6 holds with*

$$DV(h_0)[v] = \int_{\mathcal{M}_0} v(x) \frac{w(x)}{\|\nabla_x h_0(x)\|} d\mathcal{H}^{d-1}(x). \quad (34)$$

We note that the calculation of the pathwise derivative (34) is nonstandard, since it involves differentiation w.r.t. the changing boundary of the region of integration.

Theorem 5 (Minimax Rate Optimal Estimation of $V(h_0)$). *Let $\theta_0 = V(h_0)$ be defined in (33). Let Assumptions 3, 5, 11, 12 and 14 hold.*

- (1) *If $s \geq m/2 + 1 = (d + 1)/2$, then $\hat{\theta}_{SS} - \theta_0 = O_p(r_n^*)$, where $\hat{\theta}_{SS}$ is a split-sample debiased estimator*

$$\hat{\theta}_{SS} := V(\bar{h}) - \frac{1}{8} D^2 V(\bar{h}) [\hat{h}_1 - \hat{h}_2, \hat{h}_1 - \hat{h}_2], \quad (35)$$

where $D^2 V$ is the second functional derivative defined in Lemma 6 in the Appendix, $\hat{h}_1$ and $\hat{h}_2$ are split-sample sieve estimators with sieve dimension $K_n^ \asymp n^{\frac{d}{2s+1}}$, and $\bar{h} := (\hat{h}_1 + \hat{h}_2)/2$.*

- (2) *If $s \geq m/2 + 1 = (d + 1)/2$, then $\hat{\theta}_{LOO} - \theta_0 = O_p(r_n^*)$, where $\hat{\theta}_{LOO}$ is a Leave-One-Out debiased estimator:*

$$\hat{\theta}_{LOO} := V(\hat{h}) - \frac{1}{2n^2} \sum_{i=1}^n D^2 V(\hat{h}) [\hat{s}_i, \hat{s}_i] \cdot \left(\hat{\epsilon}_i^{(-i)} \right)^2, \quad (36)$$

where $\hat{h}$, $\hat{s}_i$ and $\hat{\epsilon}_i^{(-i)}$ are defined in Theorem 4(2) with sieve dim $K_n^ \asymp n^{\frac{d}{2s+1}}$.*

(3) If $s \geq m + 1 = d$, then $\hat{\theta} - \theta_0 = O_p(r_n^*)$, where $\hat{\theta} = V(\hat{h})$ is a plug-in sieve estimator with sieve dimension $K_n^* \asymp n^{\frac{d}{2s+1}}$.

In either (1) or (2) or (3), the rate $r_n^* = n^{-s/(2s+d-m)} = n^{-s/(2s+1)}$ is minimax optimal.

5 Sieve Inference on Submanifold Integrals

The minimax lower bound results from Section 3 imply that our functionals of interest, $\theta_0 = L(h_0)$, $\Gamma(h_0)$, $V(h_0)$ are all *irregular* functionals of the unknown function h_0 , where h_0 could be a nonparametric regression, a nonparametric density and a NPIV function. Consequently, the well-known $L^2(P_X)$ -Riesz representers and the efficient influence functions *do not exist* for these functionals. Nevertheless, applying the general theories in Chen, Liao and Sun (2014); Chen and Liao (2014) for sieve M-estimation and in Chen and Pouzo (2015) for penalized sieve MD-estimation,¹⁷ the sieve Riesz representers and sieve influence functions are still well-defined for these irregular functionals. The sieve Riesz representer approach enables us to show that the sieve student t statistics for these irregular functionals are still asymptotically standard normal, which can be used to construct simple confidence intervals for θ_0 .

For the sake of simplicity, in this section we specialize their general theory to our functionals of interest when h_0 is a nonparametric regression. In this section we denote $\theta_0 := \Phi(h) : \Lambda^s \mapsto \mathbb{R}$, where $\Phi(h_0) = L(h_0)$, $\Gamma(h_0)$, $V(h_0)$. By exploring the submanifold structures, we can characterize the growth rate of the norm of the sieve Riesz representer for the first-order expansions of $\Gamma(h_0)$ and $V(h_0)$, provide a tighter control of nonlinear remainder terms, and establish the asymptotic normality under mild sufficient conditions.

5.1 Asymptotic Normality via Sieve Riesz Representers

Let $\mathbb{H}_{K_n} := \{h \in \mathcal{B}_{K_n} : \|h - h_0\|_\infty < \epsilon\}$. Let $\hat{h}_n \in \mathbb{H}_{K_n}$ be any sup-norm consistent sieve estimator of the regression function $h_0 \in \Lambda^s(\mathcal{X})$. Let $\mathcal{V}_{K_n}$ denote a K_n -dimensional Hilbert space that contains $h_{0,n} := \arg \min_{h \in \mathbb{H}_{K_n}} \|h - h_0\|_2$, and can be equivalently expressed as $clsp(\bar{\psi}^{K_n}) := clsp\{\psi_k : k = 1, \dots, K_n\}$, where $\{\psi_k : k \geq 1\}$

¹⁷M-estimation includes nonparametric regression, quantile regression and density estimation as special cases. MD-estimation includes M-estimation, nonparametric IV regression and nonparametric quantile IV as special cases.

is an orthonormal basis for $L_2(P_X)$. Following [Chen, Liao and Sun \(2014\)](#); [Chen and Liao \(2014\)](#), we define the sieve Riesz representer $v_{K_n}^*$ and the sieve variance $\|v_{K_n}^*\|_{sd}^2$ in the same way as in (24) and (26).

The following Remark follows from Lemma 1, which allows us to control the growth rate of the sieve Riesz representer and the sieve variance for the nonlinear submanifold functionals $\Phi = \Gamma$ and V .

Remark 4 (Growth Rate of Sieve Riesz Representer for $D\Phi(h_0)$). *Under Assumptions 1, 3, 6 and $\lambda_{\min}(R_K) > 0$ for each K , we have:*

$$\|v_{K_n}^*\|_2^2 = \|D\Phi(h_0)(\bar{\psi}^{K_n})\|^2 \asymp K_n^{\frac{d-m}{d}} = J_n^{d-m}. \quad (37)$$

Further, if Assumptions 5 and 11 hold, then: $\|v_{K_n}^*\|_{sd}^2 \asymp \|v_{K_n}^*\|_2^2 \asymp K_n^{\frac{d-m}{d}}$.

The following is a standard Lindeberg condition for CLT.

Assumption 15 (Lindeberg). $\sup_{x \in \mathcal{X}} \mathbb{E}[\epsilon_i^2 \{|\epsilon_i| > c\} | X_i = x] \rightarrow 0$ as $c \rightarrow \infty$.

In the following, we let $\hat{\theta} = \Phi(\hat{h})$ be the plug-in sieve estimator of $\theta_0 = \Phi(h_0)$. We let $\hat{\theta}_{SS}$ and $\hat{\theta}_{LOO}$ respectively denote the Split-Sample sieve estimator and Leave-One-Out sieve estimator of $\theta_0 = \Phi(h_0)$ when $\Phi = \Gamma, V$.

Proposition 1 (Asymptotic Normality). *Let $\hat{h}$ be the sieve LS estimator of h_0 . Let Assumptions 3, 5, 11, 12 and 15 hold. Let the sieve dimension K_n satisfy $K_n \asymp K_n^*[\log(n)]^\kappa$ for some $\kappa \in (0, 1)$ and $K_n^* \asymp n^{\frac{d}{2s+d-m}}$. Then:*

$$\frac{\sqrt{n}(\tilde{\Phi} - \Phi(h_0))}{\|v_{K_n}^*\|_{sd}} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{v_{K_n}^*(X_i)}{\|v_{K_n}^*\|_{sd}} \epsilon_i + o_p(1) \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{with } \|v_{K_n}^*\|_{sd} \asymp \sqrt{K_n^{\frac{d-m}{d}}},$$

provided the following conditions hold for the specific functional forms of $\Phi : \Lambda^s \mapsto \mathbb{R}$:

- For $\Phi = L$: Assumptions 1 and 4 hold with $s > m/2$ for $\tilde{\Phi} = L(\hat{h})$.
- For $\Phi = \Gamma$: Assumptions 1 and 13 hold, $s > m/2$ for $\tilde{\Phi} = \hat{\theta}_{SS}, \hat{\theta}_{LOO}$; and $s > m$ for $\tilde{\Phi} = \Gamma(\hat{h})$.
- For $\Phi = V$: Assumption 14 holds, $s > (d+1)/2$ for $\tilde{\Phi} = \hat{\theta}_{SS}, \hat{\theta}_{LOO}$; and $s > d$ for $\tilde{\Phi} = V(\hat{h})$.

5.2 Inference based on Estimated Sieve Riesz Representer

We note that $v_{K_n}^*(\cdot)$ can be estimated by

$$\hat{v}_{K_n}^*(\cdot) := \bar{\psi}^{K_n}(\cdot)' (\bar{R}_{K_n})^{-1} D\Phi(\hat{h}) [\bar{\psi}^{K_n}], \quad \text{with } \bar{R}_{K_n} := \frac{1}{n} \sum_i \bar{\psi}^{K_n}(X_i) \bar{\psi}^{K_n}(X_i)'$$

Denote $\sigma_{*,K_n}^2 := \|v_{K_n}^*\|_{sd}^2$ and $\hat{\epsilon}_i := Y_i - \hat{h}(X_i)$. We can estimate the sieve variance as¹⁸

$$\hat{\sigma}_{*,K_n}^2 = \frac{1}{n} \sum_{i=1}^n [\hat{v}_{K_n}^*(X_i) \hat{\epsilon}_i]^2 = (D\Phi(\hat{h}) [\psi^{(K)}])' \hat{\Omega}_{K_n} (D\Phi(\hat{h}) [\psi^{(K)}]) \quad (38)$$

with $\hat{\Omega}_{K_n} := \bar{R}_{K_n}^{-1} \left(\frac{1}{n} \sum_{i=1}^n (\hat{\epsilon}_i)^2 \bar{\psi}^{K_n}(X_i) \bar{\psi}^{K_n}(X_i)' \right) \bar{R}_{K_n}^{-1}$. The consistency of the sieve variance estimator $\hat{\sigma}_{*,K_n}^2$ can be easily established under the following assumption.

Assumption 16 (Higher Error Moments). $\mathbb{E} [|\epsilon_i|^{2+\delta}] < \infty$ for some $\delta > 2d/(2s-m)$.

Proposition 2 (Sieve t statistics). *Let Assumptions for Proposition 1 and Assumption 16 hold. Let $K_n \asymp K_n^* [\log(n)]^\kappa$ for some $\kappa \in (0, 1)$ and $K_n^* \asymp n^{\frac{d}{2s+d-m}}$. Then:*

$$\left| \frac{\hat{\sigma}_{*,K_n}}{\|v_{K_n}^*\|_{sd}} - 1 \right| = o_p(1); \quad \frac{\sqrt{n} (\tilde{\Phi} - \theta_0)}{\hat{\sigma}_{*,K_n}} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{v_{K_n}^*(X_i)}{\|v_{K_n}^*\|_{sd}} \epsilon_i + o_p(1) \xrightarrow{d} \mathcal{N}(0, 1),$$

where $\tilde{\Phi}$ is the estimator specified in Proposition 1 (i.e., $L(\hat{h})$, $\hat{\theta}_{SS}$, $\hat{\theta}_{LOO}$, or $\Phi(\hat{h})$, depending on the case and smoothness regime).

As an application of Proposition 2, we can construct valid .95 confidence interval using normal critical value for $\theta_0 = \Phi(h_0)$ as follows:

$$\text{CI} := [\tilde{\Phi} - 1.96\hat{\sigma}_\theta, \tilde{\Phi} + 1.96\hat{\sigma}_\theta], \quad \text{with } \hat{\sigma}_\theta^2 = \hat{\sigma}_{*,K_n}^2/n$$

provided that the sieve variance is computed using undersmoothed sieve dimension K_n , say $K_n \asymp K_n^* [\log(n)]^{1/2}$. In practice, we implement an adaptive inference procedure that (i) selects the sieve dimension in a data-driven way (bootstrap–Lepski), with an undersmoothed choice used for valid confidence interval; and (ii) uses a multiplier bootstrap to approximate the distribution of the studentized sieve estimator. See Appendix E for details.

¹⁸In small samples, it might be more accurate to estimate the sieve variance using $\hat{\sigma}_{*,K_n}^2 = \frac{1}{n} \sum_{i=1}^n \left[\hat{v}_{K_n}^*(X_i) \hat{\epsilon}_i - \frac{1}{n-1} \sum_{j \neq i, j=1}^n \hat{v}_{K_n}^*(X_j) \hat{\epsilon}_j \right]^2$.

6 Monte Carlo Simulations

We present two Monte Carlo studies to check finite sample performance of the sieve estimation and inference of linear and nonlinear submanifold integrals. Subsection 6.1 considers a linear submanifold integral $\theta_0 = L(h_0)$, and Subsection 6.2 considers an upper contour integral $\theta_0 = V(h_0)$. In both examples, $h_0(x) = \mathbb{E}[Y|X = x]$ is a nonparametric regression, and is estimated using a tensor-product B-spline sieve LS estimator $\hat{h}$ based on a random sample $\{(Y_i, X_i)\}_{i=1}^n$, drawn from the joint distribution of (Y, X) satisfying:

$$Y_i = h_0(X_i) + \epsilon_i, \quad \text{with } \epsilon_i \sim \mathcal{N}(0, 1), \quad X_{i1} \sim_{i.i.d.} X_{i2} \sim_{i.i.d.} \text{Uniform}[-2, 2],$$

and ϵ_i is independent of the covariates $X_i = (X_{i1}, X_{i2})' \in \mathcal{X} = [-2, 2]^2$ with $d = 2$.

In both Monte Carlo studies, we implement multiplier bootstrap Lepski data-driven choice of sieve dimensions K as described in Appendix E, with $B_{Lepski} = 500$ multiplier bootstrap draws. We report simulation results over $B = 1000$ Monte Carlo simulation replications, using performance metrics in terms of Monte Carlo root-mean-squared error (RMSE), bias and standard deviation (SD), as well as coverage rates and coverage length. We report the simulation results for different sample sizes $n = 500, 1000, 2000, 4000, 8000$.

6.1 Integral on Known Submanifold

We first report simulations results for the estimation of an integral functional of h_0 over a known submanifold defined by the unit circle, i.e., $\theta_0 = L(h_0)$ with $L(h) = \int_{\mathbb{S}^1} h(x) d\mathcal{H}^1(x)$ and $h_0(x) = x_1^2 + 2 \sin(x_1) x_2$ with $x = (x_1, x_2)' \in \mathcal{X} = [-2, 2]^2$. Under the transformation $x_1 = \cos(\beta)$ and $x_2 = \sin(\beta)$, we obtain the true value

$$\theta_0 = L(h_0) = \int_0^{2\pi} (\cos(\beta)^2 + 2 \sin(\cos(\beta)) \sin(\beta)) d\beta = \pi.$$

In this exercise, $L(h_0)$ is a linear functional of h_0 on a known $m = 1$ -dimensional submanifold (i.e., the unit circle $\mathbb{S}^1$). Theorem 3 is directly applicable, and the optimal convergence rate of a spline sieve plug-in estimator $L(\hat{h}) - L(h_0)$ is given by $n^{-s/(2s+1)}$ with the optimal sieve dimension $K^* \asymp n^{2/(2s+1)}$ and $d = 2, m = 1$.

Given the spline LS estimator $\hat{h}$, the plug-in estimator $\hat{\theta} = L(\hat{h}) = \int_{\mathbb{S}^1} \hat{h}(x) d\mathcal{H}^1(x)$ is numerically computed using the sample average over $M = 5000$ Sobol sequence

points¹⁹ in the angle space $[0, 2\pi]$. We obtain the plug-in estimator $\hat{\theta}$ and construct the confidence interval $\text{CI} := [\hat{\theta} \pm 1.96\hat{\sigma}_\theta]$ with $\hat{\sigma}_\theta^2 = \hat{\sigma}_{*,K_n}^2/n$ as in (38), where the pathwise derivative $D\Phi(\hat{h})[v] = L(v) = \int_{\mathbb{S}^1} v(x) d\mathcal{H}^1(x)$ is computed numerically in the same manner as described above.

Table 1 reports the RMSE, bias, standard deviation, and the average selected sieve dimension $\bar{\hat{K}}$ for the plug-in estimator using the Lepski sieve dimension choice $\hat{K}$. Note that the rate-optimal $\hat{K}$ is chosen to minimize estimation risk, and does not involve the undersmoothing that is needed for valid confidence interval construction.

Table 1: Integral on Known Manifold: Rate-Optimal $\hat{K}$

n	RMSE	Bias	SD	$\bar{\hat{K}}$
500	0.4515	-0.0133	0.4515	17.1
1000	0.3175	-0.0097	0.3175	16.8
2000	0.2208	0.0059	0.2208	17.1
4000	0.1626	0.0061	0.1626	17.2
8000	0.1138	-0.0042	0.1138	17.4

Rate-optimal $\hat{K}$ selected by bootstrap-Lepski without undersmoothing.

Next, we turn to the inference results. Table 2 reports the finite-sample performance of the plug-in estimator and the corresponding 95% confidence interval, constructed with bootstrap-Lepski adaptive choice of the *undersmoothed* sieve dimension $\tilde{K}$ and normal critical values. The average $\bar{\tilde{K}} \approx 51$ is substantially larger than the rate-optimal $\hat{K} \approx 17$ in Table 1, reflecting the undersmoothing needed for valid inference.

In Table 2 and subsequent tables, we report the square root of the mean squared error (RMSE), the bias (Bias), the standard deviation (SD)²⁰ of the estimator, the average lower and upper bounds of the confidence interval (CI_L and CI_U), the average length of the confidence interval (U-L), and the realized coverage probability of the CI (Coverage). Overall, the plug-in estimator and the corresponding CI perform

¹⁹The Sobol sequence sampling, proposed by Sobol (1967), is a well-known quasi-random Monte Carlo sampling method that generates a deterministic sequence of points, whose distribution asymptotically converges to the uniform distribution, but achieves better finite-sample approximation of the population expectation (integral) by the sample average over M Sobol points.

²⁰The standard deviation is calculated using the standard $\frac{1}{B-1} \sum_{b=1}^B$ formula. For this technical reason, “SD” can be larger than “RMSE”, which is calculated based on the $\frac{1}{B} \sum_{b=1}^B$ formula.

Table 2: Integral over Known Manifold: Adapt+Normal

n	RMSE	Bias	SD	CI_L	CI_U	U-L	Coverage	$\tilde{K}$
500	0.6198	-0.0320	0.6193	1.9364	4.2828	2.3465	93.80	50.6
1000	0.4232	-0.0169	0.4231	2.2952	3.9541	1.6589	95.20	50.7
2000	0.3050	0.0062	0.3051	2.5580	3.7376	1.1796	95.00	51.4
4000	0.2249	0.0078	0.2249	2.7325	3.5663	0.8338	94.40	51.4
8000	0.1496	-0.0102	0.1493	2.8363	3.4264	0.5901	95.80	51.8

“Adapt+Normal”: bootstrap-Lepski sieve dimension choice and normal critical values.

very well under all five sample sizes: the RMSE shrinks (almost at $\sqrt{n}$ rate) as the sample size increases, the bias is of negligible order relative to the standard deviation, and the realized coverage probability is close to the nominal 95% level.

Table 3 reports the estimator RMSE and the CI coverage depending on: (i) whether the bootstrap-Lepski adaptive choice of sieve dimension or a fixed sieve dimension is used, and (ii) whether normal or bootstrap quantile critical values²¹ are used, when constructing the CI. The first column “Adapt+Normal” is identical to that in Table 2, while the remaining three columns contain results about the three alternative CI constructions. We observe that all four versions produce coverage close to the 95% nominal level.

Table 3: Integral on Known Manifold: RMSE and Coverage Comparison

n	RMSE		CI Coverage			
	Adapt	Fixed	Adapt+Norm	Fixed+Norm	Adapt+Boot	Fixed+Boot
500	0.6198	0.4463	93.8%	94.2%	93.8%	94.3%
1000	0.4232	0.3097	95.2%	94.7%	95.0%	94.2%
2000	0.3050	0.2160	95.0%	94.9%	94.7%	94.9%
4000	0.2249	0.1600	94.4%	94.3%	93.9%	94.2%
8000	0.1496	0.1091	95.8%	94.9%	95.6%	94.6%

“Adapt”: bootstrap-Lepski sieve dimension choice; “Fixed”: fixed sieve dimension $K = 36$.

“Norm”: normal critical values; “Boot”: bootstrap quantile critical values.

²¹The bootstrap quantiles are also computed based on $B_{Lepski} = 500$ draws.

6.2 Integral on Estimated Upper Contour Set

In this subsection, we analyze the estimation of an integral functional of a nonparametric function over the (estimated) unit disk: $\theta_0 := V(h_0)$ with

$$V(h_0) := \int \mathbb{1}\{x \in \mathcal{X} : h_0(x) \geq 0\} dx, \quad \mathcal{X} = [-2, 2]^2$$

$$h_0(x) := (1 - \|x\|^2)(4 + \sin(x_1)x_2 + \cos(x_2)).$$

Under the above construction, $h_0(x) \geq 0$ if and only if $\|x\| \leq 1$, and thus $\theta_0 = \int \mathbb{1}\{\|x\| \leq 1\} dx = \pi$. We compute the spline nonparametric regression estimator $\hat{h}(x)$ for $h_0(x) = \mathbb{E}[Y_i|X_i = x]$, and the plug-in estimator $V(\hat{h}) = \int \mathbb{1}\{\hat{h}(x) \geq 0\} dx$ using the sample average over $M = 50,000$ Sobol sequence points, and CI := $[\hat{\theta} \pm 1.96\hat{\sigma}_\theta]$ with the sieve variance $\hat{\sigma}_\theta^2 = \hat{\sigma}_{*,K_n}^2/n$ as in (38), where the pathwise derivative $DV(\hat{h})[v]$ now takes the following form:

$$DV(h)[v] = \int_{\{x \in \mathcal{X} : h(x)=0\}} \frac{v(x)}{\|\nabla_x h(x)\|} d\mathcal{H}^{d-1}(x),$$

which we numerically approximate via $\hat{DV}(h)[v] = \frac{1}{2\epsilon} \int_{\{x \in \mathcal{X} : -\epsilon < h(x) < \epsilon\}} v(x) dx$ based on the mathematical result²² that

$$\lim_{\epsilon \searrow 0} \frac{1}{2\epsilon} \int_{\{x \in \mathcal{X} : -\epsilon < h(x) < \epsilon\}} v(x) dx = \int_{\{x \in \mathcal{X} : h(x)=0\}} \frac{v(x)}{\|\nabla_x h(x)\|} d\mathcal{H}^{d-1}(x).$$

We set $\epsilon = 0.01$ in our simulation, and, given that $\{x \in \mathcal{X} : -\epsilon < h(x) < \epsilon\}$ may occur infrequently for small ϵ , we use the sample average from $M = 50,000$ Sobol sequence points to approximate $\hat{DV}(h)[v]$.

Table 4 reports the rate-optimal estimation results for the upper contour set integral. The average rate-optimal $\hat{K}$ grows with n (from approximately 17 at $n = 500$ to 36 at $n = 8000$), in contrast to the stable $\hat{K} \approx 17$ in Simulation 1. The plug-in and LOO-debiased estimators display similar RMSE at the rate-optimal $\hat{K}$ for $n \geq 1000$.

Next, we turn to the inference results. Table 5 reports the finite-sample performance of the plug-in estimator as well as the Leave-One-Out (LOO) estimator, along with the CIs constructed using bootstrap-Lepski adaptive choice of sieve dimension and normal critical values. The results are again based on $B = 1000$ Monte Carlo replications and (within each replication) $B_{Lepski} = 500$ bootstrap draws when applicable. The results in Table 5 display a similar pattern as those in Table 2: Again, the RMSE shrinks (almost at $\sqrt{n}$ rate) as the sample size increases, the bias is of

²²See Theorem 3.13.(iii) of Evans and Gariepy (2015).

Table 4: Integral on Estimated Upper Contour Set: Rate-Optimal $\hat{K}$

	(A) Plug-In				(B) Leave-One-Out			
n	RMSE	Bias	SD	$\hat{K}$	RMSE	Bias	SD	$\hat{K}$
500	0.0652	0.0351	0.0550	17.3	1.0175	0.0148	1.0179	17.1
1000	0.0483	0.0270	0.0402	18.0	0.0488	0.0275	0.0403	17.8
2000	0.0358	0.0182	0.0309	21.7	0.0364	0.0192	0.0310	21.4
4000	0.0253	0.0104	0.0231	28.2	0.0254	0.0106	0.0230	28.3
8000	0.0212	0.0086	0.0193	35.9	0.0212	0.0087	0.0193	35.9

Rate-optimal $\hat{K}$ selected by bootstrap-Lepski without undersmoothing.

Table 5: Integral on Estimated Upper Contour Set: Adaptive+Normal

n	RMSE	Bias	SD	CI_L	CI_U	U-L	Coverage	$\tilde{K}$
(A) Plug-In								
500	0.0694	0.0072	0.0690	3.0159	3.2818	0.2659	94.10%	51.1
1000	0.0470	0.0011	0.0470	3.0476	3.2378	0.1902	95.20%	55.3
2000	0.0335	0.0009	0.0335	3.0739	3.2110	0.1370	95.00%	65.7
4000	0.0237	-0.0004	0.0237	3.0916	3.1907	0.0991	95.40%	81.9
8000	0.0180	0.0009	0.0180	3.1065	3.1786	0.0721	95.50%	94.4
(B) Leave-One-Out								
500	1.0379	-0.0199	1.0383	2.9560	3.2873	0.3314	94.80%	50.5
1000	0.0471	0.0018	0.0471	3.0451	3.2417	0.1966	96.20%	54.9
2000	0.0335	0.0012	0.0335	3.0731	3.2125	0.1394	95.30%	64.9
4000	0.0238	-0.0001	0.0238	3.0913	3.1917	0.1004	95.40%	82.0
8000	0.0181	0.0012	0.0180	3.1065	3.1791	0.0726	95.60%	94.4

“Adapt+Normal”: bootstrap-Lepski sieve dimension choice and normal critical values.

smaller order relative to the standard deviation, and the realized CI coverage is close to the nominal 95% level. One noticeable difference between the two tables, however, lies in that the RMSEs in Table 5 are substantially smaller than those in Table 2. Heuristically, this might have been due to the fact that the upper contour set integral being estimated in Table 5 is itself a full-dimensional integral: even though its asymptotic behavior is theoretically driven by the lower-dimensional submanifold, in finite sample the full-dimensional nature of the integral may have made it overall easier to estimate. Lastly, for $n \geq 1000$, the LOO-debiased estimator does result in (slightly but noticeably) smaller biases relative to the plug-in estimator, yielding comparable

or better CI coverage as anticipated.²³

Table 6 again reports the RMSE and coverage comparison under the two sieve dimension choice methods and the two critical value choice methods. We observe that all four versions of CI constructions produce coverage close to the 95% nominal level, and the improvement of the LOO estimator over the plug-in one in terms of size control is consistent across all four versions of CI construction procedures.

Table 6: Integral on Estimated Upper Contour Set: RMSE and Coverage Comparison

n	RMSE		CI Coverage			
	Adapt	Fixed	Adapt+Norm	Fixed+Norm	Adapt+Boot	Fixed+Boot
(A) Plug-In						
500	0.0694	0.0671	94.1%	93.3%	93.4%	93.2%
1000	0.0470	0.0444	95.2%	96.3%	94.8%	96.1%
2000	0.0335	0.0316	95.0%	94.3%	94.7%	94.5%
4000	0.0237	0.0222	95.4%	96.1%	95.0%	95.6%
8000	0.0180	0.0165	95.5%	95.2%	95.2%	94.4%
(B) Leave-One-Out						
500	1.0379	0.0743	94.8%	95.7%	94.4%	95.4%
1000	0.0471	0.0445	96.2%	97.0%	95.4%	96.5%
2000	0.0335	0.0316	95.3%	94.8%	94.9%	94.6%
4000	0.0238	0.0223	95.4%	96.1%	95.1%	95.8%
8000	0.0181	0.0165	95.6%	95.3%	95.7%	95.4%

“Adapt”: bootstrap-Lepski sieve dimension choice; “Fixed”: fixed sieve dimension $K = 64$.

“Norm”: normal critical values; “Boot”: bootstrap quantile critical values.

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²³At $n = 500$ the LOO estimator exhibits a large RMSE outlier, likely reflecting instability the leave-out procedure at this small sample size.

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Online Appendix

This Appendix consists of eight sections. Appendix A provides an instrumental transformation of submanifold integrals wrt Hausdorff measures into sums of lower-dimensional Lebesgue integrals using basic differential geometry tools. Appendix B provides proofs of the lower bound rate results in Section 3. Appendix C provides proofs of the upper bound rate results in Section 4. Appendix D provides proofs for Section 5. The remaining four sections cover the multiplier bootstrap choice of sieve dimension, analysis of quadratic submanifold integrals, additional upper bound rate results using a kernel nonparametric regression in the first stage, and a comparison of the bias-aware confidence interval with the undersmoothing approach used in the main text.

A Decomposition of Hausdorff Integrals on Submanifolds via Partition of Unity

In this section, we explain how we can decompose a Hausdorff integral on a regular submanifold into a sum of lower-dimensional Lebesgue integrals, which can be analyzed more straightforwardly based on the existing theory on the semiparametric estimation of Lebesgue integral functionals.

This is done using a combination of standard mathematical tools from differential manifold theory and geometric measure theory: see, for example, Munkres (1991) and Lang (2002) for textbook treatments of differentiable manifolds, as well as Federer (1996) and Evans and Gariepy (2015) for textbook treatments of geometric measure theory. See also Chapter 4 and Chapter 9 of Delfour and Zolésio (2011) for a unified treatment. Specifically, the key idea is as follows:

- (i) Construct a finite open cover of the manifold $\mathcal{M}$ in question.
- (ii) On each piece of the open cover in (i), apply the implicit function theorem to obtain a parametrization (i.e., a coordinate chart) of the m -dimensional manifold by an open subset of an m -dimensional Euclidean space.
- (iii) On each piece of the open cover in (i), use the “change of variable” formula in geometric measure theory to convert Hausdorff integral on the manifold into an m -dimensional Lebesgue integral via the parametrization in (ii).

- (iv) Use the “partition of unity” method to combine the results in (iii) across different pieces of the open cover in (i).

We now proceed through (i)-(iv) formally with more details.

By the regularity of $\mathcal{M}$ (Assumption 1), for every $\bar{x} \in \mathcal{M}$, $\nabla_x g(\bar{x})$ has rank $d - m$. Hence, we can decompose $\bar{x}$, potentially with a permutation of coordinate indexes, as $(\bar{x}_{(m)}, \bar{x}_{-(m)})$, where $\bar{x}_{(m)}$ is an m -dimensional subvector and $\bar{x}_{-(m)}$ is the remaining $(d - m)$ -dimensional subvector, and correspondingly decompose $\nabla_x g(\bar{x})$ into

$$\nabla_x g(\bar{x}) = \begin{bmatrix} \underbrace{\nabla_{x_{(m)}} g(\bar{x})}_{m \times (d-m)}, & \underbrace{\nabla_{x_{-(m)}} g(\bar{x})}_{(d-m) \times (d-m)} \end{bmatrix}$$

so that $\text{rank}(\nabla_{x_{-(m)}} g(\bar{x})) = d - m$. Then, by the Implicit Function Theorem (see, e.g. Theorem 9.2 in [Munkres, 1991](#)), there exists an open neighborhood $\mathcal{U}_{\bar{x}}$ around $\bar{x}_{(m)}$ and a unique r -times continuously differentiable function $\psi_{\bar{x}} : \mathcal{U}_{\bar{x}} \rightarrow \mathbb{R}^{-(m)}$ such that

$$x_{-(m)} = \psi_{\bar{x}}(x_{(m)}), \quad \forall x_{(m)} \in \mathcal{U}_{\bar{x}}.$$

Then, define $\varphi_{\bar{x}} : \mathcal{U}_{\bar{x}} \rightarrow \mathcal{M} \subseteq \mathbb{R}^d$ by

$$\varphi_{\bar{x}}(x_{(m)}) := (x_{(m)}, \psi_{\bar{x}}(x_{(m)})).$$

Then $\varphi_{\bar{x}}$ is a diffeomorphism (r -times differentiable bijection) between $\mathcal{U}_{\bar{x}}$ and $\varphi_{\bar{x}}(\mathcal{U}_{\bar{x}}) \subseteq \mathcal{M}$: $(\mathcal{U}_{\bar{x}}, \varphi_{\bar{x}})$ is thus a local coordinate chart of $\mathcal{M}$ at $\bar{x}$, which parameterizes each point x in $\varphi_{\bar{x}}(\mathcal{U}_{\bar{x}})$ on the manifold with a vector $x_{(m)} \in \mathcal{U}_{\bar{x}}$, where $\mathcal{U}_{\bar{x}}$ is an open set in an m -dimensional Euclidean space $\mathbb{R}^m$.

Clearly $\{\varphi_{\bar{x}}(\mathcal{U}_{\bar{x}})\}_{\bar{x} \in \mathcal{M}}$ is an open cover for $\mathcal{M}$. Since $\mathcal{M}$ is compact, there exists a finite sub-cover $\{\varphi_j(\mathcal{U}_j) : j = 1, \dots, \bar{j}\}$, where, for each j , $(\mathcal{U}_j, \varphi_j) = (\mathcal{U}_{\bar{x}^{(j)}}, \varphi_{\bar{x}^{(j)}})$ for some point $\bar{x}^{(j)} \in \mathcal{M}$. From now on, we also write ψ_j as the function $\psi_{\bar{x}^{(j)}}$ associated with $(\mathcal{U}_{\bar{x}^{(j)}}, \varphi_{\bar{x}^{(j)}})$, and we write $x_{(j,m)}$ for a generic point in $\mathcal{U}_{\bar{x}^{(j)}}$. The “ j ” in the subscript “ (j, m) ” emphasizes that, while $x_{(j,m)}$ is always an m -dimensional vector that corresponds to m different coordinates in $\mathbb{R}^d$, the exact sequence of the m coordinates may differ across different j ’s.

In general, the open cover $\{\varphi_j(\mathcal{U}_j) : j = 1, \dots, \bar{j}\}$ may have nonempty intersections. The standard mathematical tool to avoid “double counting” under potentially overlapping covers is the so-called “partition of unity”, i.e., a collection of smooth real-valued functions $\{\rho_j : j = 1, \dots, \bar{j}\}$ on $\mathbb{R}^d$ such that (i) each ρ_j is nonnegative

(ii) $\sum_{j=1}^J \rho_j(x) = 1$ for all $x \in \mathcal{M}$, and (iii) $\rho_j(x) = 0$ for all $x \notin \varphi_j(\mathcal{U}_j)$. Such a “partition of unity” (ρ_j) is guaranteed to exist, say, by Lemma 25.2 in [Munkres \(1991\)](#).

Given $(\mathcal{U}_j, \varphi_j, \rho_j)_{j=1}^{\bar{j}}$, we may then decompose an integral on the manifold $\mathcal{M}$ into a sum of integrals on $\varphi_j(\mathcal{U}_j)$: for any map $\omega : \mathcal{X} \mapsto \mathbb{R}$,

$$\int_{\mathcal{M}} \omega(x) d\mathcal{H}^m(x) = \sum_{j=1}^{\bar{j}} \int_{\varphi_j(\mathcal{U}_j)} \rho_j(x) \omega(x) d\mathcal{H}^m(x) \quad (39)$$

Then, since φ_j is a bijection between $\mathcal{U}_j$ and $\varphi_j(\mathcal{U}_j)$ by construction, each $x \in \varphi_j(\mathcal{U}_j)$ can be expressed as $x = \varphi_j(x_{(j,m)})$ for $x_{(j,m)} \in \mathbb{R}^m$.

By the “change-of-variable” formula in geometric measure theory, e.g., Theorem 3.9 in [Evans and Gariepy \(2015\)](#), each of the $\bar{j}$ integrals on $\varphi_j(\mathcal{U}_j)$ can be evaluated as Lebesgue integrals on $\mathcal{U}_j \subset \mathbb{R}^m$ through the parametrization φ_j , i.e.,

$$\int_{\varphi_j(\mathcal{U}_j)} \rho_j(x) w(x) d\mathcal{H}^m(x) = \int_{\mathcal{U}_j} \rho_j(\varphi_j(x_{(j,m)})) w(\varphi_j(x_{(j,m)})) \mathcal{J}\varphi_j(x_{(j,m)}) dx_{(j,m)}, \quad (40)$$

where $\mathcal{J}\varphi_j := \sqrt{\det(\varphi_j' \varphi_j)}$ denotes the Jacobian of φ_j . Combining (39) and (40) we obtain:

$$\int_{\mathcal{M}} \omega(x) d\mathcal{H}^m(x) = \sum_{j=1}^{\bar{j}} \int_{\mathcal{U}_j} \rho_j(\varphi_j(x_{(j,m)})) \omega(\varphi_j(x_{(j,m)})) \mathcal{J}\varphi_j(x_{(j,m)}) dx_{(j,m)}. \quad (41)$$

The decomposition in (41) converts a Hausdorff integral on a m -dimensional manifold $\mathcal{M}$ into a finite sum of m -dimensional Lebesgue integrals on $\mathbb{R}^m$, the latter of which are easier to analyze using existing results on semi/nonparametric estimation. Below we show how we use (41) to establish the convergence rate and asymptotic normality of the semiparametric estimator when the first stage nonparametric function h_0 could be learned/estimated using any machine learning/AI algorithms. The key is to use a sieve Riesz representation theory to establish the asymptotic local influence function representation for $\theta = \Gamma(h)$.

B Proofs of Theoretical Results in Section 3

B.1 Proof of Results in Subsection 3.1

To establish the lower bound for the convergence rate in Theorem 1, we use Le Cam's two-point comparison approach based on KL divergence.

Lemma 4 (Le Cam's Minimax Rate Bounds based on KL divergence). *Suppose that there exist P_0, P_1 such that $KL(P_0, P_1) \leq \frac{\log 2}{n}$. Then*

$$R_n := \inf_{\tilde{\theta}} \sup_P \mathbb{E}_P \left[d(\tilde{\theta}, \theta(P)) \right] \geq \frac{1}{16} d(\theta(P_0), \theta(P_1)).$$

Proof of Theorem 1. In the lower bound proof, we can assume that the submanifold function $g \in \Lambda^{s_1}(\mathcal{X})$ for some $s_1 \geq \max\{1, s\}$. We first prove the result for the nonparametric regression case $h_0(x) = \mathbb{E}[Y_i | X_i = x]$. Given any $h_0 \in \Lambda_c^s(\mathcal{X})$, let b_n be a small positive number and define

$$h_1(x) := h_0(x) + b_n^s K_{d-m} \left(\frac{g(x)}{b_n} \right)$$

where $g(x) := (g_1(x), \dots, g_{d-m}(x))$ and $K_{d-m}((t_1, \dots, t_{d-m})) := \prod_{\ell=1}^{d-m} K(t_\ell)$ and

$$K(t) := a \exp \left(-\frac{1}{1-t^2} \right) \mathbb{1}_{\{|t| \leq 1\}} \quad (42)$$

for some sufficiently small $a > 0$ such that h_1 stays in Hölder class of smoothness order s . To see this, notice that K and K_{d-m} are infinitely differentiable with uniformly bounded derivatives, and for any $k \leq s$ we have:

$$\frac{\partial^k}{\partial x_j^k} h_1(x) = \frac{\partial^k}{\partial x_j^k} h_0(x) + b_n^{s-k} \left(K_{d-m}^{(1)}(\cdot) \frac{\partial^k}{\partial x_j^k} g(x) + \dots + K_{d-m}^{(k)}(\cdot) \frac{\partial}{\partial x_j} g(x) \right).$$

Since $h_0, g \in \Lambda_c^s(\mathcal{X})$, all derivatives of K are uniformly bounded, and b_n is small, $\frac{\partial^k}{\partial x_j^k} h_1(x)$ is uniformly bounded as well. The Hölder condition for the fractional exponent $s - [s]$ can also be similarly verified.

Note that, to derive a lower bound for the minimax rate for a class of estimation problems, it suffices to establish the lower bound under one admissible example of the problem. Specifically, we consider the example setting where X_i is uniformly distributed on $[0, 1]^d$, and $w(x) \equiv 1$. In addition, let P_0 be the joint distribution (with density p_0) of $(X_i, Y_{i0})_{i=1}^n$ with

$$Y_{i0} = h_0(X_i) + \epsilon_i, \quad \epsilon_i \sim_{i.i.d.} \mathcal{N}(0, 1).$$

and let P_1 be the joint distribution (with density p_1) of $(X_i, Y_{i1})_{i=1}^n$ with

$$Y_{i1} = h_1(X_i) + \epsilon_i, \quad \epsilon_i \sim_{i.i.d.} \mathcal{N}(0, 1).$$

Then, the KL divergence between P_0^n and P_1^n is given by:

$$\begin{aligned} KL(P_0^n, P_1^n) &= nKL(P_0, P_1) \\ &= n \int_{[0,1]^d} \int p_0(x, y) \log \left(\frac{p_0(x, y)}{p_1(x, y)} \right) dy dx \\ &= n \int_{[0,1]^d} \int p_0(y|x) \log \left(\frac{p_0(y|x)}{p_1(y|x)} \right) dy dx \\ &= n \int_{[0,1]^d} \int \phi(y) \log \left(\frac{\phi(y - h_0(x))}{\phi(y - h_1(x))} \right) dy dx \\ &= n \int_{[0,1]^d} \int KL(\mathcal{N}(h_0(x), 1), \mathcal{N}(h_1(x), 1)) dy dx \\ &= n \frac{1}{2} \int_{[0,1]^d} (h_1(x) - h_0(x))^2 dx \\ &= n \frac{1}{2} \int_{[0,1]^d} b_n^{2s} K_{d-m}^2 \left(\frac{g(x)}{b_n} \right) dx \\ &= n \frac{1}{2} b_n^{2s} \int_{[0,1]^d} K_{d-m}^2 \left(\frac{g(x)}{b_n} \right) dx \\ &\leq \text{const.} \times n b_n^{2s} \int_{[0,1]^d} \prod_{\ell=1}^{d-m} \mathbb{1} \left\{ \frac{|g_\ell(x)|}{b_n} \leq 1 \right\} dx \\ &\leq C n b_n^{2s} \mathbb{P}(\|g(X_i)\| \leq b_n) \leq C' n b_n^{2s+d-m} \end{aligned} \tag{43}$$

provided that

$$\mathbb{P}(\|g(X_i)\| \leq b_n) \leq M b_n^{d-m}. \tag{44}$$

We now show that (44) holds. Let $W_i := g(X_i) \in \mathbb{R}^{d-m}$, note that the density of W_i

$$p_W(w) := \int_{\{g(x)=w\}} p_X(x) \frac{1}{\mathcal{J}_g(x)} d\mathcal{H}^m(x) \leq M,$$

is uniformly bounded since $p_X(x) \equiv 1$ and $\mathcal{J}_g(x)$ is uniformly bounded away from zero as below by Assumption 2(i). Consequently,

$$\mathbb{P}(\|g(X_i)\| \leq b_n) = \int_{\mathbb{1}_{\{|w| \leq b_n\}}} p_W(w) dw \leq M \int_{\mathbb{1}_{\{|w| \leq b_n\}}} dw \leq M b_n^{d-m}$$

verifying condition (44).

Hence, we can set $b_n = \left(\frac{\log 2}{C'_n}\right)^{\frac{1}{2s+d-m}} \asymp n^{-\frac{1}{2s+d-m}}$ so that

$$KL(P_0, P_1) \leq C' b_n^{2s+d-m} = \frac{\log 2}{n}$$

as required in Lemma 4.

In the meanwhile, notice that

$$|L(h_1) - L(h_0)| = \int_{\{x \in \mathcal{X}: g(x)=\mathbf{0}\}} |h_1(x) - h_0(x)| d\mathcal{H}^m(x) = C'' b_n^s K(0)$$

Hence, define $\theta(P) := L(\mathbb{E}_P[Y|X = \cdot])$ and $d(\theta, \theta') = |\theta - \theta'|$, we deduce from Lemma 4 that the minimax rate

$$\begin{aligned} R_n &\geq \frac{1}{16} d(\theta(P_0), \theta(P_1)) \\ &= \frac{1}{16} |L(h_1) - L(h_0)| \\ &= \frac{1}{16} \int_{\{x \in \mathcal{X}: g(x)=\mathbf{0}\}} |h_1(x) - h_0(x)| d\mathcal{H}^m(x) \\ &= C'' b_n^s |K(0)| \asymp b_n^s \asymp n^{-\frac{s}{2s+d-m}}. \end{aligned}$$

□

Proof of Corollary 1. We now prove that the same rate r_n^* as in Theorem 1 also applies in the nonparametric density case, i.e., $h_0(x) \equiv p_0(x)$ is the probability density function of X_i .

We set $h_0(x) \equiv p_0(x) \equiv 1$ on $[0, 1]^d$, and set

$$p_1(x) := p_0(x) + b_n^s \tilde{K} \left(\frac{\|g(x)\|}{b_n} \right)$$

where

$$\tilde{K}_n(t) := a \left[\exp \left(-\frac{1}{1-t^2} \right) - c_n \right] \mathbb{1}_{\{|t| \leq 1\}}$$

with

$$c_n := \frac{\int \exp \left(-\frac{1}{1-\|g(x)\|^2/b_n^2} \right) \mathbb{1}_{\{\|g(x)\| \leq b_n\}} dx}{\int \mathbb{1}_{\{\|g(x)\| \leq b_n\}} dx}$$

so that $\tilde{K}_n(t) = 0$ whenever $|t| > 1$ and furthermore

$$\int \tilde{K}_n \left(\frac{\|g(x)\|}{b_n} \right) dx = 0.$$

This ensures that

$$\int p_1(x) dx = \int p_0(x) dx + b_n^s \int \tilde{K}_n \left(\frac{\|g(x)\|}{b_n} \right) dx = 1 + 0 = 1.$$

Furthermore, $a > 0$ can be set sufficiently small to ensure that

$$p_1(x) \geq 0,$$

so that p_1 remains a valid density function.

Then, using the inequality that $KL(P_1, P_0) \leq \chi^2(P_1, P_0)$ we have

$$\begin{aligned} KL(P_1, P_0) &\leq \chi_2(P_1, P_0) := \int_{[0,1]^d} \left(\frac{p_1(x)}{p_0(x)} - 1 \right)^2 p_0(x) dx \\ &= \int_{[0,1]^d} \left(b_n^s \tilde{K} \left(\frac{\|g(x)\|}{b_n} \right) \right)^2 dx \\ &= b_n^{2s} \int_{[0,1]^d} \tilde{K} \left(\frac{\|g(x)\|}{b_n} \right)^2 dx \\ &\leq b_n^{2s} \int_{[0,1]^d} \mathbb{1} \left\{ \frac{\|g(X_i)\|}{b_n} \leq 1 \right\} dx \\ &\leq C b_n^{2s} \mathbb{P} \{ \|g(X_i)\| \leq b_n \} \\ &\leq C^2 b_n^{2s+d-m} \end{aligned} \tag{45}$$

which coincides with the rate in (43) under (44) for the nonparametric regression case. The rest of the proof is the same as in the nonparametric regression case. $\square$

Proof of Corollary 2. The proof is adapted from the proof of Theorem 1, combined with the linearization in Assumption 6. As in the proof of Theorem 1, fix any P_0 with regression function h_0 and associated (g, w) . For a given small $b_n > 0$ small, define

$$h_1(x) := h_0(x) + b_n^s K_{d-m} \left(\frac{g(x)}{b_n} \right),$$

where K_{d-m} is defined in the proof of Theorem 1. Let P_0 denote the joint distribution of (X_i, Y_i) with X_i uniform on $[0, 1]^d$ and $Y_i = h_0(X_i) + \epsilon_i$ with $\epsilon_i \sim_{i.i.d.} \mathcal{N}(0, 1)$ and let P_1 be the analogous distribution obtained by replacing h_0 with h_1 . Then, exactly as in the proof of Theorem 1, we have

$$KL(P_0^n, P_1^n) = \frac{n}{2} \int_{[0,1]^d} \left(h_1(x) - h_0(x) \right)^2 dx \lesssim n b_n^{2s+d-m}.$$

and thus $b_n \asymp n^{-\frac{1}{2s+d-m}}$ ensures that $KL(P_0^n, P_1^n) \leq \log 2$ for all n large enough. Now,

by Assumption 6 and the definition of h_1 , we have

$$D\Gamma(h_0)[h_1 - h_0] = \int_{\mathcal{M}} (h_1(x) - h_0(x)) w(x) d\mathcal{H}^m(x) = O(b_n^s)$$

following the same argument in proof of Theorem 1, just with $D\Gamma(h_0)[h_1 - h_0]$ in place of $L(h_1 - h_0)$.

$$|D\Gamma(h_0)[h_1 - h_0]| \asymp b_n^s. \quad (46)$$

Then, writing $R_n := \Gamma(h_1) - \Gamma(h_0) - D\Gamma(h_0)[h_1 - h_0]$

$$\begin{aligned} |\Gamma(h_1) - \Gamma(h_0)| &= |D\Gamma(h_0)[h_1 - h_0] + R_n| \\ &\geq ||D\Gamma(h_0)[h_1 - h_0]| - |R_n|| \\ &\geq (1 - C_r) |D\Gamma(h_0)[h_1 - h_0]| \asymp O(b_n^s) \end{aligned}$$

where the last inequality is due to Assumption 6(ii). Hence, by Lemma 4, we have the conclusion. $\square$

B.2 Proof of Results in Subsection 3.2

Proof of Theorem 2. Let $\Pr_h$ denote the joint distribution of (Y, X, Z) satisfying $Y = Th + u$ with a known conditional expectation operator T and $u|Z \sim \mathcal{N}(0, \sigma^2)$, the so-called reduced-form nonparametric indirection regression (NPIR) model as in Chen and Reiss (2011) with a fixed variance $\sigma^2 > 0$. As pointed out in Chen and Reiss (2011) it is enough to consider the NPIR model to establish the lower bound for the NPIV model, specifically, a lower bound for the NPIR model is also a lower bound for the NPIV model.

Fix any joint law of (X, Z) that satisfies Assumption 7(i)–(ii) and yields a conditional expectation operator $Th(z) = \mathbb{E}[h(X) | Z = z]$ satisfying Assumption 9. Consider the following *submodel* of the NPIV model (19):

$$Y = (Th)(Z) + u, \quad u | Z \sim \mathcal{N}(0, \underline{\sigma}^2), \quad u \perp (X, Z), \quad (47)$$

with $\underline{\sigma}^2$ as in Assumption 7(iii). For this submodel, writing $\epsilon := Y - h(X) = (Th)(Z) - h(X) + u$, we have $\mathbb{E}[\epsilon | Z] = Th(Z) - \mathbb{E}[h(X) | Z] + \mathbb{E}[u | Z] = 0$, so (19) holds. Moreover, $\mathbb{E}[\epsilon^2 | Z] \geq \mathbb{E}[u^2 | Z] = \underline{\sigma}^2$, so Assumption 7(iii) is satisfied. Since Theorem 2 is a minimax *lower bound* (over a larger class of NPIV DGPs), it suffices to lower bound the minimax risk on the restricted submodel (47).

We again apply Le Cam two-point testing argument. Let $h_0 \equiv 0$. We now use

the CDV wavelet to construct another function h_1 . Fix a wavelet type $\tilde{G}_0$ (one of the finitely many tensor-product CDV wavelet types) such that the corresponding mother wavelet is continuous and not identically zero. Such a type exists by construction of the CDV basis. For each resolution level $j \in \mathbb{N}$, let $\{\psi_{j,k,\tilde{G}_0} : k \in \mathcal{K}_j\}$ denote the corresponding wavelets at level j . By compact support and continuity, there exist constants $c_\psi > 0$ and $r_\psi \in (0, 1)$ such that for every sufficiently large j there is a collection of indices k for which

$$|\psi_{j,k,\tilde{G}_0}(x)| \geq c_\psi 2^{jd/2} \quad \text{for all } x \in B(x_{j,k}, r_\psi 2^{-j}), \quad (48)$$

for some $x_{j,k}$ in the interior of $\text{supp}(\psi_{j,k,\tilde{G}_0})$; moreover the balls $B(x_{j,k}, r_\psi 2^{-j})$ can be chosen to be contained in $\text{supp}(\psi_{j,k,\tilde{G}_0})$.

Next, we pick a set of wavelets whose supports lie along the submanifold. Assume $L(\cdot)$ is nontrivial, i.e. $\int_{\mathcal{M}} |w(x)| d\mathcal{H}^m(x) > 0$; otherwise $L(h) \equiv 0$ and the lower bound is trivial. Choose $\underline{w} > 0$ such that the set $\mathcal{M}_w := \{x \in \mathcal{M} : |w(x)| \geq \underline{w}\}$ has $\mathcal{H}^m(\mathcal{M}_w) > 0$.

Under Assumption 1, $\mathcal{M}$ is a C^1 compact m -dimensional submanifold. Hence, there exists $c_M > 0$ such that for all $x \in \mathcal{M}$ and all small enough $r > 0$,

$$\mathcal{H}^m(\mathcal{M} \cap B(x, r)) \geq c_M r^m. \quad (49)$$

Consequently, for each large j we can select a set of indices $\mathcal{I}_j \subset \mathcal{K}_j$ with cardinality $|\mathcal{I}_j| \asymp 2^{jm}$ such that the balls $\{B(x_{j,k}, r_\psi 2^{-j})\}_{k \in \mathcal{I}_j}$ are pairwise disjoint and satisfy $x_{j,k} \in \mathcal{M}_w$ for all $k \in \mathcal{I}_j$.

For this fixed j , define the perturbation

$$h_1(x) := \delta 2^{-j(s+d/2)} \sum_{k \in \mathcal{I}_j} a_{j,k} \psi_{j,k,\tilde{G}_0}(x), \quad a_{j,k} := \text{sgn} \left(\int_{\mathcal{M}} \psi_{j,k,\tilde{G}_0}(x) w(x) d\mathcal{H}^m(x) \right), \quad (50)$$

with a constant $\delta > 0$ to be chosen sufficiently small.

By the standard CDV-wavelet characterization of Holder/Besov classes (e.g. Appendix A of [Chen and Christensen \(2018\)](#)), there exists a constant $\delta_0 > 0$ such that if $\sup_{j,k,\tilde{G}} 2^{j(s+d/2)} |\langle h, \psi_{j,k,\tilde{G}} \rangle_{L^2(\mathcal{X})}| \leq \delta_0$, then $h \in \Lambda_c^s$ (for a possibly different radius constant c). Since h_1 has wavelet coefficients equal to $\delta 2^{-j(s+d/2)}$ at level j and 0 elsewhere, choosing $\delta \leq \delta_0$ ensures $h_1 \in \Lambda_c^s(\mathcal{X})$.

Because $a_{j,k}$ aligns the sign of each contribution,

$$|L(h_1) - L(h_0)| = |L(h_1)| = \delta 2^{-j(s+d/2)} \sum_{k \in \mathcal{I}_j} \left| \int_{\mathcal{M}} \psi_{j,k,\tilde{G}_0}(x) w(x) d\mathcal{H}^m(x) \right|.$$

For each $k \in \mathcal{I}_j$, using (48), $|w| \geq \underline{w}$ on $\mathcal{M}_w$, and the lower volume bound (49) with $r = r_\psi 2^{-j}$, we obtain

$$\left| \int_{\mathcal{M}} \psi_{j,k,\tilde{G}_0}(x) w(x) d\mathcal{H}^m(x) \right| \geq \underline{w} \int_{\mathcal{M} \cap B(x_{j,k}, r_\psi 2^{-j})} |\psi_{j,k,\tilde{G}_0}(x)| d\mathcal{H}^m(x) \gtrsim 2^{jd/2} \cdot 2^{-jm} = 2^{j(d/2-m)}.$$

Therefore,

$$|L(h_1) - L(h_0)| \gtrsim \delta 2^{-j(s+d/2)} \cdot |\mathcal{I}_j| \cdot 2^{j(d/2-m)} \asymp \delta 2^{-j(s+d/2)} \cdot 2^{jm} \cdot 2^{j(d/2-m)} = \delta 2^{-js}.$$

Hence, the parameter separation is of order 2^{-js} .

Next we compute the KL divergence under the NPIR model. Consider the distributions P_0, P_1 induced by (47) with $h = h_0$ and $h = h_1$, and the same joint law of (X, Z) in both cases. Since only the conditional mean of $Y \mid Z$ changes, standard Gaussian KL calculations yield

$$KL(P_0^n, P_1^n) = n KL(P_0, P_1) = \frac{n}{2\sigma^2} \|T(h_1 - h_0)\|_{L^2(\mathcal{Z})}^2 = \frac{n}{2\sigma^2} \|Th_1\|_{L^2(\mathcal{Z})}^2. \quad (51)$$

By the link condition (20) and the fact that h_1 has nonzero wavelet coefficients only at the single level j and type $\tilde{G}_0$, we have

$$\|Th_1\|_{L^2(\mathcal{Z})}^2 \lesssim \nu(2^j)^2 \sum_{k \in \mathcal{I}_j} \langle h_1, \psi_{j,k,\tilde{G}_0} \rangle_{L^2(\mathcal{X})}^2 = \nu(2^j)^2 \cdot |\mathcal{I}_j| \cdot \delta^2 2^{-2j(s+d/2)} \asymp \delta^2 \nu(2^j)^2 2^{-j(2s+d-m)}.$$

Plugging this into (51) gives

$$KL(P_0^n, P_1^n) \lesssim n \delta^2 \nu(2^j)^2 2^{-j(2s+d-m)}. \quad (52)$$

Choose $\delta > 0$ small enough (fixed) so that the implicit constant in (52) is absorbed.

Mildly ill-posed case: $\nu(t) = t^{-\varsigma}$. Then (52) becomes $KL(P_0^n, P_1^n) \lesssim n 2^{-j(2(s+\varsigma)+d-m)}$. Choose $j = j_n$ such that $2^{j_n} \asymp n^{1/(2(s+\varsigma)+d-m)}$, which yields $KL(P_0^n, P_1^n) \leq (\log 2)$ for all large n . By Step 3, the separation satisfies $|L(h_1) - L(h_0)| \gtrsim 2^{-j_n s} \asymp n^{-s/(2(s+\varsigma)+d-m)}$.

Severely ill-posed case: $\nu(t) = \exp(-\frac{1}{2}t^\varsigma)$. Then (52) becomes $KL(P_0^n, P_1^n) \lesssim n \exp(-c 2^{\varsigma j}) 2^{-j(2s+d-m)}$ for some $c > 0$. Choose $j = j_n$ such that $2^{\varsigma j_n} \asymp \log n$, which implies $KL(P_0^n, P_1^n) \rightarrow 0$ and hence $KL(P_0^n, P_1^n) \leq (\log 2)$ for all large n . Step 3 gives $|L(h_1) - L(h_0)| \gtrsim 2^{-j_n s} \asymp (\log n)^{-s/\varsigma}$.

Finally, apply Lemma 4 with the metric $d(\theta, \theta') = |\theta - \theta'|$ and $\theta(P_h) = L(h)$. Since

$KL(P_0^n, P_1^n) \leq (\log 2)$ for the above choices, Lemma 4 yields

$$\inf_{\tilde{\theta}_n} \sup_{h \in \Lambda_c^s} \mathbb{E}_{P_h} [(\tilde{\theta}_n - L(h))^2] \gtrsim |L(h_1) - L(h_0)|^2 \gtrsim r_{NPIV,n}^2,$$

with $r_{NPIV,n}$ as stated in Theorem 2. This proves the claim. $\square$

Proof of Corollary 3. Since the minimax risk over $\Lambda_c^s(\mathcal{X})$ dominates the risk over any subset, it suffices to construct $h_0, h_1 \in \Lambda_c^s(\mathcal{X})$ (with associated distributions P_0, P_1 under the NPIR model such that (i) $KL(P_0^n, P_1^n) \leq \log 2$ and (ii) $|\Gamma(h_1) - \Gamma(h_0)| \gtrsim r_{NPIV,n}$. Lemma 4 then yields the desired lower bound.

The proof follows from those of Corollary 2 and Theorem 2. Hence we are brief here. We follow the proof of Corollary 2 to control the nonlinear remainder in the separation in the target parameter $\Gamma(h)$. Let $R_n := \Gamma(h_1) - \Gamma(h_0) - D\Gamma(h_0)[h_1 - h_0]$

$$\begin{aligned} |\Gamma(h_1) - \Gamma(h_0)| &= |D\Gamma(h_0)[h_1 - h_0] + R_n| \\ &\geq ||D\Gamma(h_0)[h_1 - h_0]| - |R_n|| \\ &\geq (1 - C_r) |D\Gamma(h_0)[h_1 - h_0]| \geq \text{const.} \times 2^{-js} \end{aligned} \quad (53)$$

where the last inequality is due to Assumption 6(ii), and the bound $|D\Gamma(h_0)[h_1 - h_0]| \geq \text{const.} \times 2^{-js}$ follows from the proof of Theorem 2 their lower bound on the linear separation of $|L[h_1 - h_0]| \geq \text{const.} \times 2^{-js}$.

From the proof of Theorem 2 we have:

$$KL(P_0^n, P_1^n) \lesssim n \delta^2 \nu(2^j)^2 2^{-j(2s+d-m)}. \quad (54)$$

We finally optimize the choice of $j = j_n$ depending on the degree of ill-posedness.

Mildly ill-posed case. If $\nu(t) = t^{-\varsigma}$ then (54) becomes $KL(P_0^n, P_1^n) \lesssim n 2^{-j(2(s+\varsigma)+d-m)}$. Choose $j = j_n$ so that $2^{j_n} \asymp n^{1/(2(s+\varsigma)+d-m)}$. Then $KL(P_0^n, P_1^n) \leq \log 2$ for all large n (after fixing $\delta > 0$ small enough). By (53),

$$|\Gamma(h_1) - \Gamma(h_0)| \gtrsim 2^{-j_n s} \asymp n^{-\frac{s}{2(s+\varsigma)+d-m}} = r_{NPIV,n}.$$

Severely ill-posed case. If $\nu(t) = \exp(-\frac{1}{2}t^\varsigma)$ then (54) gives $KL(P_0^n, P_1^n) \lesssim n \exp(-c 2^{\varsigma j}) 2^{-j(2s+d-m)}$ for some $c > 0$. Choose $j = j_n$ so that $2^{\varsigma j_n} \asymp \log n$. Then $KL(P_0^n, P_1^n) \rightarrow 0$, hence $KL(P_0^n, P_1^n) \leq \log 2$ for all large n , and (53) yields

$$|\Gamma(h_1) - \Gamma(h_0)| \gtrsim 2^{-j_n s} \asymp (\log n)^{-\frac{s}{\varsigma}} = r_{NPIV,n}.$$

Finally, Lemma 4 implies that for any estimator $\tilde{\theta}_n$,

$$\sup_{h \in \{h_0, h_1\}} \mathbb{E}_h \left[(\tilde{\theta}_n - \Gamma(h))^2 \right] \gtrsim r_{\text{NPIV}, n}^2.$$

Since $\{h_0, h_1\} \subset \Lambda_c^s(\mathcal{X})$, the same lower bound holds for the minimax risk over $\Lambda_c^s(\mathcal{X})$. Hence, we obtain Corollary 3. □

C Proofs of Theoretical Results in Section 4

C.1 Proof of Results in Subsection 4.1

Proof of Lemma 1. Write $K_n = K = J^d$. Note that

$$\|L(\bar{b}^K)\|^2 = \sum_{k=1}^K L^2(\bar{b}_k^K) = \sum_{k=1}^{K_n} \left[\int_{\mathcal{M}} \bar{b}_k^{K_n}(x) w(x) d\mathcal{H}^m(x) \right]^2.$$

Since $\bar{b}^K = G^{-1/2} b^K$ and G has eigenvalues bounded away from zero and infinity by Assumption 12(i), we have $\|L(\bar{b}^K)\|^2 = L(b^K)^\top G^{-1} L(b^K) \asymp \|L(b^K)\|^2 = \sum_{k=1}^K L^2(b_k^K)$. It therefore suffices to bound $\sum_{k=1}^K L^2(b_k^K)$. We note that the tensor-product structure holds for the original B-spline basis b^K , but *not* in general for the orthonormalized basis $\bar{b}^K$; accordingly, we carry out the following calculation for b^K .

Recall that $(b_k^{K_n})_{k=1}^{K_n}$ is constructed as tensor products of univariate basis functions

$$b_k^{K_n}(x) = \prod_{\ell=1}^d b_{k_\ell}(x_\ell)$$

for some $1 \leq k_1, \dots, k_d \leq J$: in other words, each index k is bijectively identified by the vector $(k_1, \dots, k_d)$. Hence,

$$\sum_{k=1}^K L^2(b_k^K) = \sum_{k_1, \dots, k_d=1}^J L^2\left(\prod_{\ell=1}^d b_{k_\ell}(x_\ell)\right).$$

For each $L(b_k^K)$, we apply the decomposition (41) and obtain

$$L(b_k^K) = \int_{\mathcal{M}} b_k^K(x) w(x) p(x) d\mathcal{H}^m(x) = \sum_{j=1}^{\bar{j}} T_{kj},$$

where each j corresponds to a piece from the local charts $(\mathcal{U}_j, \varphi_j)$ for $\mathcal{M}$ along with

the partition of unit function ρ_j and

$$T_{kj} := \int_{\mathcal{U}_j} \rho_j \left(\varphi_j \left(x_{(j,m)} \right) \right) b_k^K \left(\varphi_j \left(x_{(j,m)} \right) \right) w \left(\varphi_j \left(x_{(j,m)} \right) \right) \mathcal{J} \varphi_j \left(x_{(j,m)} \right) dx_{(j,m)}.$$

From now on, to simplify notation, we will suppress the subscript j whenever there is no ambiguity. Furthermore, define

$$\bar{w} \left(x_{(m)} \right) := \rho \left(\varphi \left(x_{(m)} \right) \right) w \left(\varphi \left(x_{(m)} \right) \right) \mathcal{J} \varphi \left(x_{(m)} \right),$$

so that we may write T_{kj} more succinctly as

$$T_{kj} = \int_{\mathcal{U}} b_k^K \left(\varphi \left(x_{(m)} \right) \right) \bar{w} \left(x_{(m)} \right) dx_{(m)}.$$

Again, since (b_k^K) is constructed as tensor products of univariate basis functions, we can decompose

$$b_k^K \left(\varphi \left(x_{(m)} \right) \right) = b_{k,(m)}^K \left(x_{(m)} \right) \cdot b_{k,-(m)}^K \left(\psi \left(x_{(m)} \right) \right)$$

where $b_{k,(m)}^K \left(x_{(m)} \right) = \prod_{\ell \in (m)} b_{k_\ell} \left(x_{\ell} \right)$ and $b_{k,-(m)}^K \left(x_{(m)} \right) = \prod_{\ell \notin (m)} b_{k_\ell} \left(\left[\psi \left(x_{(m)} \right) \right]_\ell \right)$ correspond to the coordinates in $x_{(m)}$ and $x_{-(m)}$, respectively. Then, we have

$$\begin{aligned} T_{kj} &= \int_{\mathcal{U}} b_{k,(m)}^K \left(x_{(m)} \right) b_{k,-(m)}^K \left(\psi \left(x_{(m)} \right) \right) \bar{w} \left(\varphi \left(x_{(m)} \right) \right) dx_{(m)} \\ &= \langle b_{(m)}^K (\cdot), b_{-(m)}^K (\psi(\cdot)) \bar{w}_j (\varphi(\cdot)) \rangle_{L_2(\mathcal{U})} \end{aligned}$$

where $\langle f_1, f_2 \rangle_{L_2(\mathcal{U})} := \int_{\mathcal{U}} f_1 \left(x_{(m)} \right) f_2 \left(x_{(m)} \right) dx_{(m)}$.

Since $\{b_{(m)}^K(\cdot)\}$ is a B-spline basis on $\mathcal{X} \subseteq \mathbb{R}^d$, its restriction to $\mathcal{U}$ satisfies the frame condition,²⁴ which gives

$$\begin{aligned} \sum_{k(m)} T_{kj}^2 &= \sum_{k_\ell: \ell \in (m)} \langle b_{(m)}^K (\cdot), b_{-(m)}^K (\psi(\cdot)) \bar{w}_j (\varphi(\cdot)) \rangle_{L_2}^2 \\ &\asymp M \cdot \left\| b_{-(m)}^K (\psi(\cdot)) \bar{w}_j (\varphi(\cdot)) \right\|_{L_2}^2 \quad \text{by the frame condition} \\ &= M \cdot \int \left[b_{-(m)}^K \left(\psi \left(x_{(m)} \right) \right) \right]^2 \bar{w}_j^2 \left(\varphi \left(x_{(m)} \right) \right) dx_{(m)} \\ &\asymp M. \end{aligned}$$

Hence,

$$\sum_{k=1}^K T_{kj}^2 = \sum_{k_{-(m)}} \left[\sum_{k(m)} T_{kj}^2 \right] \asymp \sum_{k_{-(m)}} M = J^{d-m} M$$

²⁴A (finite or countable) sequence of functions $(b_k(\cdot))$ is said to be a frame on a Hilbert space $(\mathcal{H}, \langle \cdot, \cdot \rangle)$ of functions if it satisfies the *frame condition*: there exist constants $\underline{M}, \overline{M} > 0$ s.t. $\underline{M} \|f\|^2 \leq \sum_k \langle b_k(\cdot), f(\cdot) \rangle^2 \leq \overline{M} \|f\|^2$ for every $f \in \mathcal{H}$.

Now, since $L(b_k^K) = \sum_{j=1}^{\bar{j}} T_{kj}$, and $L^2(b_k^K) = \left(\sum_{j=1}^{\bar{j}} T_{kj}\right)^2 \asymp \sum_{j=1}^{\bar{j}} T_{kj}^2$, we have

$$\|L(b^K)\|^2 = \sum_{k=1}^K L^2(b_k^K) \asymp \bar{j} \sum_{j=1}^{\bar{j}} J^{d-m} M \asymp M J^{d-m}.$$

The claimed result $\|L(\bar{b}^K)\|^2 \asymp J^{d-m}$ then follows from the eigenvalue equivalence $\|L(\bar{b}^K)\|^2 \asymp \|L(b^K)\|^2$ noted above. $\square$

Proof of Theorem 3. Note that

$$\begin{aligned} |\hat{\theta} - \theta_0| &= |L(\hat{h} - h_0)| \\ &\leq |L(\hat{h} - \tilde{h})| + |L(\tilde{h} - h_0)| \\ &\leq |L(\hat{h} - \tilde{h})| + \|\tilde{h} - h_0\|_\infty \end{aligned}$$

where $\tilde{h} := P_{K,n} h_0$ is the empirical Least Squares projection of h_0 to the linear sieve space under $P_{K,n}$, which is defined as follows:

$$P_{K,n} h_0 := b^K(x)' \hat{G}^{-1} \frac{1}{n} \sum_{i=1}^n b^K(X_i) h_0(X_i) = \bar{b}^K(x)' [\bar{G}]^{-1} \frac{1}{n} \sum_{i=1}^n \bar{b}^K(X_i) h_0(X_i),$$

where $\hat{G} := \frac{1}{n} \sum_{i=1}^n b^{K_n}(X_i) b^{K_n}(X_i)'$ and $\bar{G}_n := \frac{1}{n} \sum_{i=1}^n \bar{b}^{K_n}(X_i) \bar{b}^{K_n}(X_i)'$. Under Assumptions 1, 3, 4, 5, 11 and 12(i)(ii)(iii), following the proof of Theorem 3.1 of Chen and Christensen (2015), under the i.i.d. setting, we obtain result (1):

$$\frac{\sqrt{n} L(\hat{h} - P_{K,n} h_0)}{\|v_{K_n}^*\|_{sd}} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{v_{K_n}^*(X_i)}{\|v_{K_n}^*\|_{sd}} \epsilon_i + o_p(1) = O_p(1),$$

with $\|v_{K_n}^*\|_{sd}^2 \asymp K_n^{\frac{d-m}{d}}$ by Lemma 1.

For result (2), we note that Result (1) implies that

$$L(\hat{h} - \tilde{h}) = O_p\left(\sqrt{K_n^{\frac{d-m}{d}}/n}\right).$$

In addition, Assumption 4 implies that

$$|L(P_{K,n} h_0 - h_0)| \leq \|\tilde{h} - h_0\|_\infty \times \int_{\mathcal{M}} |w(x)| d\mathcal{H}^m(x).$$

Let $h_n^* \in \mathbb{H}_{K_n}$ solves $\inf_{h \in \mathbb{H}_{K_n}} \|(h - h_0)\|_\infty$. Then we have:

$$\begin{aligned} \|\tilde{h} - h_0\|_\infty &= \|(P_{K,n} h_0 - h_n^*) + (h_n^* - h_0)\|_\infty \\ &= \|P_{K,n}(h_0 - h_n^*) + (h_n^* - h_0)\|_\infty \end{aligned}$$

$$\begin{aligned}
&\leq \|P_{K,n}(h_0 - h_n^*)\|_\infty + \|(h_n^* - h_0)\|_\infty \\
&\leq (\|P_{K,n}\|_\infty + 1) \|(h_n^* - h_0)\|_\infty \\
&\leq O_p(K_n^{-s/d}),
\end{aligned}$$

where the last inequality is due to Assumption 12(iv)(v). Consequently,

$$\hat{\theta} - \theta_0 \equiv L(\hat{h} - h_0) = O_p\left(\sqrt{K_n^{\frac{d-m}{d}}/n} + K_n^{-s/d}\right).$$

The rate is minimized by setting K_n^* such that $\sqrt{K_n^{\frac{d-m}{d}}/n} \asymp K_n^{-s/d}$, we have

$$\hat{\theta} - \theta_0 = O_p\left((K_n^*)^{-s/d}\right) = O_p\left(n^{-\frac{s}{2s+d-m}}\right).$$

□

C.1.1 Minimax rate-optimal oracle estimator for $L(h_0)$ when density p_0 is known

If the density p_0 is indeed known and the submanifold satisfies the extra smoothness assumption 2, then one can construct a simple estimator for $\theta_0 = L(h_0)$ that attains the lower bound rate $r_n^* = n^{-\frac{s}{2s+d-m}}$ as follows. Let $c := d - m$ denote the codimension of the submanifold $\mathcal{M}$ satisfying Assumption 2. For a non-negative kernel $\mathcal{K}_c : \mathbb{R}^c \rightarrow \mathbb{R}$ with $\int \mathcal{K}_c = 1$ with $\mathcal{K}_c(0) > 0$, and a bandwidth b_n going to zero as n goes to infinity, define the tube kernel

$$K_{\mathcal{M},b_n}(x) := \frac{1}{b_n^c} \mathcal{K}_c\left(\frac{g(x)}{b_n}\right) \mathcal{J}_g(x),$$

and the tube-smoothed linear functional $L_{b_n}(h_0) := \int_{\mathcal{X}} h_0(x) w(x) K_{\mathcal{M},b_n}(x) dx$. We note that for any $h, h_0 \in \Lambda^s$ for $s > 0$,

$$L_{b_n}(h - h_0) = \mathbb{E}[v_{b_n}^*(X_i)(h(X_i) - h_0(X_i))], \quad v_{b_n}^*(x) := \frac{w(x) K_{\mathcal{M},b_n}(x)}{p_0(x)}.$$

It is easy to check that $\mathbb{E}[(v_{b_n}^*(X_i))^2] \asymp b_n^{-c}$, which goes to infinity as b_n goes to zero. Notice that $L_{b_n}(h_0) = \mathbb{E}[v_{b_n}^*(X_i)h_0(X_i)]$ we could call $v_{b_n}^*(\cdot)$ a tube Riesz representer for $\theta_0 = L(h_0)$. One can estimate $\theta_0 = L(h_0)$ by the following (oracle) tube-weighted estimator

$$\hat{\theta}_{\text{oracle}}(b_n) := \frac{1}{n} \sum_{i=1}^n v_{b_n}^*(X_i) Y_i.$$

It is easy to see that under Assumptions 2, 3, 11 and 4, we have:

$$|L_{b_n}(h_0) - L(h_0)| \lesssim b_n^s, \quad \text{and}$$

$$\mathbb{E} \left[(\hat{\theta}_{\text{oracle}}(b) - L_{b_n}(h_0))^2 \right] \lesssim \frac{1}{n} \times \mathbb{E}[(v_b^*(X_i))^2] \asymp \frac{1}{nb_n^c}.$$

By setting $b_n^* \asymp n^{-1/(2s+c)}$ we obtain $\hat{\theta}_{\text{oracle}}(b_n^*) - \theta_0 = O_p \left(n^{-\frac{s}{2s+d-m}} \right)$.

C.2 Proof of Theorem 4 in Section 4.2

Proof of Lemma 2. The pathwise derivative formula (29) follows by differentiating under the integral sign:

$$\frac{d}{dt} \Gamma(h_0 + tv) \Big|_{t=0} = \frac{d}{dt} \int_{\mathcal{M}} \phi(h_0(x) + tv(x), x) w(x) d\mathcal{H}^m(x) \Big|_{t=0} = \int_{\mathcal{M}} \phi_1(h_0(x), x) v(x) w(x) d\mathcal{H}^m(x),$$

which is justified by dominated convergence using the Lipschitz condition on ϕ . $\square$

Proof of Theorem 4(1). Define estimation errors $\delta_1 := \hat{h}_1 - h_0$ and $\delta_2 := \hat{h}_2 - h_0$. By sample splitting, δ_1 and δ_2 are independent. Recall that $\bar{h} := (\hat{h}_1 + \hat{h}_2)/2$. Consider the Taylor expansion of $\phi(\bar{h}(x), x)$ around $h_0(x)$:

$$\phi(\bar{h}, x) = \phi(h_0, x) + \phi_1(h_0, x)(\bar{h} - h_0) + \frac{1}{2} \phi_{11}(h_0, x)(\bar{h} - h_0)^2 + R_3(x),$$

where the third-order remainder satisfies

$$R_{\text{rem}} := \int_{\mathcal{M}} |R_3| w d\mathcal{H}^m = O_p \left(\|\bar{h} - h_0\|_{\infty} \cdot \int_{\mathcal{M}} (\bar{h} - h_0)^2 w d\mathcal{H}^m \right).$$

Now, since

$$(\bar{h} - h_0)^2 = \frac{1}{4}(\delta_1 + \delta_2)^2 = \frac{1}{4}(\delta_1^2 + \delta_2^2 + 2\delta_1\delta_2),$$

$$\frac{1}{4}(\hat{h}_1 - \hat{h}_2)^2 = \frac{1}{4}(\delta_1 - \delta_2)^2 = \frac{1}{4}(\delta_1^2 + \delta_2^2 - 2\delta_1\delta_2),$$

we have

$$(\bar{h} - h_0)^2 - \frac{1}{4}(\hat{h}_1 - \hat{h}_2)^2 = \delta_1\delta_2,$$

where the diagonal terms δ_1^2 and δ_2^2 are eliminated.

Write $\hat{\theta}_{SS} - \theta_0 = T_{\text{lin}} + T_{\text{quad}} + R_{\text{rem}}$, where

$$T_{\text{lin}} := \int_{\mathcal{M}} \phi_1(h_0, x)(\bar{h} - h_0) w d\mathcal{H}^m,$$

$$T_{\text{quad}} := \frac{1}{2} \int_{\mathcal{M}} \phi_{11}(h_0, x) \delta_1 \delta_2 w d\mathcal{H}^m,$$

and $R_{\text{rem}} = O_p \left(\|\bar{h} - h_0\|_{\infty} \cdot T_{\text{quad}} \right) = o_p(T_{\text{quad}})$ collects higher-order remainders.

For the linear term T_{lin} , we have, by Theorem 3, $T_{lin} = O_p(\sqrt{K_n^{(d-m)/d}/n} + K_n^{-s/d})$. At $K_n^* \asymp n^{d/(2s+d-m)}$ (and $s > m/2$), this gives $T_{lin} = O_p(n^{-s/(2s+d-m)})$.

For the quadratic term T_{quad} , following the same proof of Lemma 8(b)(c) we obtain that $T_{quad} = O_p(n^{-s/(2s+d-m)})$ at $K_n^* \asymp n^{d/(2s+d-m)}$ (and $s > m/2$).

Hence, under $s > m/2$, we obtain: $\hat{\theta}_{SS} - \theta_0 = T_{lin} + o_p(r_n^*) = O_p(r_n^*)$. $\square$

Proof of Theorem 4(2). Let $K \equiv K_n \asymp K_n^* \asymp n^{d/(2s+d-m)}$ and $r_n^* := n^{-s/(2s+d-m)}$. Recall $\epsilon_i = Y_i - h_0(X_i)$, $\tilde{h} := P_{K,n}h_0$ and the series LS representation (23):

$$\hat{h}(x) = \frac{1}{n} \sum_{i=1}^n \hat{s}_i(x) Y_i, \quad \hat{s}_i(x) := b(x)' \hat{G}^{-1} b(X_i). \quad (55)$$

Hence

$$\hat{h}(x) - \tilde{h}(x) = \frac{1}{n} \sum_{i=1}^n \hat{s}_i(x) \epsilon_i. \quad (56)$$

For each $x \in \mathcal{M}$, similar to that in the proof of Theorem 4(1), there exists a measurable function $\bar{h}_0(x)$ lying between $\hat{h}(x)$ and $h_0(x)$ such that

$$\phi(\hat{h}(x), x) = \phi(h_0(x), x) + \phi_1(h_0(x), x) \{\hat{h}(x) - h_0(x)\} + \frac{1}{2} \phi_{11}(\bar{h}_0(x), x) \{\hat{h}(x) - h_0(x)\}^2. \quad (57)$$

Integrating (57) against $w(x) d\mathcal{H}^m(x)$ yields the decomposition

$$\hat{\theta}_{LOO}^\Gamma - \Gamma(h_0) = T_{lin} + T_{quad} - C_n, \quad (58)$$

where

$$\begin{aligned} T_{lin} &:= \int_{\mathcal{M}} \phi_1(h_0(x), x) \{\hat{h}(x) - h_0(x)\} w(x) d\mathcal{H}^m(x), \\ T_{quad} &:= \frac{1}{2} \int_{\mathcal{M}} \phi_{11}(\bar{h}_0(x), x) \{\hat{h}(x) - h_0(x)\}^2 w(x) d\mathcal{H}^m(x), \\ C_n &:= \frac{1}{2n^2} \sum_{i=1}^n \int_{\mathcal{M}} \phi_{11}(\hat{h}(x), x) \hat{s}_i(x)^2 \{\hat{\epsilon}_i^{(-i)}\}^2 w(x) d\mathcal{H}^m(x) \quad (\text{the correction term in (31)}). \end{aligned}$$

Write $\hat{h} - h_0 = (\hat{h} - \tilde{h}) + (\tilde{h} - h_0)$. By Theorem 3 (applied to the linear functional with weight $x \mapsto \phi_1(h_0(x), x)w(x)$, which is bounded by Assumption 13(b)),

$$T_{lin} = O_p\left(\sqrt{\frac{K^{(d-m)/d}}{n}} + K^{-s/d}\right) = O_p(r_n^*). \quad (59)$$

Let $u(x) := \hat{h}(x) - \tilde{h}(x)$ and $b(x) := \tilde{h}(x) - h_0(x)$. Then $\hat{h} - h_0 = u + b$ and

$$(u + b)^2 = u^2 + 2ub + b^2. \quad (60)$$

Using $\|\phi_{11}(\bar{h}_0, \cdot)\|_\infty \lesssim 1$ (implied by Assumption 13(b)), boundedness of w on $\mathcal{M}$, Cauchy–Schwarz, and the standard approximation bound $\|b\|_\infty = O(K^{-s/d})$ (Assumption 12(iv)(v)), we have

$$\int_{\mathcal{M}} |u(x)b(x)| |w(x)| d\mathcal{H}^m(x) \leq \|b\|_\infty \left(\int_{\mathcal{M}} u(x)^2 |w(x)| d\mathcal{H}^m(x) \right)^{1/2} \left(\int_{\mathcal{M}} |w(x)| d\mathcal{H}^m(x) \right)^{1/2} \quad (61)$$

$$= O_p(K^{-s/d} \|u\|_{L^2(|w|)}), \quad (62)$$

and

$$\int_{\mathcal{M}} b(x)^2 |w(x)| d\mathcal{H}^m(x) = O(K^{-2s/d}). \quad (63)$$

By the same argument as for the quadratic remainder term in the proof of Theorem 4(1), $\|u\|_{L^2(|w|)} = O_p(\sqrt{K/n})$ so that the right-hand side of (61) is $o_p(r_n^*)$ when $K \asymp K_n^*$ and $s > m/2$. Therefore,

$$T_{\text{quad}} = \frac{1}{2} \int_{\mathcal{M}} \phi_{11}(\bar{h}_0(x), x) u(x)^2 w(x) d\mathcal{H}^m(x) + o_p(r_n^*). \quad (64)$$

By (56), for each $x \in \mathcal{M}$,

$$u(x)^2 = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \hat{s}_i(x) \hat{s}_j(x) \epsilon_i \epsilon_j = \underbrace{\frac{1}{n^2} \sum_{i=1}^n \hat{s}_i(x)^2 \epsilon_i^2}_{\text{diagonal}} + \underbrace{\frac{1}{n^2} \sum_{i \neq j} \hat{s}_i(x) \hat{s}_j(x) \epsilon_i \epsilon_j}_{\text{off-diagonal}}. \quad (65)$$

Plugging (65) into (64) yields

$$\frac{1}{2} \int_{\mathcal{M}} \phi_{11}(\bar{h}_0(x), x) u(x)^2 w(x) d\mathcal{H}^m(x) = T_{\text{diag}} + T_{\text{off}}, \quad (66)$$

where

$$T_{\text{diag}} := \frac{1}{2n^2} \sum_{i=1}^n \epsilon_i^2 \int_{\mathcal{M}} \phi_{11}(\bar{h}_0(x), x) \hat{s}_i(x)^2 w(x) d\mathcal{H}^m(x), \quad (67)$$

$$T_{\text{off}} := \frac{1}{2n^2} \sum_{i \neq j} \epsilon_i \epsilon_j \int_{\mathcal{M}} \phi_{11}(\bar{h}_0(x), x) \hat{s}_i(x) \hat{s}_j(x) w(x) d\mathcal{H}^m(x). \quad (68)$$

The diagonal term T_{diag} is of order K/n and, for $m/2 < s < m$, would dominate r_n^* . The correction term C_n in (58) is designed to estimate and subtract T_{diag} .

To formalize this, we use the standard leave-one-out identity for series LS. Let $h_{ii} := \hat{s}_i(X_i)/n$ denote the i th diagonal element of the hat matrix. Then the leave-one-out residual satisfies $\hat{\epsilon}_i^{(-i)} = \hat{\epsilon}_i/(1 - h_{ii})$ and, for every $x \in \mathcal{X}$,

$$\hat{h}(x) - \hat{h}^{(-i)}(x) = \frac{1}{n} \hat{s}_i(x) \hat{\epsilon}_i^{(-i)}. \quad (69)$$

(Equation (69) is an immediate consequence of the Sherman–Morrison formula; see, e.g., the standard OLS influence/LOO formula.) Hence C_n can equivalently be written as

$$C_n = \frac{1}{2} \sum_{i=1}^n \int_{\mathcal{M}} \phi_{11}(\hat{h}(x), x) \{\hat{h}(x) - \hat{h}^{(-i)}(x)\}^2 w(x) d\mathcal{H}^m(x). \quad (70)$$

Combining (69) with the decomposition $\hat{\epsilon}_i^{(-i)} = \epsilon_i + \{h_0(X_i) - \hat{h}^{(-i)}(X_i)\}$ and using that $\hat{h}^{(-i)}$ is independent of ϵ_i conditional on the designs, one obtains (by a direct expansion of the square) that

$$C_n = T_{\text{diag}} + R_{\text{diag}}, \quad (71)$$

where the remainder R_{diag} collects the cross and squared terms involving the leave-one-out first-stage error $h_0(X_i) - \hat{h}^{(-i)}(X_i)$ and also the (asymptotically negligible) difference between $\phi_{11}(\hat{h}, \cdot)$ and $\phi_{11}(\bar{h}_0, \cdot)$. A standard Cauchy–Schwarz bound combined with the same matrix trace bound used in the proof of Lemma 8(b) (to control the stochastic term $T_{3,\text{stoch}}(K)$ in (137)), with w replaced by $\phi_{11}(\bar{h}_0, \cdot)w$ yields

$$R_{\text{diag}} = O_p\left(\frac{1}{n} \sqrt{K^{(2d-m)/d}}\right) = O_p(r_n^*) \quad \text{when } s > m/2 \text{ and } K \asymp K_n^*. \quad (72)$$

Consequently, by (66)–(71),

$$T_{\text{quad}} - C_n = T_{\text{off}} + o_p(r_n^*) + R_{\text{diag}}. \quad (73)$$

Conditional on the designs, T_{off} is a (conditionally) degenerate U-statistic of order two. By exactly the same argument as in the proof of Lemma 8(b),

$$T_{\text{off}} = O_p\left(\frac{1}{n} \sqrt{K^{(2d-m)/d}}\right) = O_p(r_n^*) \quad \text{when } s > m/2 \text{ and } K \asymp K_n^*. \quad (74)$$

Combining (58), (59), (64), (73), (72) and (74) yields

$$\hat{\theta}_{\text{LOO}}^\Gamma - \Gamma(h_0) = T_{\text{lin}} + T_{\text{off}} + O_p\left(\frac{1}{n} \sqrt{K^{(2d-m)/d}}\right) + o_p(r_n^*) = O_p(r_n^*),$$

which proves the claim. $\square$

Proof of Theorem 4(3). We start by noting that

$$\begin{aligned} \hat{\theta} - \theta_0 &= \int_{\mathcal{M}} [\phi(\hat{h}, x) - \phi(h_0, x)] w d\mathcal{H}^m \\ &= \underbrace{\int_{\mathcal{M}} \phi_1(h_0, x)(\hat{h} - h_0) w d\mathcal{H}^m}_{=: T_{\text{lin}}} + \underbrace{\int_{\mathcal{M}} \phi_{11}(\bar{h}_0, x)(\hat{h} - h_0)^2 w d\mathcal{H}^m}_{=: T_{\text{quad}}}, \end{aligned} \quad (75)$$

where $\bar{h}_0(x)$ lies between $\hat{h}(x)$ and $h_0(x)$.

By Theorem 3, the linear term satisfies

$$T_{lin} = O_p \left(\sqrt{K_n^{(d-m)/d}/n} + K_n^{-s/d} \right).$$

At $K_n^* \asymp n^{d/(2s+d-m)}$ (and $s > m/2$), this gives $T_{lin} = O_p(r_n^*)$.

For the quadratic remainder, since $\|\phi_{11}\|_\infty \leq C$, we have

$$|T_{quad}| \leq C \int_{\mathcal{M}} (\hat{h} - h_0)^2 |w| d\mathcal{H}^m.$$

Decomposing $\hat{h} - h_0 = (\hat{h} - \tilde{h}) + (\tilde{h} - h_0)$ where $\tilde{h} := P_{K_n} h_0$ is the empirical LS projection onto the sieve space $\mathbb{H}_{K_n}$. Following the proof of Lemma 5(1) below, we obtain:

- *Bias term:* $\int_{\mathcal{M}} (\tilde{h} - h_0)^2 w d\mathcal{H}^m \leq \|\tilde{h} - h_0\|_\infty^2 \cdot \mathcal{H}^m(\mathcal{M}) \|w\|_\infty = O_p(K_n^{-2s/d})$.
- *Variance term:* $\mathbb{E}[\int_{\mathcal{M}} (\hat{h} - \tilde{h})^2 w d\mathcal{H}^m] = O(K_n/n)$.

Thus, $\int_{\mathcal{M}} (\hat{h} - h_0)^2 w d\mathcal{H}^m = O_p(K_n^{-2s/d} + K_n/n)$.

For the quadratic remainder to be asymptotically no larger in magnitude than the linear term, we need $T_{quad} = O_p(\sqrt{K_n^{(d-m)/d}/n})$, i.e.,

$$K_n^{-2s/d} + K_n/n = O\left(\sqrt{K_n^{(d-m)/d}/n}\right).$$

The bias condition $K_n^{-2s/d} = O(\sqrt{K_n^{(d-m)/d}/n})$ at $K_n \asymp n^{d/(2s+d-m)}$ is equivalent to $K_n^{-2s/d} = o(K_n^{-s/d})$, which always holds. The variance condition $K_n/n = O(\sqrt{K_n^{(d-m)/d}/n})$ simplifies as follows:

$$\begin{aligned} \frac{K_n}{n} &= O\left(\sqrt{\frac{K_n^{(d-m)/d}}{n}}\right) \Leftrightarrow \frac{K_n^2}{n^2} = O\left(\frac{K_n^{(d-m)/d}}{n}\right) \\ &\Leftrightarrow K_n^{2-(d-m)/d} = O(n) \\ &\Leftrightarrow K_n^{(d+m)/d} = O(n). \end{aligned}$$

At $K_n^* = n^{d/(2s+d-m)}$, we have

$$n^{(d+m)/(2s+d-m)} = O(n) \Leftrightarrow \frac{d+m}{2s+d-m} \leq 1 \Leftrightarrow d+m \leq 2s+d-m \Leftrightarrow s \geq m.$$

Hence, under $s \geq m$, $\hat{\theta} - \theta_0 = \Gamma(\hat{h}) - \Gamma(h_0) = O_p(r_n^*)$. $\square$

In the proofs of Theorems 4(3) and 5(3), as well as of Proposition 1, we use the following rate result for the first stage sieve LS estimator to control the second order remainder term of $\Phi(\hat{h}) - \Phi(h_0)$ for $\Phi = \Gamma, V$.

Lemma 5. *Let Assumptions 1, 3, 4, 11 and 12 hold, and $\hat{h}$ be the sieve LS estimator. Then:*

$$(1) \quad \|\hat{h} - h_0\|_{L_2(\mathcal{M})} = O_p \left(K_n^{-s/d} + \sqrt{\frac{K_n}{n}} \right).$$

$$(2) \quad \|\nabla_x \hat{h} - \nabla_x h_0\|_{L_2(\mathcal{M})} = O_p \left(K_n^{1/d} \times \left[K_n^{-s/d} + \sqrt{\frac{K_n}{n}} \right] \right).$$

Here $\|v\|_{L_2(\mathcal{M})}^2 := \int_{\mathcal{M}} (v(x))^2 w(x) d\mathcal{H}^m(x)$.

Proof of Lemma 5. For Result (1), recall that $\hat{h} - \tilde{h} = (\hat{h} - \tilde{h}) + (\tilde{h} - h_0)$, where $\hat{h}$ is the sieve LS estimator using a sieve space $\mathbb{H}_K$ (with the sieve dimension $K = K_n$), and $\tilde{h} := P_{K_n,n} h_0$ is the empirical LS projection of h_0 onto the sieve space $\mathbb{H}_K$. Note that

$$\int_{\mathcal{M}} (\tilde{h} - h_0)^2 d\mathcal{H}^m \leq \|\tilde{h} - h_0\|_{\infty}^2 \cdot \mathcal{H}^m(\mathcal{M}) = O_p(K_n^{-2s/d}),$$

and

$$\hat{h}(x) - \tilde{h}(x) = b^K(x)' \hat{G}^{-1} \frac{1}{n} \sum_{i=1}^n b^K(X_i) \epsilon_i$$

Write the stochastic part in orthonormalized form:

$$\hat{h}(x) - \tilde{h}(x) = \bar{b}^K(x)' \bar{G}_n^{-1} \frac{1}{n} \sum_{i=1}^n \bar{b}^K(X_i) \epsilon_i, \quad \bar{G}_n := \frac{1}{n} \sum_{i=1}^n \bar{b}^K(X_i) \bar{b}^K(X_i)'$$

We now show that under Assumptions 1, 3, 4, 11 and 12(i)(ii)(iii), we have:

$$\mathbb{E} \left[\int_{\mathcal{M}} (\hat{h}(x) - \tilde{h}(x))^2 |w(x)| d\mathcal{H}^m(x) \right] = O\left(\frac{K}{n}\right).$$

Define the (deterministic) matrix

$$A_K := \int_{\mathcal{M}} \bar{b}^K(x) \bar{b}^K(x)' |w(x)| d\mathcal{H}^m(x),$$

which is positive semidefinite. Expanding the square and using Fubini gives

$$\int_{\mathcal{M}} (\hat{h} - \tilde{h})^2 |w| d\mathcal{H}^m = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \epsilon_i \epsilon_j \bar{b}^K(X_i)' \bar{G}_n^{-1} A_K \bar{G}_n^{-1} \bar{b}^K(X_j).$$

Conditioning on $X_1^n := (X_1, \dots, X_n)$ and using $\mathbb{E}[\epsilon_i \epsilon_j | X_1^n] = 0$ for $i \neq j$,

$$\mathbb{E} \left[\int_{\mathcal{M}} (\hat{h} - \tilde{h})^2 |w| d\mathcal{H}^m \mid X_1^n \right] = \frac{1}{n^2} \sum_{i=1}^n \mathbb{E}[\epsilon_i^2 | X_i] \bar{b}^K(X_i)' \bar{G}_n^{-1} A_K \bar{G}_n^{-1} \bar{b}^K(X_i).$$

Let $\bar{\sigma}^2 := \sup_x \mathbb{E}[\epsilon_i^2 \mid X_i = x] < \infty$ (Assumption 11) and define

$$S_n := \frac{1}{n} \sum_{i=1}^n \mathbb{E}[\epsilon_i^2 \mid X_i] \bar{b}^K(X_i) \bar{b}^K(X_i)' \preceq \bar{\sigma}^2 \bar{G}_n.$$

Then the conditional expectation can be written as

$$\mathbb{E} \left[\int_{\mathcal{M}} (\hat{h} - \tilde{h})^2 |w| d\mathcal{H}^m \mid X_1^n \right] = \frac{1}{n} \text{tr}(\bar{G}_n^{-1} A_K \bar{G}_n^{-1} S_n) = \frac{1}{n} \text{tr}(A_K \bar{G}_n^{-1} S_n \bar{G}_n^{-1}).$$

Since $A_K \succeq 0$, $\text{tr}(A_K B) \leq \|B\|_{op} \text{tr}(A_K)$ for any $B \succeq 0$, hence

$$\mathbb{E} \left[\int_{\mathcal{M}} (\hat{h} - \tilde{h})^2 |w| d\mathcal{H}^m \mid X_1^n \right] \leq \frac{1}{n} \|\bar{G}_n^{-1} S_n \bar{G}_n^{-1}\|_{op} \text{tr}(A_K).$$

Using $S_n \preceq \bar{\sigma}^2 \bar{G}_n$ gives

$$\|\bar{G}_n^{-1} S_n \bar{G}_n^{-1}\|_{op} \leq \bar{\sigma}^2 \|\bar{G}_n^{-1}\|_{op}^2 \|\bar{G}_n\|_{op} = O_p(1)$$

by the assumed Gram-matrix stability. Finally,

$$\text{tr}(A_K) = \int_{\mathcal{M}} \|\bar{b}^K(x)\|^2 |w(x)| d\mathcal{H}^m(x) \leq \|w\|_{\infty} \mathcal{H}^m(\mathcal{M}) \sup_{x \in \mathcal{M}} \|\bar{b}^K(x)\|^2.$$

Because $\sup_x \|\bar{b}^K(x)\| \lesssim \sup_x \|b^K(x)\| = \zeta_K = O(\sqrt{K})$ (Assumption 12(i)(ii)), we have $\text{tr}(A_K) = O(K)$. Taking expectations yields

$$\mathbb{E} \left[\int_{\mathcal{M}} (\hat{h} - \tilde{h})^2 |w| d\mathcal{H}^m \right] = O\left(\frac{K}{n}\right).$$

For Result (2), we have:

$$\begin{aligned} \nabla_x \hat{h} - \nabla_x h_0 &= \nabla_x \hat{h} - \nabla_x \tilde{h} + (\nabla_x \tilde{h} - \nabla_x h_0) \\ \nabla_x \hat{h}(x) - \nabla_x \tilde{h}(x) &= [\nabla_x b^K(x)]' \hat{G}^{-1} \frac{1}{n} \sum_{i=1}^n b^K(X_i) \epsilon_i \end{aligned}$$

For spline or wavelet bases, we have:

$$\int_{\mathcal{M}} (\nabla_x \tilde{h} - \nabla_x h_0)^2 d\mathcal{H}^m \leq \|\nabla_x \tilde{h} - \nabla_x h_0\|_{\infty}^2 \cdot \mathcal{H}^m(\mathcal{M}) = O_p(K_n^{-2(s-1)/d}).$$

For the stochastic part, the proof follows from Result (1)'s proof except to replace A_K be

$$\bar{A}_K := \int_{\mathcal{M}} \nabla_x \bar{b}^K(x) \nabla_x \bar{b}^K(x)' |w(x)| d\mathcal{H}^m(x), \quad \text{tr}(\bar{A}_K) \lesssim \sup_x \|\nabla_x b^K(x)\|^2 = [K^{1/d} \zeta_K]^2.$$

Hence

$$\mathbb{E} \left[\int_{\mathcal{M}} (\nabla_x \hat{h} - \nabla_x \tilde{h})^2 |w| d\mathcal{H}^m \right] = O\left(K^{2/d} \frac{K}{n}\right).$$

□

C.3 Proof of Theorem 5 in Section 4.3

Proof of Lemma 3. Defining

$$\mathcal{V}_t := \{x : h_0(x) + tv(x) \geq 0\},$$

whose boundary is given by

$$\partial\mathcal{V}_t = \mathcal{M}_t := \{x : h_0(x) + tv(x) = 0\}$$

with

$$\mathcal{V}_0 = \{x : h_0(x) \geq 0\}, \quad \partial\mathcal{V}_0 = \mathcal{M}_0 = \{x : h_0(x) = 0\}.$$

By the generalized Stokes Theorem,

$$\begin{aligned} \frac{d}{dt} \int_{\mathcal{V}_t} w(x) dx \Big|_{t=0} &= \int_{\partial\mathcal{V}_0} w(x) (\dot{\mathbf{X}}(x, 0) \cdot \mathbf{n}(x, 0)) d\mathcal{H}^{d-1}(x) \\ &= \int w(x) (\dot{\mathbf{X}}(x, 0) \cdot \mathbf{n}(x, 0)) d\mathcal{H}^{d-1}(x) \end{aligned}$$

where

$$\dot{\mathbf{X}}(x, 0) := \frac{\partial}{\partial t} \mathbf{X}(x, t) \Big|_{t=0}, \quad \mathbf{n}(x, 0) := -\frac{\nabla h_0(x)}{\|\nabla h_0(x)\|}$$

with $\mathbf{n}(x, t)$ denoting the outward-pointing unit normal at $x \in \mathcal{M}_t$, and $\mathbf{X}(x, t)$ being a diffeomorphism from $\mathcal{M}_0$ to $\mathcal{M}_t$, which by definition satisfies $\mathbf{X}(x, 0) \equiv x$ and

$$h_0(\mathbf{X}(x, t)) + tv(\mathbf{X}(x, t)) = 0, \tag{76}$$

for each $x \in \mathcal{M}_0$ and $t \in [0, \epsilon]$ for some $\epsilon > 0$.

Taking derivatives of (76) with respect to t yields

$$\nabla_x h_0(\mathbf{X}(x, t)) \cdot \dot{\mathbf{X}}(x, t) + t \nabla_x v(\mathbf{X}(x, t)) \cdot \dot{\mathbf{X}}(x, t) + v(\mathbf{X}(x, t)) = 0$$

and thus, evaluating the above at $t = 0$, we have

$$\nabla_x h_0(\mathbf{X}(x, 0)) \cdot \dot{\mathbf{X}}(x, 0) + v(\mathbf{X}(x, 0)) = 0$$

or equivalently,

$$\nabla_x h_0(x) \cdot \dot{\mathbf{X}}(x, 0) = -v(x), \tag{77}$$

Hence,

$$\dot{\mathbf{X}}(x, 0) \cdot \mathbf{n}(x, 0) = -\dot{\mathbf{X}}(x, 0) \cdot \frac{\nabla h_0(x)}{\|\nabla h_0(x)\|} = \frac{v(x)}{\|\nabla_x h_0(x)\|}$$

and thus

$$\left. \frac{d}{dt} \int_{\mathcal{V}_t} w(x) dx \right|_{t=0} = \int_{\mathcal{M}_0} w(x) \frac{v(x)}{\|\nabla_x h_0(x)\|} d\mathcal{H}^{d-1}(x).$$

□

Lemma on the Pathwise Derivative of Level Set Integrals

In the proof of Theorem 5, we apply the following key lemma to control the second order remainder term of $V(\hat{h}) - V(h_0)$. Lemma 6 is established using tools for differential geometry over a moving level set.

Lemma 6. *Given h, v , define $h_t(x) := h(x) + tv(x)$ and*

$$I(t) := \int_{\{h_t(x)=0\}} w(x) d\mathcal{H}^{d-1}(x) \quad (78)$$

Then $I(t)$ is continuously differentiable in t with

$$I'(0) = - \int_{\mathcal{M}_0} \left[\frac{v(x)}{\|\nabla_x h_0(x)\|} \frac{\nabla_x h_0(x)}{\|\nabla_x h_0(x)\|} \cdot \nabla_x w(x) + w(x) \frac{v(x)}{\|\nabla_x h_0(x)\|} H(x) \right] d\mathcal{H}^{d-1}(x) \quad (79)$$

and

$$\|I'(t)\| \leq M \left(\|\nabla w\|_{L_2(\mathcal{M}_0)} \|v\|_{L_2(\mathcal{M}_0)} + \|w\|_{L_2(\mathcal{M}_0)} \|v\|_{L_2(\mathcal{M}_0)} \right) \quad (80)$$

where $\|v\|_{L_2(\mathcal{M}_0)}$ denotes the localized L_2 norm on the submanifold $\mathcal{M}_0$, defined by

$$\|v\|_{L_2(\mathcal{M}_0)} := \sqrt{\int_{\mathcal{M}} v^2(x) d\mathcal{H}^{d-1}(x)}. \quad (81)$$

Proof of Lemma 6. Let $\mathbf{X}(\cdot, t)$ be a diffeomorphism mapping $\mathcal{M}_0 = \{x : h_0(x) = 0\}$ to $\mathcal{M}_t = \{x : h_t(x) = 0\}$. Then

$$I(t) = \int_{\mathcal{M}_t} w(x) d\mathcal{H}^{d-1}(x) = \int_{\mathcal{M}_0} w(\mathbf{X}(x, t)) J_{\mathbf{X}}(x, t) d\mathcal{H}^{d-1}(x), \quad (82)$$

where $J_{\mathbf{X}}(x, t)$ is the *surface Jacobian* (i.e., the $(d-1)$ -dimensional area element ratio) of $\mathbf{X}(\cdot, t)$ on $\mathcal{M}_0$.

Since $J_{\mathbf{X}}$ is the Jacobian of a diffeomorphism between $(d-1)$ -dimensional hyper-surfaces, by Chapter 9 of [Delfour and Zolésio \(2011\)](#), we have

$$\dot{J}_{\mathbf{X}}(x, t) = J_{\mathbf{X}}(x, t) \operatorname{div}_{\mathcal{M}_t}(\dot{\mathbf{X}}(x, t)), \quad (83)$$

where $\operatorname{div}_{\mathcal{M}_t}$ denotes the *surface (tangential) divergence* on $\mathcal{M}_t$, defined for a vector field $\mathbf{V}$ by $\operatorname{div}_{\mathcal{M}_t}(\mathbf{V}) := \operatorname{div}(\mathbf{V}) - \mathbf{n} \cdot (D\mathbf{V})\mathbf{n}$, and $\mathbf{n}$ is the unit normal to $\mathcal{M}_t$.

It is without loss of generality to take $\dot{\mathbf{X}}$ to be normal:

$$\dot{\mathbf{X}}(x, t) = -\bar{u}(x, t) \mathbf{n}(x, t), \quad \bar{u}(x, t) = \frac{v(x)}{\|\nabla_x h_t(x)\|}.$$

which results in a simplification of the surface divergence²⁵ to

$$\operatorname{div}_{\mathcal{M}_t}(F \mathbf{n}) = F H_t, \quad (84)$$

where $H_t := \operatorname{div}(\mathbf{n}) = \operatorname{div}\left(\frac{\nabla_x h_t}{\|\nabla_x h_t\|}\right)$ is the mean curvature of $\mathcal{M}_t$.

Differentiating (82) and using (83):

$$\begin{aligned} I'(t) &= \int_{\mathcal{M}_0} \left[\nabla_x w(\mathbf{X}) \cdot \dot{\mathbf{X}} + w(\mathbf{X}) \operatorname{div}_{\mathcal{M}_t}(\dot{\mathbf{X}}) \right] J_{\mathbf{X}} d\mathcal{H}^{d-1}(x) \\ &= \int_{\mathcal{M}_0} \left[-\frac{v}{\|\nabla_x h_t\|} \frac{\nabla_x h_t}{\|\nabla_x h_t\|} \cdot \nabla_x w(\mathbf{X}) - w(\mathbf{X}) \frac{v}{\|\nabla_x h_t\|} H_t \right] J_{\mathbf{X}} d\mathcal{H}^{d-1}(x), \end{aligned} \quad (85)$$

where in the second line we substituted $\dot{\mathbf{X}} = -\bar{u} \mathbf{n}$ and applied (84) with $F = -\bar{u}$.

Then, at $t = 0$, $\mathbf{X}(x, 0) = x$ and $J_{\mathbf{X}}(x, 0) = 1$:

$$I'(0) = - \int_{\mathcal{M}_0} \left[\frac{v(x)}{\|\nabla_x h_0(x)\|} \frac{\nabla_x h_0(x)}{\|\nabla_x h_0(x)\|} \cdot \nabla_x w(x) + w(x) \frac{v(x)}{\|\nabla_x h_0(x)\|} H(x) \right] d\mathcal{H}^{d-1}(x). \quad (86)$$

Based on (85), by Cauchy–Schwarz with respect to $\int_{\mathcal{M}_0} \cdot d\mathcal{H}^{d-1}$, we have

$$|I'(t)| \leq \|\nabla_x w\|_{L_2(\mathcal{M}_0)} \|\dot{\mathbf{X}}\|_{L_2(\mathcal{M}_0)} + \|w\|_{L_2(\mathcal{M}_0)} \|\bar{u} H_t\|_{L_2(\mathcal{M}_0)}.$$

Since $\|\dot{\mathbf{X}}\|_{L_2(\mathcal{M}_0)} \leq M \|v\|_{L_2(\mathcal{M}_0)}$ and

$$\|\bar{u} H_t\|_{L_2(\mathcal{M}_0)} = \left\| \frac{v}{\|\nabla_x h_t\|} H_t \right\|_{L_2(\mathcal{M}_0)} \leq \frac{\|H_t\|_{\infty}}{\inf_{x \in \mathcal{M}_0} \|\nabla_x h_t(x)\|} \|v\|_{L_2(\mathcal{M}_0)} \leq M \|v\|_{L_2(\mathcal{M}_0)},$$

where the last inequality uses the bounded curvature assumption and $\|\nabla_x h_t\| \geq c > 0$ on $\mathcal{M}_t$ (Assumption 14), we obtain:

$$|I'(t)| \leq M \left(\|\nabla_x w\|_{L_2(\mathcal{M}_0)} \|v\|_{L_2(\mathcal{M}_0)} + \|w\|_{L_2(\mathcal{M}_0)} \|v\|_{L_2(\mathcal{M}_0)} \right). \quad (87)$$

□

Proof of Theorem 5(1). We show that under $s \geq m/2 + 1 = (d+1)/2$, the split-sample debiased sieve estimator $\hat{\theta}_{SS}$ achieves the optimal rate $r_n^* = n^{-s/(2s+1)}$.

²⁵Indeed, $\operatorname{div}(F \mathbf{n}) = F \operatorname{div}(\mathbf{n}) + \mathbf{n} \cdot \nabla F$ and $\mathbf{n} \cdot (D(F \mathbf{n})) \mathbf{n} = \mathbf{n} \cdot ((\nabla F) \mathbf{n}' + F D \mathbf{n} \mathbf{n}) = \mathbf{n} \cdot \nabla F$, since $D \mathbf{n} \mathbf{n} = 0$ for a unit normal; subtracting gives $F \operatorname{div}(\mathbf{n}) = F H_t$.

Let $\delta_1 := \hat{h}_1 - h_0$, $\delta_2 := \hat{h}_2 - h_0$, and $\bar{h} = (\hat{h}_1 + \hat{h}_2)/2 = h_0 + (\delta_1 + \delta_2)/2$. Expanding $V(\bar{h})$ around $V(h_0)$, we have

$$V(\bar{h}) - V(h_0) = DV(h_0)[\bar{h} - h_0] + \frac{1}{2}D^2V(h_0)[\bar{h} - h_0, \bar{h} - h_0] + R_3, \quad (88)$$

where R_3 is the third-order remainder term.

Applying the shape calculus transport formula to differentiate $DV(h_0 + tv_2)[v_1] = \int_{\mathcal{M}_{h_0+tv_2}} \frac{w v_1}{\|\nabla_x(h_0+tv_2)\|} d\mathcal{H}^{d-1}$ with respect to t at $t = 0$, we obtain the second functional derivative $D^2V(h)[v_1, v_2]$ as

$$\begin{aligned} D^2V(h)[v_1, v_2] = & - \int_{\mathcal{M}_h} \frac{v_1(x) v_2(x)}{\|\nabla_x h(x)\|^2} \left[\frac{\nabla_x h}{\|\nabla_x h\|} \cdot \nabla_x \left(\frac{w}{\|\nabla_x h\|} \right) + \frac{w}{\|\nabla_x h\|} H(x) \right] d\mathcal{H}^{d-1}(x) \\ & - \int_{\mathcal{M}_h} \frac{w(x)}{\|\nabla_x h(x)\|^2} \left[\frac{\nabla_x h}{\|\nabla_x h\|} \cdot \nabla_x v_1 \cdot v_2 + \frac{\nabla_x h}{\|\nabla_x h\|} \cdot \nabla_x v_2 \cdot v_1 \right] d\mathcal{H}^{d-1}(x). \end{aligned} \quad (89)$$

The first integral involves only pointwise values of v_1, v_2 (with a bounded integrand under Assumption 14), while the second integral involves normal derivatives $\mathbf{n} \cdot \nabla_x v_1$ and $\mathbf{n} \cdot \nabla_x v_2$. Since $D^2V(h)[v_1, v_2]$ is bilinear, we have

$$\frac{1}{4}D^2V(h_0)[\hat{h}_1 - \hat{h}_2, \hat{h}_1 - \hat{h}_2] = \frac{1}{4} \left(D^2V(h_0)[\delta_1, \delta_1] - 2D^2V(h_0)[\delta_1, \delta_2] + D^2V(h_0)[\delta_2, \delta_2] \right).$$

Taking difference of the above removes the diagonal term:

$$D^2V(h_0)[\bar{h} - h_0, \bar{h} - h_0] - \frac{1}{4}D^2V(h_0)[\hat{h}_1 - \hat{h}_2, \hat{h}_1 - \hat{h}_2] = D^2V(h_0)[\delta_1, \delta_2].$$

Under Assumption 14, the coefficient functions in (89) are bounded, and therefore by Cauchy–Schwarz:

$$\left| D^2V(\bar{h}_0)[v_1, v_2] \right| \leq C \|v_1\|_{L^2(\mathcal{M}_0)} \left(\|v_2\|_{L^2(\mathcal{M}_0)} + \|\nabla_x v_2\|_{L^2(\mathcal{M}_0)} \right) + C \|v_2\|_{L^2(\mathcal{M}_0)} \|\nabla_x v_1\|_{L^2(\mathcal{M}_0)}. \quad (90)$$

Given the above, we have

$$\hat{\theta}_{SS} - \theta_0 = \underbrace{DV(h_0)[\bar{h} - h_0]}_{T_{lin}} + \underbrace{\frac{1}{2}D^2V(h_0)[\delta_1, \delta_2]}_{T_{cross}} + R_{rem}. \quad (91)$$

Again, the linear term is a $m = (d - 1)$ -dimensional submanifold integral (over $\mathcal{M}_0$) with weight $w/\|\nabla_x h_0\|$. By Theorem 3 with $m = (d - 1)$ we obtain:

$$T_{lin} = O_p \left(\sqrt{\frac{K_n^{1/d}}{n}} + K_n^{-s/d} \right).$$

By setting $K_n^* \asymp n^{d/(2s+1)}$ (noting $d - m = 1$), we get $T_{lin} = O_p(n^{-s/(2s+1)}) = O_p(r_n^*)$.

The cross-term $T_{cross} = \frac{1}{2}D^2V(h_0)[\delta_1, \delta_2]$ involves integrals of the forms:

$$(i) \int_{\mathcal{M}_0} \frac{\delta_1(x)\delta_2(x)}{\|\nabla_x h_0\|^2} \tilde{w}(x) d\mathcal{H}^{d-1}(x), \quad (ii) \int_{\mathcal{M}_0} \frac{(\nabla_x \delta_1 \cdot \mathbf{n})\delta_2 + (\nabla_x \delta_2 \cdot \mathbf{n})\delta_1}{\|\nabla_x h_0\|^2} w d\mathcal{H}^{d-1}(x).$$

By the same proof of Lemma 8(b) Term $T_3 = O_p\left(n^{-1}K_n^{(2d-m)/(2d)}\right)$, we obtain, term (i) is of the order $O_p\left(n^{-1}K_n^{(d+1)/(2d)}\right)$ with $m = d - 1$. Notice that term (i) goes to zero strictly faster than term (ii), we now focus on bounding term (ii). Applying Lemma 7(1) below, we have, with $K_n^* \asymp n^{d/(2s+1)}$:

$$\text{Term (ii)} = O_p(n^{-s/(2s+1)}) = O_p(r_n^*) \quad \text{when } s \geq (d+1)/2.$$

Consequently, when $s \geq (d+1)/2$, at $K_n^* \asymp n^{d/(2s+1)}$, the quadratic cross term satisfies

$$T_{cross} = O_p(r_n^*).$$

Finally, since $\|\hat{h} - h_0\|_\infty = o_p(1)$, the third-order remainder term $R_3 = o_p(T_{cross})$.

In summary, under $s \geq (d+1)/2$, at $K_n^* = n^{d/(2s+1)}$, we obtain: $\hat{\theta}_{SS} - \theta_0 = O_p(r_n^*)$. $\square$

Lemma 7. Let $\hat{h}_1, \hat{h}_2$ be split-sample sieve LS estimators of h_0 computed on two independent subsamples I_1, I_2 of sizes n_1, n_2 with $n_1 \asymp n_2 \asymp n$, using sieve dimension $K \equiv K_n$. Let $\tilde{h}_r := P_{K,r}h_0$ denote the (subsample- r) empirical LS projection of h_0 onto $\mathbb{H}_K$. Assume Assumptions 14, 3, 4, 11, and 12 hold, and assume in addition that $h_0 \in \Lambda^s(\mathcal{X})$ with $s > 1$ so that $\|\nabla_x(\tilde{h}_r - h_0)\|_\infty = O_p(K^{-(s-1)/d})$ for $r = 1, 2$ (e.g. for spline/wavelet tensor-product sieves). Define

$$T_{\text{mix-cross}} := \int_{\mathcal{M}_0} (\hat{h}_1 - h_0) \nabla_x(\hat{h}_2 - h_0) w' d\mathcal{H}^{d-1},$$

where w' is a bounded (row-)vector-valued weight on $\mathcal{M}_0$ and the product $\nabla_x(\hat{h}_2 - h_0) w'$ is interpreted as the usual inner product. Then

$$T_{\text{mix-cross}} = O_p\left(K_n^{1/d} \times \left[K_n^{-2s/d} + K_n^{-s/d} \sqrt{\frac{K_n^{1/d}}{n}} + \frac{1}{n} K_n^{\frac{d+1}{2d}}\right]\right).$$

Hence, with $K_n^* \asymp n^{d/(2s+1)}$, we obtain:

$$(1) \quad T_{\text{mix-cross}} = O_p(n^{-s/(2s+1)}) = O_p(r_n^*) \quad \text{when } s \geq (d+1)/2.$$

(2) $T_{\text{mix-cross}} = o_p(n^{-s/(2s+1)}) = o_p(r_n^*)$ when $s > (d+1)/2$.

Proof of Lemma 7. Write $K := K_n$ for brevity. Decompose each split-sample error as

$$\hat{h}_r - h_0 = (\hat{h}_r - \tilde{h}_r) + (\tilde{h}_r - h_0), \quad r \in \{1, 2\}.$$

Expanding $T_{\text{mix-cross}}$ yields

$$\begin{aligned} T_{\text{mix-cross}} &= \int_{\mathcal{M}_0} (\hat{h}_1 - \tilde{h}_1) \nabla_x (\hat{h}_2 - \tilde{h}_2) w' d\mathcal{H}^{d-1} \\ &\quad + \int_{\mathcal{M}_0} (\tilde{h}_1 - h_0) \nabla_x (\hat{h}_2 - \tilde{h}_2) w' d\mathcal{H}^{d-1} \\ &\quad + \int_{\mathcal{M}_0} (\hat{h}_1 - \tilde{h}_1) \nabla_x (\tilde{h}_2 - h_0) w' d\mathcal{H}^{d-1} \\ &\quad + \int_{\mathcal{M}_0} (\tilde{h}_1 - h_0) \nabla_x (\tilde{h}_2 - h_0) w' d\mathcal{H}^{d-1} \\ &=: T_{\text{stoch}} + T_1 + T_2 + T_{\text{bias}}. \end{aligned} \tag{92}$$

By Assumption 12(iv)(v), $\|\tilde{h}_1 - h_0\|_\infty = O_p(K^{-s/d})$ and, by the stated $s > 1$ smoothness/approximation, $\|\nabla_x(\tilde{h}_2 - h_0)\|_\infty = O_p(K^{-(s-1)/d})$. Since $\|w'\|_\infty < \infty$ and $\mathcal{H}^{d-1}(\mathcal{M}_0) < \infty$ under Assumption 14(i),

$$|T_{\text{bias}}| \leq \|w'\|_\infty \mathcal{H}^{d-1}(\mathcal{M}_0) \|\tilde{h}_1 - h_0\|_\infty \|\nabla_x(\tilde{h}_2 - h_0)\|_\infty = O_p(K^{-\frac{2s-1}{d}}).$$

Let $A_{1,n} := \|\tilde{h}_1 - h_0\|_\infty$ and, on $\{A_{1,n} > 0\}$, define the (random but bounded) weight

$$w'_{1,n}(x) := w'(x) \frac{\tilde{h}_1(x) - h_0(x)}{A_{1,n}}, \quad x \in \mathcal{M}_0,$$

and set $w'_{1,n} \equiv 0$ on $\{A_{1,n} = 0\}$. Then $\|w'_{1,n}\|_\infty \leq \|w'\|_\infty$ and $w'_{1,n}$ is measurable w.r.t. the first split and hence independent of $\hat{h}_2$. With this normalization,

$$T_1 = A_{1,n} \int_{\mathcal{M}_0} \nabla_x (\hat{h}_2 - \tilde{h}_2)(x) w'_{1,n}(x) d\mathcal{H}^{d-1}(x) \equiv A_{1,n} L_{w'_{1,n}}^\nabla(\hat{h}_2 - \tilde{h}_2), \tag{93}$$

where $L_{w'}^\nabla(f) := \int_{\mathcal{M}_0} \nabla_x f(x) w'(x) d\mathcal{H}^{d-1}(x)$ is a linear functional on $\mathbb{H}_K$.

Exactly as in the proof of Lemma 8(b), Assumption 12(iv)(v) implies

$$A_{1,n} = \|\tilde{h}_1 - h_0\|_\infty = O_p(K^{-s/d}). \tag{94}$$

To bound $L_{w'_{1,n}}^\nabla(\hat{h}_2 - \tilde{h}_2)$. Condition on the first split, so $w'_{1,n}$ is fixed and bounded. For a split-sample sieve LS estimator, the centered part admits the orthonormalized

linear representation

$$\hat{h}_2(x) - \tilde{h}_2(x) = \bar{b}^K(x)' \bar{G}_2^{-1} \frac{1}{n_2} \sum_{i \in I_2} \bar{b}^K(X_i) \epsilon_i, \quad \bar{G}_2 := \frac{1}{n_2} \sum_{i \in I_2} \bar{b}^K(X_i) \bar{b}^K(X_i)',$$

hence

$$L_{w'_{1,n}}^\nabla (\hat{h}_2 - \tilde{h}_2) = \left(L_{w'_{1,n}}^\nabla (\bar{b}^K) \right)' \bar{G}_2^{-1} \eta_2, \quad \eta_2 := \frac{1}{n_2} \sum_{i \in I_2} \bar{b}^K(X_i) \epsilon_i \in \mathbb{R}^K,$$

where $L_{w'_{1,n}}^\nabla (\bar{b}^K) := \left(L_{w'_{1,n}}^\nabla (\bar{b}_1^K), \dots, L_{w'_{1,n}}^\nabla (\bar{b}_K^K) \right)'$. Since $\mathbb{E}[\epsilon_i | X_i] = 0$ and the second split is i.i.d., this quantity is centered conditional on the second-split design.

We now bound its conditional variance. Let $\bar{\sigma}^2 := \sup_x \mathbb{E}[\epsilon_i^2 | X_i = x] < \infty$ (Assumption 11). As in the proof of Lemma 8(b),

$$\mathbb{E}[\eta_2 \eta_2' | X_1^n] = \frac{1}{n_2^2} \sum_{i \in I_2} \mathbb{E}[\epsilon_i^2 | X_i] \bar{b}^K(X_i) \bar{b}^K(X_i)' \preceq \frac{\bar{\sigma}^2}{n_2} \bar{G}_2.$$

Therefore, conditional on the first split,

$$\begin{aligned} \mathbb{E} \left[\left(L_{w'_{1,n}}^\nabla (\hat{h}_2 - \tilde{h}_2) \right)^2 \mid X_1^n \right] &\leq \frac{\bar{\sigma}^2}{n_2} \left(L_{w'_{1,n}}^\nabla (\bar{b}^K) \right)' \bar{G}_2^{-1} L_{w'_{1,n}}^\nabla (\bar{b}^K) \\ &\leq \frac{\bar{\sigma}^2}{n_2} \|\bar{G}_2^{-1}\|_{op} \|L_{w'_{1,n}}^\nabla (\bar{b}^K)\|^2. \end{aligned} \quad (95)$$

Under Assumption 12(i)(iii), $\|\bar{G}_2^{-1}\|_{op} = O_p(1)$.

Claim: uniformly over bounded weights ω' on $\mathcal{M}_0$,

$$\|L_{\omega'}^\nabla (\bar{b}^K)\|^2 = \sum_{k=1}^K \left(\int_{\mathcal{M}_0} \nabla_x \bar{b}_k^K(x) \omega'(x) d\mathcal{H}^{d-1}(x) \right)^2 \lesssim \|\omega'\|_\infty^2 K^{3/d}. \quad (96)$$

We now check the claim. Write $K = J^d$ as in Lemma 1 and write $\omega'(x) = (\omega_1(x), \dots, \omega_d(x))$. Using $(\sum_{\ell=1}^d a_\ell)^2 \leq d \sum_{\ell=1}^d a_\ell^2$, it suffices to bound, for each fixed $\ell \in \{1, \dots, d\}$,

$$S_\ell := \sum_{k=1}^K \left(\int_{\mathcal{M}_0} \partial_{x_\ell} \bar{b}_k^K(x) \omega_\ell(x) d\mathcal{H}^{d-1}(x) \right)^2.$$

We show $S_\ell \lesssim \|\omega'\|_\infty^2 K^{3/d}$ uniformly in ℓ .

Apply the chart decomposition (41) for $\mathcal{M}_0$ (dimension $m = d - 1$) to write the integral as a finite sum over charts. By Cauchy–Schwarz over finitely many charts, it suffices to bound the contribution of a fixed chart uniformly. Fix a chart $(\mathcal{U}_j, \varphi_j)$ and suppress the chart index. Let $\psi(\cdot)$ denote the implicit-function map for the remaining

coordinate and define the bounded chart weight

$$\tilde{w}(u) := \rho_j(\varphi_j(u)) \omega_\ell(\varphi_j(u)) \text{Jac}_{\varphi_j}(u),$$

absorbing bounded factors into constants.

Each basis index $k \leftrightarrow (k_1, \dots, k_d)$ corresponds to the tensor-product basis function $\bar{b}_k^K(x) = \prod_{r=1}^d b_{k_r}(x_r)$. Differentiating in x_ℓ replaces the ℓ th univariate factor by its derivative, so

$$\partial_{x_\ell} \bar{b}_k^K(\varphi_j(u)) = \bar{b}_{k,(\ell)}^{K,(\ell)}(u) \cdot \bar{b}_{k,-(m)}^K(\psi(u)),$$

where $\bar{b}_{k,(\ell)}^{K,(\ell)}(\cdot)$ is an m -variate tensor-product basis on the free coordinates (with one univariate factor differentiated if $\ell \in (m)$), and $\bar{b}_{k,-(m)}^K(\cdot)$ is the $(d-m) = 1$ -variate factor on the remaining coordinate (differentiated instead if $\ell \notin (m)$). For spline/wavelet tensor-product bases at resolution J , the collection $\{J^{-1} \bar{b}_{k,(\ell)}^{K,(\ell)}\}_{k_{(m)}}$ restricted to $\mathcal{U}_j$ is again a frame with constants independent of J (this is the same frame argument used in Lemma 1, but with one derivative and hence the scaling J). Therefore, for each fixed $k_{-(m)}$, the frame inequality yields

$$\sum_{k_{(m)}} \left(\int_{\mathcal{U}_j} \bar{b}_{k,(\ell)}^{K,(\ell)}(u) \bar{b}_{k,-(m)}^K(\psi(u)) \tilde{w}(u) du \right)^2 \lesssim J^2 \int_{\mathcal{U}_j} \left(\bar{b}_{k,-(m)}^K(\psi(u)) \right)^2 \tilde{w}(u)^2 du.$$

Summing over $k_{-(m)}$ and using the tensor-product identity $\sum_{k_{-(m)}} (\bar{b}_{k,-(m)}^K(\psi(u)))^2 \lesssim J^{d-m} = J$ pointwise in u (as in the proof of Lemma 1) gives

$$S_\ell \lesssim J^2 \cdot J \int_{\mathcal{U}_j} \tilde{w}(u)^2 du \lesssim \|\omega'\|_\infty^2 J^3 = \|\omega'\|_\infty^2 K^{3/d}.$$

This proves (96).

Combining (95) and (96) gives

$$L_{w'_{1,n}}^\nabla(\hat{h}_2 - \tilde{h}_2) = O_p\left(\sqrt{\frac{K^{3/d}}{n}}\right) = O_p\left(K^{1/d} \sqrt{\frac{K^{1/d}}{n}}\right).$$

Together with (93)–(94),

$$T_1 = O_p\left(K^{-s/d} \cdot K^{1/d} \sqrt{\frac{K^{1/d}}{n}}\right) = O_p\left(K^{-(s-1)/d} \sqrt{\frac{K^{1/d}}{n}}\right). \quad (97)$$

By symmetry, swapping the roles of the two splits yields the same rate. For completeness, let $B_{2,n} := \|\nabla_x(\tilde{h}_2 - h_0)\|_\infty = O_p(K^{-(s-1)/d})$ and define

$$w_{2,n}(x) := \frac{\nabla_x(\tilde{h}_2 - h_0)(x) w'(x)}{B_{2,n}}, \quad x \in \mathcal{M}_0,$$

with $w_{2,n} \equiv 0$ on $\{B_{2,n} = 0\}$. Then $\|w_{2,n}\|_\infty \leq \|w'\|_\infty$ and $w_{2,n}$ is measurable w.r.t. the second split, hence independent of $\hat{h}_1$. We can write

$$T_2 = B_{2,n} \int_{\mathcal{M}_0} (\hat{h}_1 - \tilde{h}_1)(x) w_{2,n}(x) d\mathcal{H}^{d-1}(x) \equiv B_{2,n} L_{w_{2,n}}(\hat{h}_1 - \tilde{h}_1),$$

where $L_w(f) := \int_{\mathcal{M}_0} f(x) w(x) d\mathcal{H}^{d-1}(x)$ is the usual (non-derivative) linear submanifold functional. By Lemma 1 with $m = d - 1$ and the same centered-linear-functional argument as in Lemma 8(b),

$$L_{w_{2,n}}(\hat{h}_1 - \tilde{h}_1) = O_p\left(\sqrt{\frac{K^{1/d}}{n}}\right),$$

and hence

$$T_2 = O_p\left(K^{-(s-1)/d} \sqrt{\frac{K^{1/d}}{n}}\right). \quad (98)$$

Using the orthonormalized linear representations on each split, define

$$\eta_r := \frac{1}{n_r} \sum_{i \in I_r} \bar{b}^K(X_i) \epsilon_i, \quad \bar{G}_r := \frac{1}{n_r} \sum_{i \in I_r} \bar{b}^K(X_i) \bar{b}^K(X_i)', \quad r \in \{1, 2\}.$$

Define the deterministic $K \times K$ matrix

$$C_K := \int_{\mathcal{M}_0} \bar{b}^K(x) \left(\nabla_x \bar{b}^K(x) w' \right)' d\mathcal{H}^{d-1}(x),$$

(where $\nabla_x \bar{b}^K(x)$ is the $K \times d$ Jacobian and $\nabla_x \bar{b}^K(x) w'$ is the resulting K -vector). Then, up to sign (replace w' by $|w'|$ if desired),

$$T_{\text{stoch}} = \eta_1' \bar{G}_1^{-1} C_K \bar{G}_2^{-1} \eta_2. \quad (99)$$

Conditioning on the full design X_1^n and using split-sample independence, $\mathbb{E}[T_{\text{stoch}} | X_1^n] = 0$. Moreover, as before,

$$\mathbb{E}[\eta_r \eta_r' | X_1^n] \preceq \frac{\bar{\sigma}^2}{n_r} \bar{G}_r, \quad r = 1, 2.$$

Therefore, by the same trace argument as in Lemma 8(b) (Step 4),

$$\begin{aligned} \mathbb{E}[T_{\text{stoch}}^2 | X_1^n] &\leq \frac{\bar{\sigma}^4}{n_1 n_2} \text{tr}(\bar{G}_1^{-1} C_K \bar{G}_2^{-1} C_K') \\ &\leq \frac{\bar{\sigma}^4}{n_1 n_2} \|\bar{G}_1^{-1}\|_{op} \|\bar{G}_2^{-1}\|_{op} \text{tr}(C_K C_K'). \end{aligned} \quad (100)$$

Under Assumption 12(i)(iii), $\|\bar{G}_r^{-1}\|_{op} = O_p(1)$. It remains to control $\text{tr}(C_K C_K') = \|C_K\|_F^2$.

Now, decompose $\mathcal{M}_0$ into finitely many charts as in (41) and write $C_K = \sum_{j=1}^{\bar{j}} C_{K,j}$ with

$$C_{K,j} := \int_{\mathcal{U}_j} \bar{b}^K(\varphi_j(u)) \left(\nabla_x \bar{b}^K(\varphi_j(u)) w' \right)' \tilde{w}_j(u) du,$$

where $\tilde{w}_j(u)$ is bounded (it collects the partition of unity and the Jacobian). Since $\|C_K\|_F^2 \leq \bar{j} \sum_{j=1}^{\bar{j}} \|C_{K,j}\|_F^2$, it suffices to show $\|C_{K,j}\|_F^2 \lesssim K^{(d+3)/d}$ uniformly in j .

Fix j and suppress the chart index. Write $K = J^d$ and note $m = d - 1$ (so $d - m = 1$). For each k, k' , the (k, k') entry equals

$$(C_{K,j})_{kk'} = \int_{\mathcal{U}_j} \bar{b}_k^K(\varphi_j(u)) \left(\nabla_x \bar{b}_{k'}^K(\varphi_j(u)) w' \right)' \tilde{w}_j(u) du.$$

Using the tensor-product decomposition as in Lemma 1, $\bar{b}_k^K(\varphi_j(u)) = \bar{b}_{k,(m)}^K(u) \cdot \bar{b}_{k,-(m)}^K(\psi(u))$. Moreover, $\nabla_x \bar{b}_{k'}^K(\varphi_j(u)) w'$ is a finite sum (over $\ell = 1, \dots, d$) of tensor-product terms with one univariate factor differentiated. As in the proof of (96), squaring and summing over the m -indices yields an additional factor J^2 from this one derivative. Hence, for fixed k and fixed $k'_{-(m)}$, the same frame-inequality argument as in the proof of Lemma 8(b) gives

$$\sum_{k'_{(m)}} (C_{K,j})_{kk'}^2 \lesssim J^2 \int_{\mathcal{U}_j} \left(\bar{b}_k^K(\varphi_j(u)) \right)^2 \left(\bar{b}_{k',-(m)}^K(\psi(u)) \right)^2 \tilde{w}_j(u)^2 du.$$

Summing over $k'_{-(m)}$ and using $\sum_{k'_{-(m)}} (\bar{b}_{k',-(m)}^K(\psi(u)))^2 \lesssim J^{d-m} = J$ pointwise in u yields

$$\sum_{k'=1}^K (C_{K,j})_{kk'}^2 \lesssim J^3 \int_{\mathcal{U}_j} \left(\bar{b}_k^K(\varphi_j(u)) \right)^2 \tilde{w}_j(u)^2 du.$$

Finally, summing over k and using $\sum_{k=1}^K (\bar{b}_k^K(x))^2 = \|\bar{b}^K(x)\|^2 \lesssim K$ pointwise (Assumption 12(ii) and $\lambda_{\min}(G) > 0$) yields

$$\|C_{K,j}\|_F^2 = \sum_{k=1}^K \sum_{k'=1}^K (C_{K,j})_{kk'}^2 \lesssim J^3 \int_{\mathcal{U}_j} \left(\sum_{k=1}^K (\bar{b}_k^K(\varphi_j(u)))^2 \right) \tilde{w}_j(u)^2 du \lesssim J^3 \cdot K = J^{d+3} = K^{(d+3)/d}.$$

This proves the claim.

Substituting into (100) and using $n_1 \asymp n_2 \asymp n$ gives

$$\mathbb{E}[T_{\text{stoch}}^2] = O\left(\frac{K^{(d+3)/d}}{n^2}\right), \quad \text{hence} \quad T_{\text{stoch}} = O_p\left(\frac{1}{n} \sqrt{K^{\frac{d+3}{d}}}\right) = O_p\left(\frac{1}{n} K^{\frac{d+3}{2d}}\right).$$

Finally, combining (92), the bounds for T_{bias} , (97), (98), and the bound for T_{stoch}

yields

$$T_{\text{mix-cross}} = O_p \left(K^{-\frac{2s-1}{d}} + K^{-\frac{s-1}{d}} \sqrt{\frac{K^{1/d}}{n}} + \frac{1}{n} \sqrt{K^{\frac{d+3}{d}}} \right),$$

as claimed. $\square$

Proof of Theorem 5(2). The proof is very similar to the proofs of Theorem 4(2) and Theorem 5(1), hence we will be brief here. Let $K \equiv K_n \asymp K_n^* \asymp n^{\frac{d}{2s+1}}$ and $r_n^* := n^{-\frac{s}{2s+1}}$. Recall $\tilde{h} := P_{K,n} h_0$ and the series LS representation (23):

$$\hat{h}(x) = b^K(x)' \hat{G}^{-1} \left(\frac{1}{n} \sum_{i=1}^n b^K(X_i) Y_i \right) = \frac{1}{n} \sum_{i=1}^n \hat{s}_i(x) Y_i, \quad \hat{s}_i(x) := b^K(x)' \hat{G}^{-1} b^K(X_i), \quad (101)$$

so that

$$\hat{h}(x) - \tilde{h}(x) = \frac{1}{n} \sum_{i=1}^n \hat{s}_i(x) \epsilon_i. \quad (102)$$

Exactly as in the mean-value expansion used in (118), there exists a (random) function $\bar{h}_0$ that lies on the line segment between $\hat{h}$ and h_0 such that

$$V(\hat{h}) - V(h_0) = DV(h_0)[\hat{h} - h_0] + \frac{1}{2} D^2 V(\bar{h}_0)[\hat{h} - h_0, \hat{h} - h_0]. \quad (103)$$

Consequently, with $\hat{\theta}_{\text{LOO}}^V$ defined in (36),

$$\hat{\theta}_{\text{LOO}}^V - V(h_0) = T_{\text{lin}} + T_{\text{quad}} - C_n, \quad (104)$$

where

$$\begin{aligned} T_{\text{lin}} &:= DV(h_0)[\hat{h} - h_0], \\ T_{\text{quad}} &:= \frac{1}{2} D^2 V(\bar{h}_0)[\hat{h} - h_0, \hat{h} - h_0], \\ C_n &:= \frac{1}{2n^2} \sum_{i=1}^n D^2 V(\hat{h})[\hat{s}_i, \hat{s}_i] \cdot (\hat{\epsilon}_i^{(-i)})^2 \quad (\text{the LOO correction term in (36)}). \end{aligned}$$

By (34), T_{lin} is an $m = (d-1)$ -dimensional submanifold integral on $\mathcal{M}_0$ with bounded weight $w/\|\nabla_x h_0\|$. Applying Theorem 3 with $m = d-1$ gives

$$T_{\text{lin}} = O_p \left(\sqrt{\frac{K^{1/d}}{n}} + K^{-s/d} \right) = O_p(r_n^*), \quad (105)$$

at $K \asymp K_n^* \asymp n^{d/(2s+1)}$.

Now, let $u := \hat{h} - \tilde{h}$ and $b := \tilde{h} - h_0$, so that $\hat{h} - h_0 = u + b$. By bilinearity of

$$D^2V(\bar{h}_0)[\cdot, \cdot],$$

$$T_{\text{quad}} = \frac{1}{2}D^2V(\bar{h}_0)[u, u] + D^2V(\bar{h}_0)[u, b] + \frac{1}{2}D^2V(\bar{h}_0)[b, b]. \quad (106)$$

From the explicit representation of D^2V (cf. (89)) and Assumption 14 (bounded w , ∇w , bounded curvature, and $\|\nabla h_0\|$ bounded away from zero on $\mathcal{M}_0$), there exists a finite constant $C > 0$ (uniform in n) such that, on the event $\|\hat{h} - h_0\|_\infty < \epsilon$,

$$\begin{aligned} & \left| D^2V(\bar{h}_0)[v_1, v_2] \right| \\ & \leq C \|v_1\|_{L^2(\mathcal{M}_0)} \left(\|v_2\|_{L^2(\mathcal{M}_0)} + \|\nabla_x v_2\|_{L^2(\mathcal{M}_0)} \right) + C \|v_2\|_{L^2(\mathcal{M}_0)} \|\nabla_x v_1\|_{L^2(\mathcal{M}_0)}. \end{aligned} \quad (107)$$

Now use (102) and Lemma 5 (applied to $\hat{h}$ and to $\tilde{h}$) to obtain

$$\|u\|_{L^2(\mathcal{M}_0)} = O_p\left(\sqrt{\frac{K}{n}}\right), \quad \|\nabla_x u\|_{L^2(\mathcal{M}_0)} = O_p\left(K^{1/d}\sqrt{\frac{K}{n}}\right). \quad (108)$$

Moreover, by the standard approximation property of spline/wavelet tensor-product sieves for $h_0 \in \Lambda^s(\mathcal{X})$ with $s > 1$ (cf. the discussion in Lemma 7),

$$\|b\|_\infty = O_p(K^{-s/d}), \quad \|\nabla_x b\|_\infty = O_p(K^{-(s-1)/d}), \quad (109)$$

and hence $\|b\|_{L^2(\mathcal{M}_0)} = O_p(K^{-s/d})$ and $\|\nabla_x b\|_{L^2(\mathcal{M}_0)} = O_p(K^{-(s-1)/d})$. Plugging (108)–(109) into (107) yields

$$D^2V(\bar{h}_0)[u, b] = O_p\left(\sqrt{\frac{K}{n}} K^{-(s-1)/d} + K^{-s/d} K^{1/d}\sqrt{\frac{K}{n}}\right) = O_p(r_n^*) \quad \text{if } s \geq (d+1)/2, \quad (110)$$

and

$$D^2V(\bar{h}_0)[b, b] = O_p(K^{-(2s-1)/d}) = O_p(r_n^*), \quad (111)$$

at $K \asymp K_n^* \asymp n^{d/(2s+1)}$. Therefore,

$$T_{\text{quad}} = \frac{1}{2}D^2V(\bar{h}_0)[u, u] + O_p(r_n^*). \quad (112)$$

By (102) and bilinearity of $D^2V(\bar{h}_0)[\cdot, \cdot]$,

$$\begin{aligned} D^2V(\bar{h}_0)[u, u] &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \epsilon_i \epsilon_j D^2V(\bar{h}_0)[\hat{s}_i, \hat{s}_j] \\ &= \underbrace{\frac{1}{n^2} \sum_{i=1}^n \epsilon_i^2 D^2V(\bar{h}_0)[\hat{s}_i, \hat{s}_i]}_{=: T_{\text{diag}}} + \underbrace{\frac{1}{n^2} \sum_{i \neq j} \epsilon_i \epsilon_j D^2V(\bar{h}_0)[\hat{s}_i, \hat{s}_j]}_{=: T_{\text{off}}}. \end{aligned} \quad (113)$$

Without debiasing, T_{diag} contains the “diagonal” variance contribution of order

$K^{1+1/d}/n$ (cf. (119)), which is too large when $(d+1)/2 \leq s < d$. The correction term C_n is designed to estimate and subtract T_{diag} .

Similar to the proof of Theorem 4(2), we may rewrite

$$C_n = \frac{1}{2} \sum_{i=1}^n D^2 V(\hat{h}) [\hat{h} - \hat{h}^{(-i)}, \hat{h} - \hat{h}^{(-i)}] = T_{\text{diag}} + R_{\text{diag}}, \quad (114)$$

where R_{diag} collects: (i) cross and squared terms involving the leave-one-out first-stage error $h_0(X_i) - \hat{h}^{(-i)}(X_i)$, and (ii) the (asymptotically negligible) difference between $D^2 V(\hat{h})$ and $D^2 V(\bar{h}_0)$. A standard Cauchy–Schwarz bound combined with the same matrix-trace argument used to control the main stochastic term in Lemma 7 (replace its bounded weight w' by the bounded weights that appear in $D^2 V$; cf. (89)) yields

$$R_{\text{diag}} = O_p\left(\frac{1}{n} \sqrt{K^{\frac{d+3}{d}}}\right). \quad (115)$$

Consequently, combining (113) and (114),

$$\frac{1}{2} D^2 V(\bar{h}_0)[u, u] - C_n = \frac{1}{2} T_{\text{off}} + R_{\text{diag}}. \quad (116)$$

We now consider the off-diagonal term T_{off} . Conditional on the designs, T_{off} is a (conditionally) degenerate U-statistic of order two. As in Lemma 7 (main stochastic term), its conditional variance is bounded by

$$\text{Var}(T_{\text{off}} \mid X_1^n) \lesssim \frac{1}{n^2} \text{tr}(C_K C_K'),$$

where C_K is a $K \times K$ matrix of the form $\int_{\mathcal{M}_0} \bar{b}^K(x) (\nabla_x \bar{b}^K(x) \omega(x))' d\mathcal{H}^{d-1}(x)$ for a bounded vector-valued weight ω (coming from (89)). By the claim proved in Lemma 7, $\text{tr}(C_K C_K') \lesssim K^{(d+3)/d}$. Therefore,

$$T_{\text{off}} = O_p\left(\frac{1}{n} \sqrt{K^{\frac{d+3}{d}}}\right). \quad (117)$$

Finally, combining (104), (105), (112), (116), (115) and (117) yields

$$\hat{\theta}_{\text{LOO}}^V - V(h_0) = T_{\text{lin}} + O_p(r_n^*) + O_p\left(\frac{1}{n} \sqrt{K^{\frac{d+3}{d}}}\right).$$

At $K \asymp K_n^* \asymp n^{d/(2s+1)}$ and $s \geq (d+1)/2$,

$$\frac{1}{n} \sqrt{K^{\frac{d+3}{d}}} = \frac{1}{n} K^{\frac{d+3}{2d}} = n^{-1+\frac{d+3}{2(2s+1)}} = O\left(n^{-\frac{s}{2s+1}}\right) = O(r_n^*),$$

so $\hat{\theta}_{\text{LOO}}^V - V(h_0) = O_p(r_n^*)$. □

Proof of Theorem 5(3). We write

$$\hat{\theta} - \theta_0 = V(\hat{h}) - V(h_0) = DV(h_0)[\hat{h} - h_0] + \frac{1}{2}D^2V(\bar{h}_0)[\hat{h} - h_0, \hat{h} - h_0], \quad (118)$$

where $\bar{h}_0$ lies between $\hat{h}$ and h_0 .

As before, the linear term $DV(h_0)[\hat{h} - h_0] = O_p(r_n^*)$ at optimal K_n . The second-order term D^2V involves:

- Terms of the form $\int_{\mathcal{M}_0} (\hat{h} - h_0)^2 \tilde{w} d\mathcal{H}^{d-1}$;
- Terms involving $\int_{\mathcal{M}_0} (\hat{h} - h_0) \cdot (\nabla(\hat{h} - h_0) \cdot \mathbf{n}) \tilde{w} d\mathcal{H}^{d-1}$.

By Lemma 6 we obtain:

$$|D^2V(\bar{h}_0)[\hat{h} - h_0, \hat{h} - h_0]| \leq C \|\hat{h} - h_0\|_{L^2(\mathcal{M}_0)} \left(\|\nabla_x(\hat{h} - h_0)\|_{L^2(\mathcal{M}_0)} + \|\hat{h} - h_0\|_{L^2(\mathcal{M}_0)} \right).$$

By Lemma 5(1) we have $\|\hat{h} - h_0\|_{L^2(\mathcal{M}_0)}^2 = O_p(K_n^{-2s/d} + K_n/n)$. By Lemma 5(2), we also have:

$$\|\nabla_x(\hat{h} - h_0)\|_{L^2(\mathcal{M}_0)}^2 = O_p \left(K_n^{2/d} \times [K_n^{-2s/d} + K_n/n] \right).$$

Hence

$$|D^2V(\bar{h}_0)[\hat{h} - h_0, \hat{h} - h_0]| = O_p \left(K_n^{1/d} \times [K_n^{-2s/d} + K_n/n] \right). \quad (119)$$

We need $D^2V = O_p(r_n^*)$ at $K_n = K_n^* \asymp n^{d/(2s+1)}$, where the dominant contribution is again from the variance part. The condition we require is then

$$K_n^{1/d} \times K_n/n = O(r_n^*) = O(n^{-s/(2s+1)}) \quad \text{at } K_n = K_n^* \asymp n^{d/(2s+1)},$$

which is equivalent to

$$n^{(d+1)/(2s+1)-1} = O(n^{-s/(2s+1)}) \Leftrightarrow d \leq s.$$

Hence, under $s \geq d = m + 1$, the plug-in sieve estimator $\hat{\theta} = V(\hat{h})$ for $\theta_0 = V(h_0)$ achieves the minimax optimal rate of $\hat{\theta} - \theta_0 = O_p(n^{-s/(2s+1)})$, which is the same optimal rate of r_n^* for the linear $m = (d - 1)$ -submanifold integrals. $\square$

D Proofs of Theoretical Results in Section 5

We first recall the following two high level conditions from [Chen, Liao and Sun \(2014\)](#) that are applicable to any linear and nonlinear irregular functionals for sieve M estimation problems.

Assumption 17. (i) $\hat{h}$, $P_{K,n}h_0 \in \mathbb{H}_{K_n}$ wpa1, and

$$\sup_{h \in \mathbb{H}_{K_n}} \frac{\sqrt{n} |\Phi(h) - \Phi(h_0) - D\Phi(h_0)[h - h_0]|}{\|v_{K_n}^*\|_2} = o_p(1). \quad (120)$$

$$(ii) \frac{\sqrt{n} |D\Phi(h_0)[P_{K,n}h_0 - h_0]|}{\|v_{K_n}^*\|_2} = o_p(1).$$

Assumption 18 (CLT of Sieve Influence Function). $\frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{v_{K_n}^*(X_i)}{\|v_{K_n}^*\|_{sd}} \epsilon_i \xrightarrow{d} \mathcal{N}(0, 1)$.

The following Remark specializes Theorem 3.1 of [Chen, Liao and Sun \(2014\)](#) (or Theorem 3.1 of [Chen and Christensen \(2015\)](#)) to our irregular functionals of a nonparametric regression model with i.i.d. data.

Remark 5. Let $\hat{\theta} = \Phi(\hat{h})$ be the plug-in sieve estimator of $\theta_0 = \Phi(h_0)$. Let Assumptions 5, 11, 17 and 18 hold. Then:

$$\frac{\sqrt{n}(\hat{\theta} - \theta_0)}{\|v_{K_n}^*\|_{sd}} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{v_{K_n}^*(X_i)}{\|v_{K_n}^*\|_{sd}} \epsilon_i + o_p(1) \xrightarrow{d} \mathcal{N}(0, 1).$$

When $\Phi(h) = L(h)$ is linear, Assumption 17(i) is trivially satisfied and its sieve variance grows at the order $\|v_{K_n}^*\|_{sd}^2 \asymp K_n^{\frac{d-m}{d}}$ by Lemma 1. For a general nonlinear irregular functional such as $\Phi = \Gamma, V$, we have the following result controlling the growth rate of the sieve Riesz representer and the sieve variance as well

Proof of Proposition 1. The asymptotic normality result follows from Theorem 3.1 of [Chen and Christensen \(2015\)](#) [CC15 for short] with its stated assumptions and additional conditions verified (for i.i.d. setting). Specifically, Assumptions 1(i) and 2(i) in CC15 are satisfied under the i.i.d. setting with $\mathcal{X} = [0, 1]^d$, Assumption 2(ii)(iv)(v) in CC15 is satisfied by our Assumptions 11, 5, and 15. Their Assumption 4(iii) is implied by our Assumption 12(i). Assumption 5 in CC15 is satisfied by our Assumption 12(i)(ii)(iii). Assumptions 9(i)(ii) in CC15 are trivially satisfied when $\Phi = L$, and are satisfied by our Assumptions 6 and 17(i) when $\Phi = \Gamma, V$. Their Assumption 9(iii) is our Assumption 17(ii), which is in turn satisfied by our Assumption 12(iv)(v) with the “undersmoothing” choice of K_n s.t.

$$|D\Gamma(h_0)[\tilde{h} - h_0]| \leq \|\tilde{h} - h_0\|_\infty = O_p(K_n^{-s/d}) = o_p\left(\sqrt{\frac{K_n^{\frac{d-m}{d}}}{n}}\right),$$

where the first inequality follows from Assumption 6 and the uniform boundedness of $\bar{w}$. Finally, Lemma 1 and Remark 4 imply that growth rate of $\|v_{K_n}^*\|_{sd} \asymp K_n^{\frac{d-m}{d}}$. It remains to check that our Assumption 17(i) is satisfied under Case 2 for $\Phi = \Gamma$ and under Case 3 for $\Phi = V$ below.

(1) $\hat{\theta}_{SS}$ in Case 2 and Case 3:

Under **Case 2** for $\Phi = \Gamma$, from the proof of Theorem 4(1) and the bound on the second order remainder term and Lemma 8(b), with $s > m/2$, $K_n \asymp K_n^*[\log(n)]^\kappa$ for some $\kappa \in (0, 1]$, $K_n^* = n^{d/(2s+d-m)}$ and $r_n^* = n^{-s/(2s+d-m)}$, we have:

$$\begin{aligned} \hat{\theta}_{SS} - \Gamma(h_0) &= \frac{1}{2} \int_{\mathcal{M}} \phi_1(h_0, x) (\hat{h}_1 - h_0) w d\mathcal{H}^m \\ &\quad + \frac{1}{2} \int_{\mathcal{M}} \phi_1(h_0, x) (\hat{h}_2 - h_0) w d\mathcal{H}^m + o_p \left(\sqrt{\frac{K_n^{(d-m)/d}}{n}} \right), \end{aligned}$$

hence Assumption 17(i) is satisfied.

Under **Case 3** for $\Phi = V$ with $m = d - 1$, from the proof of Theorem 5(1) and the bound on the second order remainder term (119) and Lemma 7(2), with $s > (m/2) + 1 = (d + 1)/2$, $K_n \asymp K_n^*[\log(n)]^\kappa$ for some $\kappa \in (0, 1]$, $K_n^* = n^{d/(2s+1)}$ and $r_n^* = n^{-s/(2s+1)}$, we have:

$$\hat{\theta}_{SS} - V(h_0) = \frac{1}{2} DV(h_0) [\hat{h}_1 - h_0] + \frac{1}{2} DV(h_0) [\hat{h}_2 - h_0] + o_p \left(\sqrt{\frac{K_n^{1/d}}{n}} \right),$$

hence Assumption 17(i) is satisfied.

(2) $\hat{\theta}_{LOO}$ in Case 2 and Case 3:

Under **Case 2** for $\Phi = \Gamma$, from the proof of Theorem 4(2) we have:

$$\hat{\theta}_{LOO} - \Gamma(h_0) = \int_{\mathcal{M}} \phi_1(h_0, x) [\hat{h} - h_0] w d\mathcal{H}^m + o_p \left(\sqrt{\frac{K_n^{(d-m)/d}}{n}} \right),$$

hence Assumption 17(i) is satisfied.

Under **Case 3** for $\Phi = V$ with $m = d - 1$, from the proof of Theorem 5(2), with $s > (m/2) + 1 = (d + 1)/2$, $K_n \asymp K_n^*[\log(n)]^\kappa$ for some $\kappa \in (0, 1]$, $K_n^* = n^{d/(2s+1)}$ and

$r_n^* = n^{-s/(2s+1)}$, we have:

$$\hat{\theta}_{LOO} - V(h_0) = DV(h_0)[\hat{h} - h_0] + o_p\left(\sqrt{\frac{K_n^{1/d}}{n}}\right),$$

hence Assumption 17(i) is satisfied.

(3) Plug-in $\hat{\theta} = \Phi(\hat{h})$ in Case 2 and Case 3:

Under **Case 2** for $\Phi = \Gamma$, from the proof of Theorem 4(3) and the bound on the second order remainder term, we see that Assumption 17(i) is satisfied provided that

$$|D^2\Gamma(\bar{h}_0)[\hat{h} - h_0, \hat{h} - h_0]| = O_p\left(K_n^{-2s/d} + K_n/n\right) = o_p\left(\sqrt{\frac{K_n^{(d-m)/d}}{n}}\right),$$

which is satisfied provided that $K_n^{(d+m)/d} = o(n)$ and $K_n^* = n^{d/(2s+d-m)} = o(K_n)$, which is in turn satisfied with $s > m$.

Under **Case 3** for $\Phi = V$ with $m = d - 1$, from the proof of Theorem 5(3) and the bound on the second order remainder term (119), we see that Assumption 17(i) is satisfied provided that

$$|D^2V(\bar{h}_0)[\hat{h} - h_0, \hat{h} - h_0]| = O_p\left(K_n^{1/d} \times [K_n^{-2s/d} + K_n/n]\right) = o_p\left(\sqrt{\frac{K_n^{1/d}}{n}}\right),$$

which is satisfied provided that $K_n^{(2d+1)/d} = o(n)$ and $K_n^* = n^{d/(2s+1)} = o(K_n)$, which is in turn satisfied with $s > m + 1 = d$.

□

The following Assumption 19 corresponds to Assumption 3.1(iii) in Chen, Liao and Sun (2014) or Assumption 10 in Chen and Christensen (2015).

Assumption 19. $\|v_{K_n}^*\|_2^{-1} \|D\Phi(h)[\bar{\psi}^{K_n}] - D\Phi(h_0)[\bar{\psi}^{K_n}]\| = o(1)$ uniformly in $h \in \mathbb{H}_{K_n}$.

Proposition 3 (Consistent Variance). *Let Assumptions 3, 5, 11, 12(i)(ii), 19 and 16 hold with $K_n^{(2+\delta)/\delta} \log K_n/n = o(1)$. Then:*

$$(1) \quad \left| \frac{\hat{\sigma}_{*,K_n}}{\|v_{K_n}^*\|_{sd}} - 1 \right| = o_p(1);$$

(2) Further, let Assumptions 12(iv)(v) and 15 hold with $K_n \asymp K_n^*[\log(n)]^\kappa$ for some $\kappa \in (0, 1)$ where $K_n^* \asymp n^{\frac{d}{2s+d-m}}$. Then:

$$\frac{\sqrt{n}(\Phi(\hat{h}) - \Phi(h_0))}{\hat{\sigma}_{*, K_n}} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{v_{K_n}^*(X_i)}{\|v_{K_n}^*\|_{sd}} \epsilon_i + o_p(1) \xrightarrow{d} \mathcal{N}(0, 1),$$

under the additional mild conditions stated for Cases 1, 2 and 3 of Proposition 1

Proof of Proposition 3. Result (1) follows from Lemmas 3.1 and 3.2 of Chen and Christensen (2015) for i.i.d. setting. Result (2) follows from Proposition 1 and Result (1). $\square$

Proof of Proposition 2. Proposition 2 follows from Proposition 3 by noting that Assumption 19 is automatically satisfied when $\Phi = L$ (linear). It is also satisfied by our conditions imposed to control for second order remainder terms in the convergence rate Theorems 4(b) and 5(b) for nonlinear functionals $\Phi = \Gamma, V$ with $K_n \asymp K_n^*[\log(n)]^\kappa$ for some $\kappa \in (0, 1]$ where $K_n^* \asymp n^{\frac{d}{2s+d-m}}$. In particular, under Assumption 16 (with $\delta > 2d/(2s - m)$), the choice of $K_n \asymp K_n^*[\log(n)]^\kappa$ satisfies $K_n^{(2+\delta)/\delta} \log K_n/n = o(1)$. $\square$

E Multiplier Bootstrap Choice of Sieve Dimension

This section proposes a *multiplier bootstrap* and a *data-driven* choice of sieve dimension K for the various sieve estimators $\tilde{\Phi}(K)$ for $\theta_0^\Phi = \Phi(h_0)$, where $\Phi \in \{L, \Gamma, V\}$. The selection rule is adapted from the bootstrap-Lepski procedure in Chen, Christensen and Kankanala (2025b, Procedure 1), but specialized to our *scalar* functionals. The purpose of the procedure is to: (i) select a *rate-adaptive* “estimation” dimension $\hat{K}_\Phi$, and (ii) convert it into an *undersmoothed* “inference” dimension $\tilde{K}_\Phi$, so that the sieve bias is negligible relative to the standard deviation and the sieve t statistic is asymptotically centered.

Estimated influence function and sieve t statistic

Fix $\Phi \in \{L, \Gamma, V\}$ and a sieve dimension K . Let $\hat{v}_{K,\Phi}^*(x)$ denote the *sieve Riesz representer estimator* of $v_{K,\Phi}^*(x)$ (cf. Proposition 2), i.e.

$$\hat{v}_{K,\Phi}^*(x) := \bar{b}^K(x)' \left(\frac{1}{n} \sum_{i=1}^n \bar{b}^K(X_i) \bar{b}^K(X_i)' \right)^{-1} D\Phi(\hat{h}_K) [\bar{b}^K] = b^K(x)' \hat{G}^{-1} \widehat{D\Phi} [b^K], \quad (121)$$

where $\bar{b}^K := G^{-1/2} b^K$, $\hat{G} := \frac{1}{n} \sum_{i=1}^n b^K(X_i) b^K(X_i)'$ and $\widehat{D\Phi} [b^K] := D\Phi(\hat{h}_K) [b^K]$. Let $\hat{\epsilon}_i := Y_i - \hat{h}_K(X_i)$ and define the estimated sieve influence function as

$$\hat{\psi}_{i,\Phi}(K) := \hat{v}_{K,\Phi}^*(X_i) \hat{\epsilon}_i, \quad \hat{\sigma}_\Phi^2(K) := \frac{1}{n} \sum_{i=1}^n \left(\hat{\psi}_{i,\Phi}(K) \right)^2. \quad (122)$$

The (studentized) sieve t statistic based on the sieve estimator $\tilde{\Phi}(K)$ for $\theta_0^\Phi = \Phi(h_0)$ is

$$T_{n,\Phi}(K) := \frac{\sqrt{n}(\tilde{\Phi}(K) - \theta_0^\Phi)}{\hat{\sigma}_\Phi(K)}. \quad (123)$$

Multiplier bootstrap for a fixed sieve dimension

Let $(\xi_i)_{i=1}^n$ be i.i.d. multipliers, independent of the data, with $\mathbb{E}[\xi_i] = 0$, $\mathbb{E}[\xi_i^2] = 1$, and $\mathbb{E}[|\xi_i|^{2+\delta_\xi}] < \infty$ for some $\delta_\xi > 0$ (e.g. $\xi_i \sim N(0, 1)$ or Rademacher). Define the *multiplier bootstrap* analogue of (123) by

$$T_{n,\Phi}^*(K) := \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i \frac{\hat{\psi}_{i,\Phi}(K)}{\hat{\sigma}_\Phi(K)}. \quad (124)$$

Let $c_{1-\alpha}^*(K)$ denote the conditional $(1 - \alpha)$ quantile of $|T_{n,\Phi}^*(K)|$ given the sample. Then a two-sided $(1 - \alpha)$ MB confidence interval for θ_0^Φ at a *fixed* K is

$$\text{CI}_{n,\Phi}(K) := \left[\tilde{\Phi}(K) \pm c_{1-\alpha}^*(K) \frac{\hat{\sigma}_\Phi(K)}{\sqrt{n}} \right]. \quad (125)$$

Bootstrap-Lepski choice of sieve dimension K

Let $\mathcal{K}_n = \{K_1 < \dots < K_{M_n}\}$ be a dyadic (or geometric) grid of sieve dimensions, with $M_n \lesssim \log n$ as in Chen, Christensen and Kankanala (2025b). Let $K_{\max} := K_{M_n}$ satisfy the usual series regularity

$$K_{\max} \log K_{\max} / n = o(1), \quad (126)$$

so that all sieve estimators and the matrix inverses involved are well-defined wpa1.

For $K, K' \in \mathcal{K}_n$, define the empirical s.d. of the *contrast* $\tilde{\Phi}(K) - \tilde{\Phi}(K')$ by the influence-score analogue of [Chen, Christensen and Kankanala \(2025b, eq. \(9\)\)](#):

$$\hat{\sigma}_\Phi^2(K, K') := \frac{1}{n} \sum_{i=1}^n \left(\hat{\psi}_{i,\Phi}(K) - \hat{\psi}_{i,\Phi}(K') \right)^2. \quad (127)$$

The multiplier bootstrap analogue of the contrast is

$$T_{n,\Phi}^*(K, K') := \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i \frac{\hat{\psi}_{i,\Phi}(K) - \hat{\psi}_{i,\Phi}(K')}{\hat{\sigma}_\Phi(K, K')}. \quad (128)$$

Let $\hat{\vartheta}_{1-\hat{\alpha}_n}^*$ be the conditional $(1 - \hat{\alpha}_n)$ quantile of $\max_{K < K' \in \mathcal{K}_n} |T_{n,\Phi}^*(K, K')|$ and

$$\hat{\alpha}_n := \min \left\{ 0.5, \sqrt{\frac{\log K_{\max}}{K_{\max}}} \right\}, \quad (129)$$

The choice of $\hat{\alpha}_n$ is the scalar analogue of [Chen, Christensen and Kankanala \(2025b, Step 2 of Procedure 1\)](#) and ensures that $\hat{\vartheta}_{1-\hat{\alpha}_n}^*$ diverges slowly.

Then, define the data-driven sieve dimension for estimation

$$\hat{K}_\Phi := \min \left\{ K \in \mathcal{K}_n : \max_{K' > K, K' \in \mathcal{K}_n} \frac{|\tilde{\Phi}(K) - \tilde{\Phi}(K')|}{\hat{\sigma}_\Phi(K, K')} \leq 1.1 \hat{\vartheta}_{1-\hat{\alpha}_n}^* \right\}. \quad (130)$$

This is the scalar version of [Chen, Christensen and Kankanala \(2025b, Step 3 of Procedure 1\)](#). Intuitively, $\hat{K}_\Phi$ is the smallest sieve dimension such that increasing the dimension further does not change the functional estimator beyond its estimated sampling noise.

For inference, to ensure the sieve bias is negligible, we set an “inference” dimension $\tilde{K}_\Phi$ as a mild enlargement of $\hat{K}_\Phi$:

$$\tilde{K}_\Phi := \min \left\{ K \in \mathcal{K}_n : K \geq \hat{K}_\Phi (\log n)^\kappa \right\} \wedge K_{\max}, \quad \text{with } \kappa \in (0, 1) \text{ fixed.} \quad (131)$$

The logarithmic inflation is a standard “undersmoothing” device in series inference: if $\hat{K}_\Phi$ is of the same order as the MSE-optimal dimension, then $\tilde{K}_\Phi$ makes the sieve approximation bias $K^{-s/d}$ smaller than the standard deviation by a factor that vanishes polynomially in $\log n$. Given $\tilde{K}_\Phi$, the final CI is $\text{CI}_{n,\Phi}(\tilde{K}_\Phi)$ as in [\(125\)](#).

The theoretical justification of the inference procedure is similar to CCK and hence is omitted.

Remark 6 (Practical implementation). *In practice, the suprema/maxima in [\(129\)](#) and [\(130\)](#) are taken over the finite grid $\mathcal{K}_n$, and the quantile $\hat{\vartheta}_{1-\hat{\alpha}_n}^*$ is approximated using a large number B of independent MB draws of $(\xi_i)_{i=1}^n$. Following [Chen et al.](#)*

(2025b), a dyadic grid (so $M_n \lesssim \log n$) is recommended for stability. We take $\kappa = 1/2$ in the undersmoothing CI construction in the Monte Carlo studies.

F Analysis of Quadratic Submanifold Integrals

In this section, we provided some detailed analysis of the minimax convergence rate for the estimation of quadratic submanifold integrals, which features a fascinating “elbow phenomenon”.

Lemma 8 (Characterization of Minimax Rates for Quadratic Submanifold Integral Functionals). *Let the linear functional $DQ(h_0)[v] = 2 \int_{\mathcal{M}} h_0(x)v(x)w(x)d\mathcal{H}^m(x)$ satisfies Assumption 4 with weight $2h_0(x)w(x)$ and $\int_{\mathcal{M}} h_0(x)w(x)d\mathcal{H}^m(x) \neq 0$. Then:*

- (a) Under Assumptions 1, 2, 3, 5 and $h_0 \in \Lambda_c^s(\mathcal{X})$, a lower bound for the minimax rate is given by

$$\underline{r}_{Quad,n} := \max \left\{ n^{-\frac{s}{2s+d-m}}, n^{-\frac{4s}{4s+2d-m}} \right\} \quad (132)$$

- (b) Under Assumptions 1, 3, 5, 11 and 12, an upper bound for the minimax rate is given by

$$\bar{r}_{Quad,n} := \max \left\{ n^{-\frac{s}{2s+d-m}}, n^{-\frac{2s}{2s+2d-m}} \right\} \quad (133)$$

- (c) Under Assumptions for part(b) and $s > m/2$, the minimax optimal rate for the estimation of $Q(h_0)$ is given by

$$r_{Quad,n} := n^{-\frac{s}{2s+d-m}} = \underline{r}_{Quad,n} = \bar{r}_{Quad,n}.$$

In addition, this rate can be attained by the split-sample sieve estimator

$$\hat{\theta}_{SS} := \int_{\mathcal{M}} \hat{h}_1(x) \hat{h}_2(x) w(x) d\mathcal{H}^m(x),$$

where $\hat{h}_1$ and $\hat{h}_2$ are two first-stage linear sieve estimators of h_0 estimated from two split subsamples, each with sample size $n/2$ and sieve dimension $K_n^* \asymp n^{d/(2s+d-m)}$.

Remark 7. The lower bound in part (a) is a global minimax lower bound over the full Hölder ball $\Lambda_c^s(\mathcal{X})$: the first term $n^{-s/(2s+d-m)}$ is derived via the linearization at a fixed h_0 with $\int_{\mathcal{M}} h_0 w d\mathcal{H}^m \neq 0$ (so that $DQ(h_0) \neq 0$), while the second term $n^{-4s/(4s+2d-m)}$ is obtained from a multi-hypothesis construction centered at $h \equiv 0$

(where $DQ(0) = 0$ and the quadratic structure drives the rate). The overall lower bound is the maximum of these two contributions, taken over different regions of the parameter space. Note that for $s \leq m/2$, the nonlinear exponents in the lower bound ($4s/(4s+2d-m)$) and the upper bound ($2s/(2s+2d-m)$) do not coincide, so in that regime the precise minimax rate for $Q(h_0)$ is not fully pinned down by parts (a)–(b); part (c) resolves this only under $s > m/2$.

Proof of Lemma 8(a). By Corollary 2, the linear rate $n^{-\frac{s}{2s+d-m}}$ is a valid lower bound with the linearization remainder $R_n = O(b_n^{2s})$, which can be shown explicitly using the two-point construction $h_0 \equiv 1$ and $h_1(x) = h_0(x) + b_n^s K_{d-m}\left(\frac{g(x)}{b_n}\right)$, which implies:

$$\begin{aligned}
\Gamma(h_1) - \Gamma(h_0) &= \int_{\mathcal{M}} \left((1 + b_n^s K_{d-m}(0))^2 - 1^2 \right) w(x) d\mathcal{H}^m(x) \\
&= \int_{\mathcal{M}} \left(2b_n^s K_{d-m}(0) + b_n^{2s} K_{d-m}^2(0) \right) w(x) d\mathcal{H}^m(x) \\
&= \underbrace{2b_n^s \int_{\mathcal{M}} K_{d-m}(0) w(x) d\mathcal{H}^m(x)}_{D_h(h_0)[h_1-h_0]} + \underbrace{b_n^{2s} \int_{\mathcal{M}} K_{d-m}^2(0) w(x) d\mathcal{H}^m(x)}_{R_n} \\
&= 2b_n^s \int_{\mathcal{M}} K_{d-m}(0) w(x) d\mathcal{H}^m(x) + O(b_n^{2s}) \\
&\asymp b_n^s
\end{aligned}$$

We now seek to establish the other lower bound $n^{-\frac{4s}{4s+2d-m}}$, based on the multi-hypothesis construction in Chapter 2.7.5 of [Tsybakov \(2009\)](#). Specifically, let μ_0 be the Dirac measure concentrated on $h_0(x) \equiv 0$ and let μ_1 be a discrete measure supported on the finite set of functions

$$h_\omega(x) := \sum_{k=1}^M \omega_k \varphi_k(x),$$

where:

- $M_n := b_n^{-m}$
- $\varphi_k(x) := b_n^s K\left(\frac{x-x_k}{b_n}\right)$ with (x_k) is a mesh grid on $\mathcal{M} = \{x : g(x) = 0\}$
- $\omega_k \sim_{i.i.d.} 2\text{Bernoulli}(0.5) - 1$, i.e., taking values ± 1 with equal probabilities 0.5.

Under the above construction, we have

$$\begin{aligned}
\Gamma(h_\omega) &= \int_{\{g(x)=0\}} \left(\sum_{k=1}^{M_n} \omega_k \varphi_k(x) \right)^2 d\mathcal{H}^m(x) \\
&= \sum_{k=1}^{M_n} \int_{\{g(x)=0\}} \varphi_k^2(x) d\mathcal{H}^m(x) \\
&\asymp M_n b_n^{2s+m} \\
&\asymp b_n^{-m} b_n^{2s+m} \\
&\asymp b_n^{2s}
\end{aligned} \tag{134}$$

For $k = 0, 1$, let P_k^n denote the joint probability distribution of the random sample $\{(X_i, Y_i)\}_{i=1}^n$ from the nonparametric regression model with h_k specified above. Then the χ^2 -divergence can be derived similar to those in Chapter 2.7.5 of [Tsybakov \(2009\)](#), with

$$\begin{aligned}
\chi^2(P_1^n, P_0^n) &\leq \exp\left(n^2 b_n^{2(2s+d)} M_n\right) \\
&= \exp\left(n^2 b_n^{2(2s+d)} b_n^{-m}\right) \\
&= \exp\left(n^2 b_n^{4s+2d-m}\right)
\end{aligned}$$

To ensure $\chi^2(P_1^n, P_0^n) \leq c$, we set

$$b_n \asymp n^{-\frac{2}{4s+2d-m}}. \tag{135}$$

Hence, by Theorem 2.15 of [Tsybakov \(2009\)](#), we obtain the following lower bound based on (134) and (135):

$$b_n^{2s} \asymp n^{-\frac{4s}{4s+2d-m}}$$

Combining the above rate with the previous $n^{-\frac{s}{2s+d-m}}$ above, we obtain a valid lower bound

$$r_{Quad,n} := \max\left\{n^{-\frac{s}{2s+d-m}}, n^{-\frac{4s}{4s+2d-m}}\right\}.$$

□

Proof of Lemma 8(b). Recall the split-sample sieve estimator

$$\hat{\theta} := \int_{\mathcal{M}} \hat{h}_1(x) \hat{h}_2(x) w(x) d\mathcal{H}^m(x),$$

where $\hat{h}_r$ is the sieve LS estimator computed on subsample I_r of size $n_r := |I_r|$ ($r = 1, 2$), with $n_1 \asymp n_2 \asymp n$ and sieve dimension K_n (write $K := K_n$ for brevity).

Let $\tilde{h}_r := P_{K,r}h_0$ denote the (subsample- r) empirical LS projection of h_0 onto $\mathbb{H}_K$, i.e.,

$$\tilde{h}_r(x) = \bar{b}^K(x)' \bar{G}_r^{-1} \frac{1}{n_r} \sum_{i \in I_r} \bar{b}^K(X_i) h_0(X_i), \quad \bar{G}_r := \frac{1}{n_r} \sum_{i \in I_r} \bar{b}^K(X_i) \bar{b}^K(X_i)'. \quad (135)$$

Then the stochastic part admits the orthonormalized linear representation

$$\hat{h}_r(x) - \tilde{h}_r(x) = \bar{b}^K(x)' \bar{G}_r^{-1} \frac{1}{n_r} \sum_{i \in I_r} \bar{b}^K(X_i) \epsilon_i, \quad r \in \{1, 2\}. \quad (136)$$

Write $\Delta_r := \hat{h}_r - h_0$. Then

$$\begin{aligned} \hat{\theta} - \theta_0 &= \int_{\mathcal{M}} (\hat{h}_1 \hat{h}_2 - h_0^2) w d\mathcal{H}^m \\ &= \underbrace{\int_{\mathcal{M}} (\hat{h}_1 - h_0) h_0 w d\mathcal{H}^m}_{=: T_1} + \underbrace{\int_{\mathcal{M}} (\hat{h}_2 - h_0) h_0 w d\mathcal{H}^m}_{=: T_2} + \underbrace{\int_{\mathcal{M}} (\hat{h}_1 - h_0)(\hat{h}_2 - h_0) w d\mathcal{H}^m}_{=: T_3}. \end{aligned} \quad (137)$$

Linear terms T_1, T_2 . Each T_r is a linear submanifold functional of $\hat{h}_r - h_0$ with bounded weight $h_0 w$. Applying Theorem 3 on each subsample (size $n_r \asymp n$) yields

$$T_1 = O_p \left(K^{-\frac{s}{d}} + \sqrt{\frac{K^{\frac{d-m}{d}}}{n}} \right), \quad T_2 = O_p \left(K^{-\frac{s}{d}} + \sqrt{\frac{K^{\frac{d-m}{d}}}{n}} \right). \quad (138)$$

Quadratic term T_3 : bias/cross/stochastic pieces. Decompose $\Delta_r = (\hat{h}_r - \tilde{h}_r) + (\tilde{h}_r - h_0)$ and expand

$$\begin{aligned} T_3 &= \int_{\mathcal{M}} (\hat{h}_1 - \tilde{h}_1)(\hat{h}_2 - \tilde{h}_2) w d\mathcal{H}^m + \int_{\mathcal{M}} (\tilde{h}_1 - h_0)(\hat{h}_2 - \tilde{h}_2) w d\mathcal{H}^m \\ &\quad + \int_{\mathcal{M}} (\hat{h}_1 - \tilde{h}_1)(\tilde{h}_2 - h_0) w d\mathcal{H}^m + \int_{\mathcal{M}} (\tilde{h}_1 - h_0)(\tilde{h}_2 - h_0) w d\mathcal{H}^m \\ &=: T_{3,\text{stoch}} + T_{3,1} + T_{3,2} + T_{3,\text{bias}}. \end{aligned} \quad (139)$$

Bias part. By Assumption 12(iv)(v), $\|\tilde{h}_r - h_0\|_{\infty} = O_p(K^{-s/d})$. Hence

$$|T_{3,\text{bias}}| \leq \|w\|_{\infty} \mathcal{H}^m(\mathcal{M}) \|\tilde{h}_1 - h_0\|_{\infty} \|\tilde{h}_2 - h_0\|_{\infty} = O_p(K^{-\frac{2s}{d}}).$$

Cross parts. Conditional on the subsample designs, $\hat{h}_r - \tilde{h}_r$ is centered and has no approximation bias. Moreover, the weights $(\tilde{h}_r - h_0)w$ are bounded by $O_p(K^{-s/d})$. Applying Theorem 3 to the centered linear functional on each subsample gives

$$T_{3,1} = O_p \left(K^{-\frac{s}{d}} \sqrt{\frac{K^{\frac{d-m}{d}}}{n}} \right), \quad T_{3,2} = O_p \left(K^{-\frac{s}{d}} \sqrt{\frac{K^{\frac{d-m}{d}}}{n}} \right).$$

These cross terms are of smaller order than $\max\{K^{-s/d}, \sqrt{K^{(d-m)/d}/n}\}$ and can be absorbed.

Cross terms $T_{3,1}$ and $T_{3,2}$. Recall

$$T_{3,1} := \int_{\mathcal{M}} (\tilde{h}_1 - h_0)(\hat{h}_2 - \tilde{h}_2) w d\mathcal{H}^m, \quad T_{3,2} := \int_{\mathcal{M}} (\hat{h}_1 - \tilde{h}_1)(\tilde{h}_2 - h_0) w d\mathcal{H}^m.$$

We treat $T_{3,1}$; the bound for $T_{3,2}$ is identical by symmetry (swap the roles of the two splits). Let

$$A_{1,n} := \|\tilde{h}_1 - h_0\|_{\infty}.$$

On the event $\{A_{1,n} > 0\}$ define the *random but bounded* weight

$$w_{1,n}(x) := w(x) \frac{\tilde{h}_1(x) - h_0(x)}{A_{1,n}}, \quad x \in \mathcal{M},$$

and set $w_{1,n} \equiv 0$ on $\{A_{1,n} = 0\}$. Then $\|w_{1,n}\|_{\infty} \leq \|w\|_{\infty}$ and, *crucially*, $w_{1,n}$ is measurable w.r.t. the first split and hence independent of the second-split estimator $\hat{h}_2$. With this normalization,

$$T_{3,1} = A_{1,n} \int_{\mathcal{M}} (\hat{h}_2 - \tilde{h}_2)(x) w_{1,n}(x) d\mathcal{H}^m(x) \equiv A_{1,n} L_{w_{1,n}}(\hat{h}_2 - \tilde{h}_2), \quad (140)$$

where $L_{w_{1,n}}(f) := \int_{\mathcal{M}} f(x) w_{1,n}(x) d\mathcal{H}^m(x)$ is a linear submanifold functional.

Step 1: bound $A_{1,n} = \|\tilde{h}_1 - h_0\|_{\infty}$. By Assumption 12(v), there exists $h_{0,K} \in \mathbb{H}_K$ with $\|h_{0,K} - h_0\|_{\infty} \lesssim K^{-s/d}$. Since $\tilde{h}_1 = P_{K,1}h_0$ is the empirical LS projection onto $\mathbb{H}_K$ based on split 1,

$$\|\tilde{h}_1 - h_0\|_{\infty} \leq \|P_{K,1}(h_0 - h_{0,K})\|_{\infty} + \|h_{0,K} - h_0\|_{\infty} \leq \|P_{K,1}\|_{\infty} \|h_0 - h_{0,K}\|_{\infty} + \|h_{0,K} - h_0\|_{\infty}.$$

Assumption 12(iv) gives $\|P_{K,1}\|_{\infty} = O_p(1)$, so

$$A_{1,n} = \|\tilde{h}_1 - h_0\|_{\infty} = O_p(K^{-s/d}). \quad (141)$$

Step 2: bound the centered linear functional $L_{w_{1,n}}(\hat{h}_2 - \tilde{h}_2)$. Condition on the first split (so that $w_{1,n}$ is fixed) and apply Theorem 3(1) to the second-split sieve LS estimator and the (now deterministic) weight $w_{1,n}$:

$$\sqrt{n_2} L_{w_{1,n}}(\hat{h}_2 - \tilde{h}_2) = \frac{1}{\sqrt{n_2}} \sum_{i \in I_2} v_{K,2}^*(w_{1,n})(X_i) \epsilon_i + o_p\left(\|v_{K,2}^*(w_{1,n})\|_{sd}\right),$$

and hence

$$L_{w_{1,n}}(\hat{h}_2 - \tilde{h}_2) = O_p\left(\frac{\|v_{K,2}^*(w_{1,n})\|_{sd}}{\sqrt{n}}\right). \quad (142)$$

By the growth-rate lemma for sieve Riesz representers (Lemma 1), and using

$\|w_{1,n}\|_\infty \leq \|w\|_\infty$, we have the *uniform* upper bound

$$\|v_{K,2}^*(w_{1,n})\|_{sd} \lesssim \|w_{1,n}\|_\infty K^{(d-m)/(2d)} \leq \|w\|_\infty K^{(d-m)/(2d)}. \quad (143)$$

Combining (142) and (143) yields

$$L_{w_{1,n}}(\hat{h}_2 - \tilde{h}_2) = O_p\left(\sqrt{\frac{K^{(d-m)/d}}{n}}\right). \quad (144)$$

Step 3: conclude for $T_{3,1}$ (and $T_{3,2}$). Combining (140), (141), and (144),

$$T_{3,1} = O_p\left(K^{-s/d} \sqrt{\frac{K^{(d-m)/d}}{n}}\right).$$

The same argument with indices swapped gives

$$T_{3,2} = O_p\left(K^{-s/d} \sqrt{\frac{K^{(d-m)/d}}{n}}\right).$$

Step 4 (main stochastic term $T_{3,\text{stoch}}$). Define the deterministic $K \times K$ matrix

$$A_K := \int_{\mathcal{M}} \bar{b}^K(x) \bar{b}^K(x)' |w(x)| d\mathcal{H}^m(x).$$

Using (136) and Cauchy–Schwarz, we may bound $|T_{3,\text{stoch}}|$ by replacing w with $|w|$. Let

$$\eta_r := \frac{1}{n_r} \sum_{i \in I_r} \bar{b}^K(X_i) \epsilon_i \in \mathbb{R}^K.$$

Then

$$T_{3,\text{stoch}} = \eta_1' \bar{G}_1^{-1} A_K \bar{G}_2^{-1} \eta_2 \quad (\text{up to sign}). \quad (145)$$

Conditioning on X_1^n and using the independence between the two subsamples (and $\mathbb{E}[\epsilon_i | X_i] = 0$), we have $\mathbb{E}[T_{3,\text{stoch}} | X_1^n] = 0$. Moreover,

$$\mathbb{E}[\eta_r \eta_r' | X_1^n] = \frac{1}{n_r^2} \sum_{i \in I_r} \mathbb{E}[\epsilon_i^2 | X_i] \bar{b}^K(X_i) \bar{b}^K(X_i)' \preceq \frac{\bar{\sigma}^2}{n_r} \bar{G}_r,$$

where $\bar{\sigma}^2 := \sup_x \mathbb{E}[\epsilon_i^2 | X_i = x] < \infty$ by Assumption 11. Therefore, by the tower property and cyclicity of trace,

$$\begin{aligned} \mathbb{E}[T_{3,\text{stoch}}^2 | X_1^n] &\leq \frac{\bar{\sigma}^4}{n_1 n_2} \text{tr}(\bar{G}_1^{-1} A_K \bar{G}_2^{-1} A_K) \\ &\leq \frac{\bar{\sigma}^4}{n_1 n_2} \|\bar{G}_1^{-1}\|_{op} \|\bar{G}_2^{-1}\|_{op} \text{tr}(A_K^2). \end{aligned}$$

Under Assumption 12(i)(iii), $\|\bar{G}_r^{-1}\|_{op} = O_p(1)$. It remains to bound $\text{tr}(A_K^2) =$

$$\|A_K\|_F^2.$$

Claim: $\text{tr}(A_K^2) \lesssim K^{\frac{2d-m}{d}}.$

Justification of the claim. Apply the manifold decomposition (41) (as in the proof of Lemma 1) to write $A_K = \sum_{j=1}^{\bar{j}} A_{K,j}$, where $\bar{j} < \infty$ is the number of charts and

$$A_{K,j} := \int_{\mathcal{U}_j} \bar{b}^K(\varphi_j(u)) \bar{b}^K(\varphi_j(u))' \bar{w}_j(u) du,$$

with bounded $\bar{w}_j(u)$. Using $\|\sum_j A_{K,j}\|_F^2 \leq \bar{j} \sum_j \|A_{K,j}\|_F^2$, it suffices to show $\|A_{K,j}\|_F^2 \lesssim K^{(2d-m)/d}$ uniformly in j . Fix j . For each k, k' , write

$$(A_{K,j})_{kk'} = \int_{\mathcal{U}_j} \bar{b}_k^K(\varphi_j(u)) \bar{b}_{k'}^K(\varphi_j(u)) \bar{w}_j(u) du.$$

Using the same tensor-product decomposition as in the proof of Lemma 1, $\bar{b}_k^K(\varphi_j(u)) = \bar{b}_{k,(m)}^K(u) \cdot \bar{b}_{k,-(m)}^K(\psi_j(u))$. Then, for fixed k and fixed $k'_{-(m)}$, the frame inequality for $\{\bar{b}_{k',(m)}^K\}$ on $\mathcal{U}_j$ yields

$$\sum_{k'_{(m)}} (A_{K,j})_{kk'}^2 \lesssim \int_{\mathcal{U}_j} \left(\bar{b}_k^K(\varphi_j(u)) \right)^2 \left(\bar{b}_{k',-(m)}^K(\psi_j(u)) \right)^2 \bar{w}_j(u)^2 du.$$

Summing over $k'_{-(m)}$ and using the tensor-product structure gives $\sum_{k'_{-(m)}} (\bar{b}_{k',-(m)}^K(\psi_j(u)))^2 \lesssim J^{d-m} = K^{(d-m)/d}$ pointwise in u . Hence,

$$\sum_{k'=1}^K (A_{K,j})_{kk'}^2 \lesssim K^{\frac{d-m}{d}} \int_{\mathcal{U}_j} \left(\bar{b}_k^K(\varphi_j(u)) \right)^2 \bar{w}_j(u)^2 du.$$

Finally, summing over k and using $\sum_{k=1}^K (\bar{b}_k^K(x))^2 = \|\bar{b}^K(x)\|^2 \lesssim K$ (Assumption 12(ii) and $\lambda_{\min}(G) > 0$) yields

$$\|A_{K,j}\|_F^2 = \sum_{k=1}^K \sum_{k'=1}^K (A_{K,j})_{kk'}^2 \lesssim K^{\frac{d-m}{d}} \int_{\mathcal{U}_j} \left(\sum_{k=1}^K (\bar{b}_k^K(\varphi_j(u)))^2 \right) \bar{w}_j(u)^2 du \lesssim K^{\frac{d-m}{d}} \cdot K = K^{\frac{2d-m}{d}}.$$

This proves the claim.

Returning to $T_{3,\text{stoch}}$, we thus have $\mathbb{E}[T_{3,\text{stoch}}^2] = O(K^{(2d-m)/d}/n^2)$ and hence

$$T_{3,\text{stoch}} = O_p\left(\frac{1}{n} \sqrt{K^{\frac{2d-m}{d}}}\right).$$

Step 5 (collect rates and optimize over K). Combining (137), (138), (139) and the

bounds above gives

$$\hat{\theta} - \theta_0 = O_p \left(K^{-\frac{s}{d}} + \sqrt{\frac{K^{\frac{d-m}{d}}}{n}} + \frac{1}{n} \sqrt{K^{\frac{2d-m}{d}}} \right).$$

We now choose K to minimize the maximal order among

$$T_1(K) := K^{-s/d}, \quad T_2(K) := \sqrt{\frac{K^{\frac{d-m}{d}}}{n}}, \quad T_3(K) := \frac{1}{n} \sqrt{K^{\frac{2d-m}{d}}}.$$

Balancing T_1 and T_2 yields $K^{1/d} \asymp n^{1/(2s+d-m)}$ and hence the “linear” rate

$$K^{-\frac{s}{d}} \asymp \sqrt{\frac{K^{\frac{d-m}{d}}}{n}} \asymp n^{-\frac{s}{2s+d-m}}. \quad (146)$$

Balancing T_1 and T_3 yields $K^{1/d} \asymp n^{2/(2s+2d-m)}$ and hence

$$K^{-\frac{s}{d}} \asymp \frac{1}{n} \sqrt{K^{\frac{2d-m}{d}}} \asymp n^{-\frac{2s}{2s+2d-m}}. \quad (147)$$

Combining (146) and (147), we obtain

$$\bar{r}_{Quad,n} := \max \left\{ n^{-\frac{s}{2s+d-m}}, n^{-\frac{2s}{2s+2d-m}} \right\} \equiv \begin{cases} n^{-\frac{s}{2s+d-m}}, & \text{if } s \geq \frac{m}{2}, \\ n^{-\frac{2s}{2s+2d-m}}, & \text{if } s < \frac{m}{2}. \end{cases} \quad (148)$$

Finally, to verify the regime split: if $K^{1/d} \asymp n^{1/(2s+d-m)}$, then $T_3(K) = O(T_2(K))$ is equivalent to $K \lesssim n$, i.e. $n^{d/(2s+d-m)} \lesssim n$, which holds iff $s \geq m/2$. If $K^{1/d} \asymp n^{2/(2s+2d-m)}$, then $T_2(K) = O(T_3(K))$ is equivalent to $n \lesssim K$, i.e. $n \lesssim n^{2d/(2s+2d-m)}$, which holds iff $s \leq m/2$. This matches (148). $\square$

Proof of Lemma 8(c). Recalling that

$$\bar{r}_{Quad,n} := \max \left\{ n^{-\frac{s}{2s+d-m}}, n^{-\frac{2s}{2s+2d-m}} \right\} \equiv \begin{cases} n^{-\frac{s}{2s+d-m}}, & \text{if } s \geq \frac{m}{2}, \\ n^{-\frac{2s}{2s+2d-m}}, & \text{if } s \leq \frac{m}{2}. \end{cases}$$

and

$$\underline{r}_{Quad,n} := \max \left\{ n^{-\frac{s}{2s+d-m}}, n^{-\frac{4s}{4s+2d-m}} \right\} = \begin{cases} n^{-\frac{s}{2s+d-m}}, & \text{if } s \geq \frac{3m-2d}{4}, \\ n^{-\frac{4s}{4s+2d-m}}, & \text{if } s \leq \frac{3m-2d}{4}. \end{cases}$$

Note that

$$\frac{m}{2} > \frac{m}{4} - \frac{d-m}{2} = \frac{3m-2d}{4}$$

Hence, whenever $s \geq \frac{m}{2}$, we have

$$\bar{r}_{Quad,n} = \underline{r}_{Quad,n} = n^{-\frac{s}{2s+d-m}}$$

based on which we may conclude that the minimax optimal rate $n^{-\frac{s}{2s+d-m}}$ is attained by the split-sample sieve estimator. $\square$

G Rate Analysis for Nadaraya-Watson First Stage

In this section, we show that the key rate-acceleration result in the previous subsection, as one may expect, also holds for kernel-based nonparametric methods. To illustrate this point, we consider the case where h_0 is given by a conditional expectation and $\hat{h}$ is given by the Nadaraya-Watson kernel estimator and show that our key result continues to hold.

Formally, let $h_0(x) = \mathbb{E}[Y_i | X_i = x]$ and

$$\hat{h}(x) = \frac{\frac{1}{nb_n^d} \sum_{i=1}^n K_d\left(\frac{X_i - x}{b_n}\right) Y_i}{\frac{1}{nb_n^d} \sum_{i=1}^n K_d\left(\frac{X_i - x}{b_n}\right)}$$

where $b_n \searrow 0$ is a bandwidth parameter and K_d is a multivariate kernel of smoothness order s .

We consider the convergence of the plug-in estimator

$$\hat{\theta} := \int_{\mathcal{M}} \hat{h}(x) w(x) p(x) d\mathcal{H}^m(x)$$

to $\theta_0 = \int_{\mathcal{M}} h_0(x) w(x) p_0(x) d\mathcal{H}^m(x)$, treating $p(x)$ in the integrand as known. Under this simplification, $\hat{\theta} = L(\hat{h})$ is a plug-in estimator of a linear functional as in Section 4.1.

Our results can also be extended to analyze the composite estimator $\Gamma(\hat{h}, \hat{p}) = \int_{\mathcal{M}} \hat{h}(x) w(x) \hat{p}(x) d\mathcal{H}^m(x)$ for $\Gamma(h_0, p_0)$, treated as a nonlinear integral functional $\Gamma(h_0, p_0)$ of the vector-valued first stage (h_0, p_0) . This can be easily handled using the similar kernel arguments as below in combination of the linearization arguments in Section 4.2, and the incorporation of the density estimator does not affect the rate of the estimator.

Assumption 20 (Kernel Smoothness). *K_d is a d -dimensional product kernel*

$$K_d(x) = \prod_{j=1}^d K(x_j),$$

where K is a univariate kernel of smoothness order s , i.e., (i) $K(u) = K(-u)$, (ii) $\int K(u) du = 1$, (iii) $|K(u)| \leq M < \infty$, (iv) $\int u^j K(u) du = 0$ for $j = 1, \dots, s-1$, and (v) $\kappa_s := \int x^s K(x) dx \in (0, \infty)$.

Theorem 6 (Kernel Nadaraya-Watson First Stage). *Let $h_0(\cdot) = \mathbb{E}[Y_i | X_i = \cdot] \in \Lambda^s(\mathcal{X})$. Under Assumptions 1 and 20, we have*

$$\text{Bias}(\hat{\theta}) = O(b_n^s), \quad \text{Var}(\hat{\theta}) = O\left(\frac{1}{nb_n^{d-m}}\right)$$

and thus

$$\|\hat{\theta} - \theta_0\| = O_p\left(n^{-\frac{s}{2s+d-m}}\right).$$

Proof of Theorem 6. Write $a(x) := h_0(x)p(x)$, we have

$$\begin{aligned} \hat{\theta} - \theta_0 &= \int_{\mathcal{M}} [\hat{h}(x) - h_0(x)] w(x) p(x) d\mathcal{H}^m(x) \\ &= \int_{\mathcal{M}} \left[\frac{\hat{a}(x)}{\hat{p}(x)} - \frac{a(x)}{p(x)} \right] w(x) p(x) d\mathcal{H}^m(x) \\ &= \int_{\mathcal{M}} \left[\frac{\hat{a}(x) - a(x)}{p(x)} - \frac{a(x)}{p^2(x)} (\hat{p}(x) - p(x)) \right] w(x) p(x) d\mathcal{H}^m(x) + R_1 \\ &= \int_{\mathcal{M}} [\hat{a}(x) - a(x) - h_0(x) (\hat{p}(x) - p(x))] w(x) d\mathcal{H}^m(x) + R_1 \\ &= \int_{\mathcal{M}} [\hat{a}(x) - h_0(x) \hat{p}(x)] w(x) d\mathcal{H}^m(x) + R_1 \\ &= \int_{\mathcal{M}} \frac{1}{nb_n^d} \sum_{i=1}^n K\left(\frac{x - X_i}{b_n}\right) (Y_i - h_0(x)) w(x) d\mathcal{H}^m(x) + R_1 \\ &= \underbrace{\frac{1}{nb_n^d} \sum_{i=1}^n \int_{\mathcal{M}} K\left(\frac{x - X_i}{b_n}\right) (Y_i - h_0(x)) w(x) d\mathcal{H}^m(x)}_{T_1} + R_1, \end{aligned} \tag{149}$$

where the remainder term R_1 is asymptotically negligible.

By (41), we may write

$$T_1 = \sum_{j=1}^{\bar{j}} \frac{1}{n} \sum_{i=1}^n T_{1ij}$$

with

$$T_{1ij} := \frac{1}{b_n^d} \int_{\varphi_j(\mathcal{U}_j)} K\left(\frac{\varphi_j(x_{(j,m)}) - X_i}{b_n}\right) (Y_i - h_0(\varphi_j(x_{(j,m)}))) \bar{w}_j(Y_i, \varphi_j(x_{(j,m)})) dx_{(j,m)}$$

where

$$\bar{w}_j(Y_i, x) := \rho_j(x) w(x) \mathcal{J}\varphi_j(x)$$

Subsequently, we consider each T_{1ij} separately and suppress the subscript j for simpler notation.

We carry out the kernel change of variables from $x_{(m)}$ to u by setting

$$u := \frac{x_{(m)} - X_{i,(m)}}{b_n}, \quad \Leftrightarrow \quad x_{(m)} = X_{i,(m)} + b_n u.$$

We write $\mathcal{V}_u := \frac{\mathcal{V} - X_{i,(m)}}{b_n}$.

$$\begin{aligned} T_{1ij} &= \frac{1}{b_n^d} \int_{\mathcal{V}_u} K \left(u, \frac{\psi(X_{i,(m)} + b_n u) - X_{i,-(m)}}{b_n} \right) (Y_i - h_0(\varphi(X_{i,(m)} + b_n u))) \\ &\quad \bar{w}(Y_i, \varphi(X_{i,(m)} + b_n u)) d(X_{i,(m)} + b_n u) \\ &= \frac{b_n^m}{b_n^d} \int_{\mathcal{V}_u} K_{(m)}(u) K_{-(m)} \left(\frac{\psi(X_{i,(m)} + b_n u) - X_{i,-(m)}}{b_n} \right) (Y_i - h_0(\varphi(X_{i,(m)} + b_n u))) \\ &\quad \bar{w}(\varphi(X_{i,(m)} + b_n u)) du \end{aligned}$$

Since $h_0(\varphi(X_{i,(m)} + b_n u)) = h_0(\varphi(X_{i,(m)})) + O(b_n)$ and $\bar{w}(\varphi(X_{i,(m)} + b_n u)) = \bar{w}(\varphi(X_{i,(m)})) + O(b_n)$ on the support of $K_{(m)}$, these smooth terms may be evaluated at $u = 0$ with $O(b_n)$ corrections absorbed into R_2 . However, the transverse kernel argument

$$\frac{\psi(X_{i,(m)} + b_n u) - X_{i,-(m)}}{b_n} = \frac{\psi(X_{i,(m)}) - X_{i,-(m)}}{b_n} + \nabla \psi(X_{i,(m)}) u + O(b_n)$$

varies at $O(1)$ in u , so $K_{-(m)}$ cannot simply be evaluated at $u = 0$. Define the *effective transverse kernel*

$$\tilde{K}_{-(m)}(\xi; x_{(m)}) := \int K_{(m)}(u) K_{-(m)}(\xi + \nabla \psi(x_{(m)}) u) du. \quad (150)$$

Then

$$T_{1ij} = \underbrace{\frac{1}{b_n^{d-m}} \tilde{K}_{-(m)} \left(\frac{\psi(X_{i,(m)}) - X_{i,-(m)}}{b_n}; X_{i,(m)} \right) (Y_i - h_0(\varphi(X_{i,(m)}))) \bar{w}(\varphi(X_{i,(m)}))}_{T_{2,ij}} + R_2,$$

where R_2 collects the $O(b_n/b_n^{d-m})$ corrections from the smooth-term evaluations and

the $O(b_n)$ Taylor remainder in ψ .

We claim that $\tilde{K}_{-(m)}$ is a valid $(d-m)$ -dimensional kernel of the same order as $K_{-(m)}$. First, by a change of variable $\eta = \xi + \nabla\psi \cdot u$,

$$\int \tilde{K}_{-(m)}(\xi; x_{(m)}) d\xi = \int K_{(m)}(u) \underbrace{\int K_{-(m)}(\eta) d\eta}_{=1} du = 1.$$

Second, for any multi-index α with $|\alpha| \leq s-1$, the same change of variable gives $\int \xi^\alpha \tilde{K}_{-(m)}(\xi) d\xi = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} (-1)^{|\alpha-\beta|} \left[\int K_{(m)}(u) (\nabla\psi \cdot u)^{\alpha-\beta} du \right] \left[\int K_{-(m)}(\eta) \eta^\beta d\eta \right]$. Since $K_{-(m)}$ has vanishing moments up to order $s-1$ and $K_{(m)}$ is a symmetric product kernel (so $\int u^j K_{(m)}(u) du = 0$ for odd j), each term in the sum vanishes. Hence $\tilde{K}_{-(m)}$ has the same moment conditions as $K_{-(m)}$, i.e., it is a kernel of order s . Third, $\int \tilde{K}_{-(m)}^2(\xi) d\xi < \infty$ since $K_{(m)}$ and $K_{-(m)}$ are bounded and compactly supported.

The term $\frac{1}{n} \sum_i T_{2,ij}$ is therefore a $(d-m)$ -dimensional kernel estimator (with the effective kernel $\tilde{K}_{-(m)}$ in place of $K_{-(m)}$) and shares the same asymptotic bias and variance rates. We formalize this via Lemmas 9 and 10 below, whose proofs apply with $\tilde{K}_{-(m)}$ replacing $K_{-(m)}$ throughout, yielding $\text{Bias} = O(b_n^s)$ and $\text{Var} = O(1/(nb_n^{d-m}))$. $\square$

Lemma 9 (Bias). $\mathbb{E} \left[\frac{1}{n} \sum_i T_{2,ij} \right] = \mathbb{E} [T_{2,ij}] = O(b_n^s)$.

Proof. Clearly,

$$\begin{aligned} & \mathbb{E} [T_{2ij}] \\ &= \frac{1}{b_n^{d-m}} \int \left[K_{-(m)} \left(\frac{\psi(z_{(m)}) - z_{-(m)}}{b_n} \right) (h_0(z) - h_0(\varphi(z_{(m)}))) \bar{w}(\varphi(z_{(m)})) \right] p(z) dz \\ &= \frac{1}{b_n^{d-m}} \int \int \left[K_{-(m)} \left(\frac{\psi(z_{(m)}) - z_{-(m)}}{b_n} \right) (h_0(z_{(m)}, z_{-(m)}) - h_0(\varphi(z_{(m)}))) \bar{w}(\varphi(z_{(m)})) \right] \\ & \quad p(z_{-(m)} | z_{(m)}) dz_{-(m)} p(z_{(m)}) dz_{(m)} \end{aligned}$$

Define the change of variable from $z_{-(m)}$ to ζ as

$$\zeta := \frac{z_{-(m)} - \psi_j(z_{(m)})}{b_n}, \quad z_{-(m)} = \psi_j(z_{(m)}) + b_n \zeta.$$

We have

$$\mathbb{E} [T_{2ij}] = \frac{1}{b_n^{d-m}} \int \int \left[K_{-(m)}(\zeta) (h_0(z_{(m)}, \psi_j(z_{(m)}) + b_n \zeta) - h_0(\varphi(z_{(m)}))) \right]$$

$$\begin{aligned}
& p\left(\psi_j\left(z_{(m)}\right)+b_n \zeta_{-(m)}\left|z_{(m)}\right.\right) b_n^{d-m} d \zeta p\left(z_{(m)}\right) \bar{w}\left(\varphi\left(z_{(m)}\right)\right) d z_{(m)} \\
&= \int\left[\int K_{-(m)}(\zeta)\left(h_0\left(z_{(m)}, \psi_j\left(z_{(m)}\right)+b_n \zeta\right)-h_0\left(\varphi\left(z_{(m)}\right)\right)\right)\left(\psi_j\left(z_{(m)}\right)+b_n \zeta_{-(m)}\left|z_{(m)}\right.\right) d \zeta\right] \\
&\quad p\left(z_{(m)}\right) \bar{w}\left(\varphi\left(z_{(m)}\right)\right) d z_{(m)} \\
&= \int\left[\int K_{-(m)}(\zeta)\left(B\left(z_{(m)}\right) b_n^s \zeta^s+o\left(b_n^s\right)\right) d \zeta\right] p\left(z_{(m)}\right) \bar{w}\left(\varphi\left(z_{(m)}\right)\right) d z_{(m)} \\
&= b_n^s \cdot \int K_{-(m)}(\zeta) \zeta^s d \zeta \cdot \int B\left(z_{(m)}\right) p\left(z_{(m)}\right) \bar{w}\left(\varphi\left(z_{(m)}\right)\right) d z_{(m)}+o\left(b_n^s\right) \\
&= O\left(b_n^s\right)
\end{aligned}$$

Hence, we also have $\mathbb{E}\left[\frac{1}{n} \sum_i T_{2,ij}\right]=\mathbb{E}\left[T_{2,ij}\right]=O\left(b_n^s\right)$. $\square$

Lemma 10 (Variance). $\text{Var}\left[\frac{1}{n} \sum_i T_{2,ij}\right]=O\left(\frac{1}{n b_n^{d-m}}\right)$

Proof. Next,

$$\begin{aligned}
& \mathbb{E}\left(T_{2ij}^2\right) \\
&= \frac{1}{b_n^{2(d-m)}} \int K_{-(m)}^2\left(\frac{\psi_j\left(z_{(m)}\right)-z_{-(m)}}{b_n}\right)\left(Y_i-h_0\left(\varphi\left(z_{(m)}\right)\right)\right)^2 \bar{w}^2\left(\varphi\left(z_{(m)}\right)\right) \\
&\quad p\left(z_{(m)}, z_{-(m)}\right) d z_{-(m)} d z_{(m)} \\
&= \frac{1}{b_n^{2(d-m)}} \int K_{-(m)}^2\left(\frac{\psi_j\left(z_{(m)}\right)-z_{-(m)}}{b_n}\right)\left[\sigma_0^2(z)+\left(h_0(z)-h_0\left(\varphi\left(z_{(m)}\right)\right)\right)^2\right] \bar{w}^2\left(\varphi\left(z_{(m)}\right)\right) \\
&\quad p\left(z_{(m)}, z_{-(m)}\right) d z_{-(m)} d z_{(m)} \\
&= \frac{1}{b_n^{2(d-m)}} \int\left[\underbrace{\int K_{-(m)}^2\left(\frac{\psi_j\left(z_{(m)}\right)-z_{-(m)}}{b_n}\right)\left[\sigma_0^2(z)+\left(h_0(z)-h_0\left(\varphi\left(z_{(m)}\right)\right)\right)^2\right] p\left(z_{-(m)}\left|z_{(m)}\right.\right) d z_{-(m)}}_{T_{3j}}\right] \\
&\quad \bar{w}^2\left(\varphi\left(z_{(m)}\right)\right) p\left(z_{(m)}\right) d z_{(m)}
\end{aligned}$$

For T_{3j} , we can carry out the change of variable from $z_{-(m)}$ to v as below:

$$v:=\frac{z_{-(m)}-\psi_j\left(z_{(m)}\right)}{b_n} \Leftrightarrow z_{-(m)}=\psi_j\left(z_{(m)}\right)+b_n v$$

and thus

$$\begin{aligned}
T_{3j} &= \int K_{-(m)}^2(v)\left[\sigma_0^2\left(z_{(m)}, \psi_j\left(z_{(m)}\right)+b_n v\right)+\left(h_0\left(z_{(m)}, \psi_j\left(z_{(m)}\right)+b_n v\right)-h_0\left(\varphi\left(z_{(m)}\right)\right)\right)^2\right] \\
&\quad p\left(\psi_j\left(z_{(m)}\right)+b_n v\left|z_{(m)}\right.\right) b_n^{d-m} d v
\end{aligned}$$

$$\begin{aligned}
&= b_n^{d-m} \int K_{-(m)}^2(v) \left[\sigma_0^2(z_{(m)}, \psi_j(z_{(m)})) + (h_0(z_{(m)}, \psi_j(z_{(m)})) - h_0(\varphi(z_{(m)})))^2 + O(b_n) \right] \\
&\quad \left[p(\psi_j(z_{(m)}) | z_{(m)}) + O(b_n) \right] dv \\
&= b_n^{d-m} \int K_{-(m)}^2(v) dv \cdot \sigma_0^2(z_{(m)}, \psi_j(z_{(m)})) p(\psi_j(z_{(m)}) | z_{(m)}) + o(b_n^{d-m}) \\
&= b_n^{d-m} \cdot C \cdot \sigma_0^2(z_{(m)}, \psi_j(z_{(m)})) p(\psi_j(z_{(m)}) | z_{(m)}) + o(b_n^{d-m})
\end{aligned}$$

Substituting T_{3j} back into the expression for $\mathbb{E}(T_{2ij}^2)$, we obtain

$$\begin{aligned}
&\mathbb{E}(T_{2ij}^2) \\
&= \frac{1}{b_n^{2(d-m)}} \int \left[b_n^{d-m} \cdot C \cdot \sigma_0^2(z_{(m)}, \psi_j(z_{(m)})) p(\psi_j(z_{(m)}) | z_{(m)}) + o(b_n^{d-m}) \right] \bar{w}^2(\varphi(z_{(m)})) p(z_{(m)}) dz_{(m)} \\
&= \frac{1}{b_n^{d-m}} C \cdot \int \sigma_0^2(z_{(m)}, \psi_j(z_{(m)})) p(\psi_j(z_{(m)}) | z_{(m)}) \bar{w}^2(\varphi(z_{(m)})) p(z_{(m)}) dz_{(m)} + o\left(\frac{1}{b_n^{d-m}}\right) \\
&= O\left(\frac{1}{b_n^{d-m}}\right)
\end{aligned}$$

Hence,

$$\text{Var} \left[\frac{1}{n} \sum_i T_{2,ij} \right] \leq \frac{C}{n^2} \sum_{i=1}^n O\left(\frac{1}{b_n^{d-m}}\right) + R_3 = O\left(\frac{1}{nb_n^{d-m}}\right).$$

□

H Bias-Aware versus Undersmoothing Confidence Intervals

In the Monte Carlo simulations of Section 6, the confidence intervals are constructed via the undersmoothing approach: the bootstrap-Lepski procedure selects the rate-optimal sieve dimension $\hat{K}$, which is then inflated to $\tilde{K} = \lceil \sqrt{\log n} \cdot \hat{K} \rceil$, and the CI is constructed at $\tilde{K}$ using the standard normal critical value $z_{0.975} = 1.96$. In this appendix, we compare this approach with the bias-aware confidence interval construction of [Chen, Christensen and Kankanala \(2025b\)](#), which retains the rate-optimal $\hat{K}$ and instead inflates the critical value with an explicit bias buffer:

$$\text{CI}_{\text{BA}} := \left[\hat{\theta}(\hat{K}) \pm \left(z_{0.975} + \hat{A} \cdot \vartheta_{1-\hat{\alpha}}^* \right) \cdot \widehat{se}(\hat{K}) \right],$$

where $\hat{A} = \log \log \hat{J}$, $\vartheta_{1-\hat{\alpha}}^*$ is the $(1 - \hat{\alpha})$ -quantile of the bootstrap-Lepski maximal contrast statistic (see Appendix E), and $\hat{\alpha} = \min(0.5, \sqrt{\log K_{\max}/K_{\max}})$. The bias-

aware construction builds on the honest confidence interval framework of [Armstrong and Kolesár \(2018, 2020\)](#) and was developed by [Chen, Christensen and Kankanala \(2025b\)](#) for linear functionals of the nonparametric estimator; no analogous construction for the general class of nonlinear functionals has been developed in the literature. We examine both designs from Section 6 to compare the two approaches.

Design 1: Linear integral on the unit circle. Table 7 compares the undersmoothing (US) and bias-aware (BA) confidence intervals for the linear integral functional considered in Section 6.1. The bias-aware CI is shorter than the undersmoothing CI at every sample size, with the reduction ranging from 3.2% at $n = 500$ to 4.5% at $n = 1000$. At the same time, the bias-aware CI achieves valid coverage at all sample sizes, though somewhat conservative (98.8%–99.2%) relative to the 95% nominal level. This is consistent with the fact that for this linear functional, the sieve bias at the rate-optimal $\hat{K} \approx 17$ is already of negligible order relative to the standard deviation, so the bias buffer adds an unnecessary but harmless inflation to the critical value. Notably, the rate-optimal $\hat{K} \approx 17$ is substantially smaller than the undersmoothed $\tilde{K} \approx 51$, which means the bias-aware estimator operates with lower variance, contributing to the shorter CI length.

Table 7: Design 1 (Linear Integral): Undersmoothing vs. Bias-Aware CI

	CI Length (U–L)		Coverage		Mean Sieve Dim.	
n	US	BA	US	BA	$\tilde{K}$	$\hat{K}$
500	2.3465	2.2704	93.8%	98.8%	50.6	17.1
1000	1.6589	1.5847	95.2%	98.9%	50.7	16.8
2000	1.1796	1.1295	95.0%	98.9%	51.4	17.1
4000	0.8338	0.7980	94.4%	99.2%	51.4	17.2
8000	0.5901	0.5657	95.8%	99.2%	51.8	17.4

“US”: undersmoothing with $\tilde{K} = \lceil \sqrt{\log n} \cdot \hat{K} \rceil$ and normal CV $z_{0.975} = 1.96$. “BA”: bias-aware CI at rate-optimal $\hat{K}$ with inflated CV $z_{0.975} + \hat{A} \cdot \vartheta_{1-\hat{\alpha}}^*$.

Design 2: Nonlinear integral on the estimated unit disk. Table 8 presents the analogous comparison for the nonlinear upper contour set integral considered in Section 6.2. Although the bias-aware CI construction of [Chen, Christensen and](#)

Kankanala (2025b) was not designed for nonlinear functionals, we apply it here to examine its empirical behavior outside the scope of its theoretical guarantees.

In contrast to Design 1, the bias-aware CI is *wider* than the undersmoothing CI at all sample sizes. At $n = 500$, the difference is modest (7–8%), but the gap grows substantially with n , reaching approximately 25% at $n = 8000$. This occurs because the nonlinear functional exhibits substantial sieve bias at the rate-optimal $\hat{K} \approx 17$ –36: the bias is 25–60% of the standard deviation, far from negligible. The bias buffer $\hat{A} \cdot \vartheta_{1-\hat{\alpha}}^*$ inflates the standard error at $\hat{K}$, which is itself already larger than the standard error at the undersmoothed $\tilde{K} \approx 51$ –94. While the bias-aware CI maintains valid coverage (95.5%–96.4%), the undersmoothing approach achieves comparable or better coverage with shorter intervals, because it directly reduces the bias by increasing the sieve dimension.

That the bias-aware approach performs less well for the nonlinear functional is expected to some extent, given that the construction was designed for linear functionals. However, the precise magnitudes—the 7–25% increase in CI length relative to undersmoothing, growing with n —were not obvious a priori. This exercise highlights the robustness of the undersmoothing approach: it achieves valid and efficient confidence intervals for both linear and nonlinear functionals, whereas the bias-aware approach, while theoretically well-founded and effective for linear functionals, produces wider confidence intervals when applied to nonlinear functionals where substantial sieve bias persists at the rate-optimal sieve dimension.

Table 8: Design 2 (Nonlinear Integral): Undersmoothing vs. Bias-Aware CI

	CI Length (U-L)		Coverage		Mean Sieve Dim.	
n	US	BA	US	BA	$\tilde{\hat{K}}$	$\hat{\hat{K}}$
(A) Plug-In						
500	0.2659	0.2852	94.1%	96.2%	51.1	17.3
1000	0.1902	0.2044	95.2%	95.5%	55.3	18.0
2000	0.1370	0.1507	95.0%	96.0%	65.7	21.7
4000	0.0991	0.1148	95.4%	96.4%	81.9	28.2
8000	0.0721	0.0901	95.5%	96.2%	94.4	35.9
(B) Leave-One-Out						
500	0.3314	0.3584	94.8%	96.0%	50.5	17.1
1000	0.1966	0.2055	96.2%	96.0%	54.9	17.8
2000	0.1394	0.1510	95.3%	95.7%	64.9	21.4
4000	0.1004	0.1150	95.4%	96.4%	82.0	28.3
8000	0.0726	0.0906	95.6%	96.3%	94.4	35.9

“US”: undersmoothing with $\tilde{\hat{K}} = \lceil \sqrt{\log n} \cdot \hat{K} \rceil$ and normal CV $z_{0.975} = 1.96$. “BA”: bias-aware CI at rate-optimal $\hat{K}$ with inflated CV $z_{0.975} + \hat{A} \cdot \vartheta_{1-\hat{\alpha}}^*$.